UPSC ISS · STATISTICS PAPER-I · OBJECTIVE

Statistical Methods

Every previous-year question from 2018 to 2026, with answer, exam shortcut, tips & tricks and a step-by-step solution.

Questions in this volume	**164** of 720 across the four units
Papers covered	9 — 2018 to 2026
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Layout of every entry	QUESTION → OPTIONS → ANSWER → EXAM SHORTCUT → TIPS & TRICKS → STEP-BY-STEP SOLUTION
Dataset version	1.0

What is authentic and what is not. The questions and their four options are reproduced word for word from the original UPSC ISS Statistics Paper-I booklets. Wording, option text, option order and question numbers are unaltered.

None of the source booklets carried an official answer key. Every ANSWER, EXAM SHORTCUT, TIPS & TRICKS note and STEP-BY-STEP SOLUTION in this volume was worked out for this project and is marked *AI-derived explanation* where it appears. They are a study aid, not an official UPSC solution.

Source defects. 5 question(s) in this volume have a defect in the printed paper. None has been silently corrected: the original text is reproduced exactly and the problem is described under the question.

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S1 · STATISTICAL METHODS

Collection, Compilation & Presentation of Data; Frequency Distributions

2 questions · 2018: 1 2019: 0 2020: 0 2021: 0 2022: 0 2023: 1 2024: 0 2025: 0 2026: 0

Weight. 2 questions (1.2% of Statistical Methods) · appeared in 2 of 9 papers · peak year 2018 (1)

Examiner's pattern. Almost dead — two items in nine years, both conceptual: scales of measurement (nominal/ordinal/interval/ratio) and what counts as data condensation. Do not spend preparation time here beyond knowing the four scales.

Must-know. Nominal < ordinal < interval < ratio in permitted operations; only ratio scale has a true zero, so only it permits meaningful ratios.

1. 2018 · Q21 t-Distribution · Conceptual

QUESTION

Consider the following statements: 1. The weight of a strawberry box is measured on an interval scale. 2. The number of students admitted to each IIM in 2007 is cross-sectional data. 3. Barcodes for items in a departmental store are measured on a nominal scale.

OPTIONS

(a) 1 and 2 only

(b) 1 and 3 only

(c) 2 and 3 only

(d) 1, 2 and 3

ANSWER (AI-derived — no official key exists for this paper)

(c) 2 and 3 only

EXAM SHORTCUT (AI-derived explanation)

Weight has a true zero and meaningful ratios, so it is RATIO scale, not interval - statement 1 fails. Statements 2 and 3 are textbook-correct, so the answer is 2 and 3 only.

TIPS & TRICKS (AI-derived explanation)

- Scale ladder: nominal (labels) < ordinal (ranks) < interval (equal gaps, arbitrary zero) < ratio (true zero). Weight, length, income and counts are ratio; temperature in Celsius and calendar dates are interval.
- The single discriminator for interval vs ratio is 'is zero meaningful?'. Zero weight means no weight, so weight is ratio.
- Cross-sectional = many units at ONE time point; time series = one unit over MANY time points; panel = both. 'Each IIM in 2007' is many units, one year, so cross-sectional.
- Barcodes, PIN codes, jersey numbers and Aadhaar numbers are all nominal even though they look numeric - arithmetic on them is meaningless.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement 1: the weight of a strawberry box is a physical measurement with an absolute zero point, and ratios such as 'twice as heavy' are meaningful. That is the definition of a RATIO scale, not an interval scale. FALSE.

2. Statement 2: the number of students admitted to each IIM in 2007 records many different units (the IIMs) at a single point in time. That is cross-sectional data. TRUE.
3. Statement 3: a barcode is only an identifying label; the numbers cannot be ordered or added meaningfully, so the measurement is on a nominal scale. TRUE.
4. Only statements 2 and 3 are correct, so option (c) is right.

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2. 2023 · Q21 Frequency Distribution & Presentation of Data · Conceptual

QUESTION

Which of the following are examples of data condensing? 1. Data array 2. Frequency distribution 3. Histogram 4. Ogives Select the correct answer using the code given below:

OPTIONS

(a) 1, 2 and 4 only

(b) 1 and 3 only

(c) 2, 3 and 4 only

(d) 1, 2, 3 and 4

ANSWER (AI-derived — no official key exists for this paper)

(c) 2, 3 and 4 only

EXAM SHORTCUT (AI-derived explanation)

Condensation means reducing bulk. A frequency distribution, a histogram and an ogive all summarise the data into classes; a data array merely re-orders the same n observations without reducing them.

TIPS & TRICKS (AI-derived explanation)

- Test each device by asking whether the number of items reported FALLS. A data array still lists every observation, so nothing is condensed.
- A frequency distribution replaces n raw values by a handful of class frequencies - the primary act of condensation.
- Histograms and ogives are drawn FROM the frequency distribution, so they inherit its condensed class structure and present it visually.
- Keep the vocabulary apart: arraying = ordering, classification and tabulation = condensing, diagrams and graphs = presenting condensed data.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Condensation of data means reducing a mass of raw observations to a compact summary that is easier to comprehend.
2. Item 1, data array: the raw observations are simply arranged in ascending or descending order. All n values are still listed, so the bulk is unchanged - this is organisation, not condensation.
3. Item 2, frequency distribution: the observations are grouped into classes and only the class frequencies are reported, which genuinely reduces the volume of data. CONDENSING.
4. Item 3, histogram: a graphical display of the grouped frequency distribution, presenting the condensed class information. CONDENSING.
5. Item 4, ogive: the cumulative frequency curve, again built from the grouped distribution. CONDENSING.

6. Hence items 2, 3 and 4 are examples of data condensing, option (c).

S2 · STATISTICAL METHODS

Measures of Location & Dispersion

10 questions · 2018: 0 2019: 2 2020: 1 2021: 3 2022: 1 2023: 1 2024: 0 2025: 0 2026: 2

Weight. 10 questions (6.1% of Statistical Methods) · appeared in 6 of 9 papers · peak year 2021 (3)

Examiner's pattern. Mean deviation is minimum about the median — asked directly and also embedded in a ratio. Other repeats: effect of y = ax + b on mean, SD and MD; comparing variability across two variables via the coefficient of variation; pooling two groups' means and variances.

Must-know. MD(ax+b) = |a| MD(x); SD(ax+b) = |a| SD(x); CV = 100sigma/mean; combined variance = [n1(s1^2 + d1^2) + n2(s2^2 + d2^2)]/(n1+n2) with di = mean_i - combined mean.

1. 2019 · Q1

Measures of Dispersion · Conceptual

QUESTION

Consider the following statements: 1. For ordinal and ratio scale data, median is a recommended measure of location. 2. Pie diagrams and bar diagrams are recommended graphs for nominal data. 3. For interval scale data, the recommended graph is histogram and the recommended measure of variability is standard deviation. Which of the above statements are correct?

OPTIONS

(a) 1 and 3 only

(b) 2 and 3 only

(c) 1 and 2 only

(d) 1, 2 and 3

ANSWER (AI-derived — no official key exists for this paper)

(b) 2 and 3 only

EXAM SHORTCUT (AI-derived explanation)

Match each scale to its recommended summary: nominal - mode, bar/pie; ordinal - median; interval/ratio - mean and SD, histogram. Statement 1 pairs the median with the RATIO scale, where the mean is recommended, so it fails. Statements 2 and 3 are correct.

TIPS & TRICKS (AI-derived explanation)

- Learn the scale-to-statistic table once: nominal gives mode, frequency, bar and pie; ordinal gives median, quartiles; interval and ratio give mean, SD and histogram.
- The permissible/recommended distinction is what the examiner exploits. The median is PERMISSIBLE at ratio level but the mean is RECOMMENDED there, because the ratio scale carries more information.
- Pie charts require categories that partition a whole, so they suit nominal data; histograms require a continuous numeric axis, so they suit interval/ratio data. Never draw a histogram of nominal categories.
- Standard deviation needs equal intervals to be meaningful, which is exactly why it is recommended from the interval scale upwards and not for ordinal data.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement 1: the median is the recommended measure of location for ORDINAL data, but for ratio-scale data the arithmetic mean is recommended because the scale supports addition and ratios. Pairing the median with the ratio scale is therefore not the recommended choice. FALSE.

2. Statement 2: nominal data consist of unordered categories with frequencies; the standard displays are bar diagrams and pie diagrams. TRUE.
3. Statement 3: interval-scale data are numeric with equal intervals, so a histogram is the appropriate graph and the standard deviation is the appropriate measure of variability. TRUE.
4. Only statements 2 and 3 are correct.
5. Hence option (b) is correct.

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2. 2019 · Q3 Sampling Distribution of Sample Mean · Numerical

QUESTION

For a group of 7-year-old boys: Body Height $\overline{X}_1 = 115$ cm, $s_1 = 6.5$ cm; Body Weight $\overline{X}_2 = 23$ kg, $s_2 = 2.7$ kg, where $\overline{X}$ is the mean of the group and s is the group's standard deviation. Which one of the following is correct?

OPTIONS

(a) Boys are five times as variable in body weight than in body height

(b) Boys are twice as variable in body weight than in body height

(c) Boys are $\frac{2}{5}$ times as variable in body weight than in body height

(d) Boys are $\frac{2}{5}$ times as variable in body height than in body weight

ANSWER (AI-derived — no official key exists for this paper)

(b) Boys are twice as variable in body weight than in body height

EXAM SHORTCUT (AI-derived explanation)

Different units force a RELATIVE comparison. CV(height) = 6.5/115 = 5.65% and CV(weight) = 2.7/23 = 11.74%. The ratio is 11.74/5.65 = 2.08, so weight is about twice as variable.

TIPS & TRICKS (AI-derived explanation)

- Whenever two variables have different units (cm and kg), never compare standard deviations directly - compute the coefficient of variation, CV = s/mean.
- Estimate the ratio mentally: 2.7/23 is a little over 11% while 6.5/115 is a little under 6%. Roughly 11 against 6 is about 2, which selects option (b) without a calculator.
- The CV is dimensionless precisely so that such comparisons are legitimate; it is also the reason the answer is a pure number with no units.
- Options (c) and (d) invert the comparison. Fix in your mind which variable is the SUBJECT of the sentence ('variable in body weight') before choosing.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Because height is in centimetres and weight in kilograms, the two standard deviations cannot be compared directly; we use the coefficient of variation CV = (s/mean) x 100%.
2. For body height: CV1 = (6.5/115) x 100 = 5.65%.
3. For body weight: CV2 = (2.7/23) x 100 = 11.74%.
4. The ratio CV2/CV1 = 11.74/5.65 = 2.08, which is approximately 2.
5. Hence the boys are about twice as variable in body weight as in body height.

6. Option (b) is correct.

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3. 2020 · Q11 Measures of Dispersion · Conceptual

QUESTION

Consider the following statements: 1. If original scores are multiplied by a constant, mean and standard deviation are also multiplied by that constant. 2. If original scores are multiplied by a constant, the SD is multiplied by the absolute value of the constant. 3. If a constant is added to each score, the SD remains unchanged.

OPTIONS

(a) 1 and 2 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

ANSWER (AI-derived — no official key exists for this paper)

(b) 2 and 3 only

EXAM SHORTCUT (AI-derived explanation)

Statement 1 is imprecise: multiplying by a NEGATIVE constant multiplies the mean by it but multiplies the SD by its absolute value. Statements 2 and 3 state the rules correctly.

TIPS & TRICKS (AI-derived explanation)

- Scaling rules: mean(cX) = c mean(X) but SD(cX) = |c| SD(X). The absolute value is compulsory because a standard deviation can never be negative.
- Shift rules: mean(X + k) = mean(X) + k but SD(X + k) = SD(X). Dispersion is unaffected by a change of origin.
- Statements 1 and 2 make claims about the same operation, and one is a corrected version of the other - a sure sign that the imprecise one is the false statement.
- Variance scales as c^2: $Var(cX) = c^2 Var(X)$. Taking the square root recovers |c|, which is where the absolute value comes from.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let Y = cX for a constant c. Then mean(Y) = c mean(X) and $Var(Y) = c^2 Var(X)$, so SD(Y) = |c| SD(X).
2. Statement 1 asserts that both the mean AND the standard deviation are multiplied by the constant. This is true for the mean but false for the SD when c is negative, since a standard deviation is non-negative. FALSE as stated.
3. Statement 2 asserts that the SD is multiplied by the ABSOLUTE VALUE of the constant, which is exactly correct. TRUE.
4. Statement 3: for Y = X + k, Var(Y) = Var(X), so the SD is unchanged by a change of origin. TRUE.
5. Only statements 2 and 3 are correct.
6. Option (b) is correct.

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4. 2021 · Q13 Measures of Dispersion · Conceptual

QUESTION

Consider the following statements: 1. Mean deviation is minimum when deviations are taken from the median. 2. If the units of two series differ, their variabilities can still be compared by standard deviations. 3. A series with smaller coefficient of variation is more consistent. Which is/are correct?

OPTIONS

(a) 1 and 2 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 3 only

ANSWER (AI-derived — no official key exists for this paper)

(c) 1 and 3 only

EXAM SHORTCUT (AI-derived explanation)

Statement 2 is false because standard deviations carry units and cannot be compared across different units; the coefficient of variation must be used. Statements 1 and 3 are standard results.

TIPS & TRICKS (AI-derived explanation)

- Mean deviation is minimised about the MEDIAN; the sum of SQUARED deviations is minimised about the MEAN. Keep the pair straight - each is a classic one-mark fact.
- SD has the same units as the data, so comparing an SD in kilograms with one in centimetres is meaningless. The CV = (SD/mean) x 100 is dimensionless and fixes this.
- Smaller CV means less relative variability, hence greater consistency or uniformity - the standard interpretation in comparison problems.
- Statements 2 and 3 are effectively about the same issue from opposite sides, so deciding one settles the other.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement 1: for any constant a, the sum of |x_i - a| is minimised when a is the median of the data. This is a standard property of the mean deviation. TRUE.
2. Statement 2: the standard deviation is measured in the same units as the original data, so two series in different units cannot be compared by their SDs. The correct tool is the coefficient of variation, which is unit-free. FALSE.
3. Statement 3: the coefficient of variation measures relative dispersion, so the series with the smaller CV shows less relative variability and is described as more consistent. TRUE.
4. Only statements 1 and 3 are correct.
5. Hence option (c) is correct.

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5. 2021 · Q31 Measures of Dispersion · Numerical

QUESTION

Frequency distribution:

Class	1-5	6-10	11-15	16-20	21-25	26-30
Frequency	10	15	25	25	15	10

Match List I with List II: List I: A. First decile B. First quartile C. Median D. Quartile deviation List II: 1. 5 2. 5.5 3. 10 4. 10.5 5. 15 6. 15.5

OPTIONS

(a) A-2, B-4, C-6, D-1

(b) A-1, B-3, C-5, D-2

(c) A-2, B-4, C-5, D-3

(d) A-1, B-3, C-4, D-2

ANSWER (AI-derived — no official key exists for this paper)

(a) A-2, B-4, C-6, D-1

EXAM SHORTCUT (AI-derived explanation)

Cumulative frequencies 10, 25, 50, 75, 90, 100 land exactly on the class boundaries, so D1 = 5.5, Q1 = 10.5, Median = 15.5 and Q3 = 20.5, giving QD = (20.5-10.5)/2 = 5. Matching gives A-2, B-4, C-6, D-1.

TIPS & TRICKS (AI-derived explanation)

- Convert INCLUSIVE classes (1-5, 6-10, ...) to exclusive boundaries (0.5-5.5, 5.5-10.5, ...) before using any positional formula - the 0.5 correction is what produces the .5 endings in the options.
- The cumulative frequencies here hit 10, 25, 50 and 75 exactly at class boundaries, so every quantile sits on a boundary and no interpolation within a class is needed.
- Quartile deviation = (Q3 - Q1)/2, a SEMI-interquartile range. Forgetting to halve gives 10, which is also on the list as a distractor.
- The distribution is symmetric (10, 15, 25, 25, 15, 10), so the median must sit at the centre of the range, 15.5 - a useful confirmation.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Convert to exclusive boundaries: 0.5-5.5, 5.5-10.5, 10.5-15.5, 15.5-20.5, 20.5-25.5, 25.5-30.5, each of width h = 5, with N = 100.
2. Cumulative frequencies: 10, 25, 50, 75, 90, 100.
3. First decile: N/10 = 10 falls at the end of the first class, so D1 = 0.5 + (10 - 0)/10 x 5 = 5.5 (List II item 2).
4. First quartile: N/4 = 25 falls at the end of the second class, so Q1 = 5.5 + (25 - 10)/15 x 5 = 10.5 (item 4).
5. Median: N/2 = 50 falls at the end of the third class, so M = 10.5 + (50 - 25)/25 x 5 = 15.5 (item 6). Similarly Q3 = 15.5 + (75 - 50)/25 x 5 = 20.5.
6. Quartile deviation = (Q3 - Q1)/2 = (20.5 - 10.5)/2 = 5 (item 1). Hence A-2, B-4, C-6, D-1, option (a).

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6. 2021 · Q39 Measures of Dispersion · Conceptual

QUESTION

If the mean deviation of x from its mean is 5, the mean deviation of $y=2x+3$ from its mean is

OPTIONS

(a) 17

(b) 13

(c) 10

(d) 5

ANSWER (AI-derived — no official key exists for this paper)

(c) 10

EXAM SHORTCUT (AI-derived explanation)

Mean deviation scales by the absolute value of the multiplier and is unaffected by a shift, so MD(2x + 3) = 2 x 5 = 10.

TIPS & TRICKS (AI-derived explanation)

- All measures of DISPERSION (range, MD, SD, QD) are unchanged by a change of origin and scale by |a| under a change of scale. Only the mean and other location measures absorb the +3.
- The +3 shifts both the values and their mean by 3, so the deviations are identical - the constant cancels exactly.
- Use |a|, not a: a multiplier of -2 would also give 10, since deviations are measured in absolute value.
- Option (b) 13 = 2(5) + 3 is the trap for candidates who apply the whole transformation to the dispersion measure instead of just the scale factor.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let y = 2x + 3. Then the mean of y is ybar = 2 xbar + 3.
2. The deviation of each observation from its mean is y_i - ybar = (2x_i + 3) - (2 xbar + 3) = 2(x_i - xbar).
3. Taking absolute values, |y_i - ybar| = 2|x_i - xbar|.
4. Averaging, MD(y) = 2 MD(x).
5. With MD(x) = 5, MD(y) = 2 x 5 = 10.
6. Option (c) is correct.

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7. 2022 · Q51 Sampling Distribution of Sample Mean · Theoretical

QUESTION

For n observations $x_1,\ldots,x_n$, consider: 1. $R=\frac{\sum|x_i-M|}{\sum|x_i-\overline{X}|}$, M(median)$\neq\overline{X}$(mean) $\Rightarrow$ $R<1$. 2. $R=\frac{\sum(x_i-A)^2}{\sum(x_i-\overline{X})^2}$, A(assumed mean)$\neq\overline{X}$ $\Rightarrow$ $R>1$. 3. $R=\frac{\sum(x_i-M)^2}{\sum(x_i-\overline{X})^2}$, M(median)$\neq\overline{X}$ $\Rightarrow$ $R>1$. (all sums run i=1 to n.) Which statements are correct?

OPTIONS

(a) 1 and 2 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

ANSWER (AI-derived — no official key exists for this paper)

(d) 1, 2 and 3

EXAM SHORTCUT (AI-derived explanation)

The sum of absolute deviations is minimised at the MEDIAN and the sum of squared deviations at the MEAN. Statement 1 follows from the first, statements 2 and 3 from the second. All correct.

TIPS & TRICKS (AI-derived explanation)

- Two minimisation facts drive every question in this family: sum |x_i - a| is smallest at a = median; sum (x_i - a)^2 is smallest at a = mean.
- Statement 1: the numerator is taken about the median (the minimiser) and the denominator about the mean, so the numerator is smaller and R < 1.
- Statements 2 and 3: the DENOMINATOR is now the minimum, so any other centre - an assumed mean A or the median M - makes the numerator larger and R > 1.
- Useful identity: sum(x_i - A)^2 = sum(x_i - xbar)^2 + n(xbar - A)^2, which makes the strict inequality obvious whenever A is not xbar.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Fact 1: for any constant a, sum |x_i - a| attains its minimum when a is the median M. Fact 2: sum (x_i - a)^2 attains its minimum when a is the mean xbar, by the identity sum(x_i - a)^2 = sum(x_i - xbar)^2 + n(xbar - a)^2.
2. Statement 1: the numerator uses the minimising centre M and the denominator the non-minimising centre xbar (given M is not equal to xbar), so the numerator is smaller and R < 1. CORRECT.
3. Statement 2: here the DENOMINATOR uses the minimising centre xbar for squared deviations, while the numerator uses A which is not xbar. By the identity, the numerator exceeds the denominator by n(xbar - A)^2 > 0, so R > 1. CORRECT.
4. Statement 3: this is the same situation with A replaced by the median M, which is assumed different from xbar, so again R > 1. CORRECT.
5. All three statements follow from the two minimisation properties.
6. Hence option (d) is correct.

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8. **2023 · Q22** Measures of Dispersion · Numerical

QUESTION

For the variable X, following observations are taken: X=2,5,0,6,3,8,1,3,0,2. What is $\mathrm{Var}(2X+2)$?

OPTIONS

(a) 24.8

(b) 26.8

(c) 28.8

(d) 29.6

ANSWER (AI-derived — no official key exists for this paper)

(a) 24.8

EXAM SHORTCUT (AI-derived explanation)

Mean = 30/10 = 3 and sum of squares = 152, so Var(X) = 15.2 - 9 = 6.2. Then Var(2X+2) = 4(6.2) = 24.8.

TIPS & TRICKS (AI-derived explanation)

- $Var(aX + b) = a^2 Var(X)$ - the constant b vanishes and the multiplier is SQUARED. Adding 2 changes nothing.
- Use the computing formula Var = (sum x^2)/n - $xbar^2$ rather than deviations: 152/10 - 3^2 = 6.2 in one line.
- Check the two totals separately: sum x = 30 and sum x^2 = 4+25+0+36+9+64+1+9+0+4 = 152. Recomputing these is where marks are lost.
- This is the population (divisor n) variance; with divisor n - 1 the answer would be 6.889 and 27.6, so note which convention the options match - here it is n.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The ten observations are 2, 5, 0, 6, 3, 8, 1, 3, 0, 2.
2. Sum = 30, so the mean is xbar = 30/10 = 3.
3. Sum of squares = 4 + 25 + 0 + 36 + 9 + 64 + 1 + 9 + 0 + 4 = 152.
4. Var(X) = (sum x^2)/n - $xbar^2$ = 152/10 - 9 = 15.2 - 9 = 6.2.
5. For a linear transformation, Var(2X + 2) = 2^2 Var(X) = 4(6.2) = 24.8, option (a).

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9. **2026 · Q28** Measures of Dispersion · Numerical

QUESTION

Sample 1 has 100 items, mean 15, sd 3. The whole group (250 items) has mean 15.6, sd $\sqrt{13.44}$. What is the variance of the second group?

OPTIONS

(a) 16

(b) 12

(c) 4

(d) 2

ANSWER (AI-derived — no official key exists for this paper)

(a) 16

EXAM SHORTCUT (AI-derived explanation)

Second group size = 150; combined mean gives 1500 + 150 m2 = 3900, so m2 = 16. Deviations are -0.6 and +0.4. Then 250(13.44) = 100(9 + 0.36) + 150($s2^2$ + 0.16), i.e. 3360 = 960 + 150 $s2^2$ + 24, so $s2^2$ = 16.

TIPS & TRICKS (AI-derived explanation)

- Two-step routine: combined mean first (to get the missing group mean), then the combined-variance formula. You cannot do the variance without the mean.
- Combined variance formula: N*$sigma^2$ = sum over groups of n_i(sigma_i^2 + d_i^2), where d_i = mean_i - combined mean.
- Compute the deviations d_1 = 15 - 15.6 = -0.6 and d_2 = 16 - 15.6 = 0.4 and note n_1 d_1 + n_2 d_2 = -60 + 60 = 0 - a free check that the group mean is right.
- The question asks for the variance, not the standard deviation: 16, not 4. Option (c) is the SD trap.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Group sizes: n_1 = 100, n_2 = 250 - 100 = 150, N = 250.

2. Combined mean: (n_1 m_1 + n_2 m_2)/N = 15.6, so 100(15) + 150 m_2 = 250(15.6) = 3900. Thus 150 m_2 = 3900 - 1500 = 2400 and m_2 = 16.
3. Deviations from the combined mean: d_1 = 15 - 15.6 = -0.6 and d_2 = 16 - 15.6 = 0.4. (Check: 100(-0.6) + 150(0.4) = -60 + 60 = 0.)
4. Combined variance formula: N sigma^2 = n_1(sigma_1^2 + d_1^2) + n_2(sigma_2^2 + d_2^2).
5. Substituting: 250(13.44) = 100(3^2 + 0.36) + 150(sigma_2^2 + 0.16).
6. 3360 = 100(9.36) + 150 sigma_2^2 + 24 = 936 + 24 + 150 sigma_2^2 = 960 + 150 sigma_2^2.
7. 150 sigma_2^2 = 2400, so sigma_2^2 = 16.
8. Hence option (a) is correct.

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10. 2026 · Q40 Measures of Location · Numerical

QUESTION

What is the mean of the numbers $1\cdot n^2$, $2\cdot(n\text{-}1)^2$, $3\cdot(n\text{-}2)^2$, …, $n\cdot 1^2$?

OPTIONS

(a) $(n+1)^2(n+2)/12$

(b) $(n+1)^2/4$

(c) $n(n+1)^2/12$

(d) $(n+2)^2(n+4)/12$

ANSWER (AI-derived — no official key exists for this paper)

(a) $(n+1)^2(n+2)/12$

EXAM SHORTCUT (AI-derived explanation)

The k-th term is k(n-k+1)^2. Reversing the index turns the sum into sum of (n+1-j)j^2 = (n+1)*sum j^2 - sum j^3 = n(n+1)^2(n+2)/12. Dividing by n gives (n+1)^2(n+2)/12.

TIPS & TRICKS (AI-derived explanation)

- Write the general term first: the k-th number is k * (n - k + 1)^2, since the second factor counts down as the first counts up.
- Substituting j = n - k + 1 converts the awkward product into (n + 1 - j) j^2, which splits into the standard sums of squares and cubes.
- Standard sums: sum j^2 = n(n+1)(2n+1)/6 and sum j^3 = [n(n+1)/2]^2. Factor n(n+1)^2 out of both before combining.
- Verify with n = 2: the numbers are 1*4 = 4 and 2*1 = 2, mean 3; the formula gives (3^2)(4)/12 = 36/12 = 3. Correct - a ten-second check that also kills options (b), (c) and (d).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The k-th number in the list is a_k = k (n - k + 1)^2, for k = 1, 2, ..., n. (Check: k = 1 gives 1 *n^2 and k = n gives n* 1^2.)
2. Sum S = sum from k = 1 to n of k(n - k + 1)^2.
3. Substitute j = n - k + 1, so k = n + 1 - j and j runs from n down to 1: S = sum from j = 1 to n of (n + 1 - j) j^2.

4. Split: $S = (n + 1) \text{ sum } j^2 - \text{sum } j^3 = (n + 1) * n(n+1)(2n+1)/6 - [n(n+1)/2]^2 = n(n+1)^2 (2n+1)/6 - n^2 (n+1)^2/4 = n(n+1)^2 [(2n+1)/6 - n/4]$.
5. Inside the bracket: $(2(2n+1) - 3n)/12 = (4n + 2 - 3n)/12 = (n + 2)/12$.
6. So $S = n(n+1)^2 (n+2)/12$.
7. Mean $= S/n = (n + 1)^2 (n + 2)/12$.
8. Verification at $n = 2$: numbers are 4 and 2 with mean 3; formula gives $9 * 4/12 = 3$. Correct.
9. Hence option (a) is correct.

S3 · STATISTICAL METHODS

Skewness & Kurtosis (Moments)

4 questions · 2018: 1 2019: 0 2020: 0 2021: 1 2022: 0 2023: 1 2024: 0 2025: 1 2026: 0

Weight. 4 questions (2.4% of Statistical Methods) · appeared in 4 of 9 papers · peak year 2018 (1)

Examiner's pattern. Small but consistent: back out sigma from Pearson's coefficient of skewness, the third central moment of a binomial, recovering (n,p) from gamma1 and gamma2, and extracting a fourth central moment from a characteristic function.

Must-know. Sk = (mean - mode)/sigma = 3(mean - median)/sigma; binomial mu3 = npq(q-p), mu4 = 3n^2q^2p^2 + npq(1-6pq); gamma1 = mu3/sigma^3, gamma2 = mu4/sigma^4 - 3.

1. 2018 · Q23 Skewness · Numerical

QUESTION

For a frequency distribution: Mean $=62$, Median $=65$, Coefficient of skewness $=-0.3$. The standard deviation is

OPTIONS

(a) 10

(b) 30

(c) 90

(d) 300

ANSWER (AI-derived — no official key exists for this paper)

(b) 30

EXAM SHORTCUT (AI-derived explanation)

Use the empirical (median) form of Pearson's skewness: Sk = 3(Mean - Median)/SD. So -0.3 = 3(62 - 65)/SD = -9/SD, giving SD = 30.

TIPS & TRICKS (AI-derived explanation)

- Two forms of Pearson's coefficient: (Mean - Mode)/SD and 3(Mean - Median)/SD. When the MEDIAN is supplied, the factor 3 is compulsory - dropping it gives SD = 10, which is planted as option (a).
- The sign works out automatically: mean below median gives a negative numerator and hence negative skewness, consistent with the given -0.3.
- The relation comes from the empirical mode formula Mode = 3 Median - 2 Mean, which is worth memorising in its own right.
- Sanity check the magnitude: |Mean - Median| = 3 and the coefficient is small (0.3), so the SD must be an order of magnitude larger than 3 - that alone points to 30.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Pearson's coefficient of skewness in its median form is Sk = 3(Mean - Median)/SD.
2. Substitute the given values: -0.3 = 3(62 - 65)/SD.
3. The numerator is 3 x (-3) = -9, so -0.3 = -9/SD.
4. Solving, SD = -9/(-0.3) = 30.

5. Hence the standard deviation is 30, option (b).

---

2. 2021 · Q30 Skewness · Numerical

QUESTION

$X\sim$ Binomial(n,p) with skewness γ_1=1/6 and kurtosis γ_2=-1/12. What are n,p?

OPTIONS

(a) 9, 2/3

(b) 18, 1/3

(c) 18, 2/3

(d) 9, 1/3

ANSWER (AI-derived — no official key exists for this paper)

(b) 18, 1/3

EXAM SHORTCUT (AI-derived explanation)

Write A = npq. Then (1-4pq) = A/36 and 1 - 6pq = -A/12. Solving gives pq = 2/9 and A = 4, so n = 18; the positive skewness forces q > p, i.e. p = 1/3.

TIPS & TRICKS (AI-derived explanation)

- Binomial shape measures: gamma_1 = (q - p)/sqrt(npq) and gamma_2 = (1 - 6pq)/(npq). Both have npq in the denominator, which is what lets you eliminate n.
- Use gamma_1^2 = (q-p)^2/(npq) = (1 - 4pq)/(npq) - squaring converts the skewness into an equation in pq alone once combined with the kurtosis equation.
- The SIGN of the skewness decides between p and q: gamma_1 > 0 means q > p, so p is the smaller root 1/3. Options (a) and (c) take p = 2/3 and fail this test.
- Verify at the end: n = 18, p = 1/3 gives npq = 4, so gamma_1 = (1/3)/2 = 1/6 and gamma_2 = (1 - 4/3)/4 = -1/12, both matching.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For the binomial, gamma_1 = (q - p)/sqrt(npq) and gamma_2 = (1 - 6pq)/(npq). Write A = npq and t = pq.
2. Squaring the skewness: (q - p)^2/A = 1/36. Since (q-p)^2 = (p+q)^2 - 4pq = 1 - 4t, this gives 1 - 4t = A/36.
3. From the kurtosis: (1 - 6t)/A = -1/12, so 1 - 6t = -A/12, i.e. A = 12(6t - 1) = 72t - 12.
4. Substituting into the first equation: 1 - 4t = (72t - 12)/36 = 2t - 1/3, so 4/3 = 6t and t = pq = 2/9.
5. Then A = 72(2/9) - 12 = 16 - 12 = 4, and since A = n t = 2n/9 = 4, we get n = 18. With p + q = 1 and pq = 2/9, p and q are 1/3 and 2/3.
6. Since gamma_1 = +1/6 > 0 we need q > p, so p = 1/3. Hence n = 18 and p = 1/3, option (b).

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3. 2023 · Q23 Skewness · Numerical

QUESTION

What is the 3rd central moment of Binomial distribution with parameters n=10 and p=1/2?

OPTIONS

(a) 25

(b) 5

(c) 2.5

(d) 0

ANSWER (AI-derived — no official key exists for this paper)

(d) 0

EXAM SHORTCUT (AI-derived explanation)

The third central moment of B(n,p) is npq(q - p). With p = q = 1/2 the factor (q - p) is 0, so the moment is 0.

TIPS & TRICKS (AI-derived explanation)

- Formula: mu_3 = npq(q - p). Its sign is the sign of the skewness, and it vanishes exactly when p = 1/2.
- A binomial with p = 1/2 is SYMMETRIC about n/2, and every odd central moment of a symmetric distribution is zero - no formula is needed.
- Do not confuse mu_3 (third central moment) with the coefficient of skewness mu_3/sigma^3; both are 0 here, but they are different quantities.
- For comparison, mu_2 = npq = 2.5 here, which is planted as option (c) to catch a mis-read of the moment order.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For the binomial distribution the third central moment is mu_3 = n p q (q - p).
2. Here n = 10 and p = 1/2, so q = 1 - p = 1/2.
3. Therefore q - p = 1/2 - 1/2 = 0.
4. Hence mu_3 = 10 (1/2)(1/2)(0) = 0.
5. This is consistent with the fact that B(n, 1/2) is symmetric about n/2, so all its odd central moments vanish.
6. Option (d) is correct.

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4. 2025 · Q50 Skewness · Numerical

QUESTION

The characteristic function of X is $\Phi_X(t) = e^{2t(i\text{-}t)}$; $|t| \leq 1$. What is the fourth-order central moment of the distribution?

OPTIONS

(a) 12

(b) 24

(c) 36

(d) 48

ANSWER (AI-derived — no official key exists for this paper)

(d) 48

EXAM SHORTCUT (AI-derived explanation)

Rewrite the characteristic function as e^(2it - 2t^2) = e^(i(2)t - (4)t^2/2), identifying N(2, 4). The fourth central moment of a normal is 3 sigma^4 = 3(16) = 48.

TIPS & TRICKS (AI-derived explanation)

- Normal characteristic function: phi(t) = exp(i mu t - sigma^2 t^2/2). Expand the given exponent 2t(i - t) = 2it - 2t^2 and match term by term.
- From -sigma^2/2 = -2 we get sigma^2 = 4, NOT 2 - the factor of 2 in the denominator is where marks are lost.
- Fourth central moment of N(mu, sigma^2) is 3 sigma^4 = 3(4^2) = 48. Remember the sequence 0, sigma^2, 0, 3 sigma^4 for the first four central moments.
- Option (a) 12 would come from 3 sigma^2 and option (b) 24 from mis-reading sigma^2 as 2 (giving 3 x 4 x 2) - both are near-misses built on the same slip.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Expand the exponent: 2t(i - t) = 2it - 2t^2.
2. The characteristic function of N(mu, sigma^2) is phi(t) = exp(i mu t - sigma^2 t^2/2).
3. Matching the linear term: i mu t = 2it, so mu = 2.
4. Matching the quadratic term: -sigma^2 t^2/2 = -2t^2, so sigma^2 = 4.
5. For a normal distribution the fourth central moment is mu_4 = 3 sigma^4 = 3(4)^2 = 3(16) = 48, option (d).

S4 · STATISTICAL METHODS

Bivariate & Multivariate Data; Association & Contingency

12 questions · 2018: 2 2019: 2 2020: 2 2021: 1 2022: 1 2023: 1 2024: 0 2025: 3 2026: 0

Weight. 12 questions (7.3% of Statistical Methods) · appeared in 7 of 9 papers · peak year 2025 (3)

Examiner's pattern. 2x2 attribute tables are the whole topic. Three asks recur: Yule's coefficient of association Q, the consistency conditions on class frequencies, and the positive/negative/independent verdict by comparing (AB) with (A)(B)/N. The coefficient of contingency from a chi-square value shows up once.

Must-know. Q = (ad - bc)/(ad + bc); independence iff (AB) = (A)(B)/N; consistency requires every ultimate class frequency >= 0; C = sqrt(chi2/(chi2 + N)).

1. 2018 · Q24

Association & Contingency · Numerical

QUESTION

Smoking habits (father, son): 1. Father smoker, son smoker: 50 2. Father smoker, son not: 10 3. Father not, son smoker: 10 4. Neither smoker: 30 The coefficient of association between smoking habits of father and son is

OPTIONS

(a) 0

(b) $\frac{7}{8}$

(c) $-\frac{7}{8}$

(d) $\frac{1}{4}$

ANSWER (AI-derived — no official key exists for this paper)

(b) $\frac{7}{8}$

EXAM SHORTCUT (AI-derived explanation)

Yule's Q = (ad - bc)/(ad + bc) with a = 50, b = 10, c = 10, d = 30. So Q = (1500 - 100)/(1500 + 100) = 1400/1600 = 7/8.

TIPS & TRICKS (AI-derived explanation)

- Yule's coefficient of association Q = (ad - bc)/(ad + bc) always lies between -1 and +1; it is 0 exactly when ad = bc, i.e. under independence.
- Set the 2x2 table out as a = both present, b and c = one present, d = both absent. Here the diagonal product ad = 1500 dominates the off-diagonal bc = 100, so the association must be strongly POSITIVE - that alone eliminates (c).
- Do not confuse Yule's Q with the coefficient of colligation Y = (sqrt(ad) - sqrt(bc))/(sqrt(ad) + sqrt(bc)), which would give a different value here.
- Sons of smokers smoking 50:10 while sons of non-smokers smoke 10:30 is a clear positive association - a quick common-sense check on the sign before computing.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Arrange the 2x2 contingency table: (AB) = 50 (both smoke), (A beta) = 10 (father smokes, son does not), (alpha B) = 10 (father does not, son does), (alpha beta) = 30 (neither smokes).

2. Yule's coefficient of association is Q = [(AB)(alpha beta) - (A beta)(alpha B)] / [(AB)(alpha beta) + (A beta)(alpha B)].
3. Diagonal product: (AB)(alpha beta) = 50 x 30 = 1500.
4. Off-diagonal product: (A beta)(alpha B) = 10 x 10 = 100.
5. Q = (1500 - 100)/(1500 + 100) = 1400/1600 = 7/8.
6. The value is positive and close to 1, indicating strong positive association. Option (b) is correct.

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2. 2018 · Q25 Association & Contingency · Definition

QUESTION

Consider the following statements: 1. Attributes A,B are positively associated if the presence of A is accompanied by the presence of B. 2. A,B are negatively associated if the presence of A ensures the absence of B or vice versa. 3. A,B are independent if the presence/absence of one has a linear relationship with the other.

OPTIONS

(a) 1, 2 and 3

(b) 1 and 2 only

(c) 1 and 3 only

(d) 2 and 3 only

ANSWER (AI-derived — no official key exists for this paper)

(b) 1 and 2 only

EXAM SHORTCUT (AI-derived explanation)

Statements 1 and 2 are the standard verbal definitions of positive and negative association. Statement 3 misdefines independence as a 'linear relationship' - independence means NO relationship at all. So 1 and 2 only.

TIPS & TRICKS (AI-derived explanation)

- Independence of attributes means (AB) = (A)(B)/N, i.e. the observed joint frequency equals the product of the marginals. Nothing about linearity is involved - linearity belongs to correlation of variables, not association of attributes.
- The examiner's favourite trap in this chapter is importing correlation language (linear, positive, negative) into attribute theory. Read every option for that swap.
- Association compares (AB) with (A)(B)/N: greater means positive association, smaller means negative association, equal means independence. Three cases, one formula.
- Whenever one statement in a set is a definition that has been subtly reworded, that is almost always the false one - here statement 3.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Two attributes A and B are said to be INDEPENDENT when (AB) = (A)(B)/N, where N is the total frequency.
2. Statement 1: A and B are positively associated when the presence of A tends to be accompanied by the presence of B, i.e. (AB) > (A)(B)/N. This is the standard verbal definition. TRUE.
3. Statement 2: A and B are negatively associated (disassociated) when the presence of one tends to exclude the other, i.e. (AB) < (A)(B)/N. TRUE.

4. Statement 3: independence is defined by the frequency identity above, not by any 'linear relationship'. Linearity is a concept of correlation between quantitative variables, not of association between attributes. FALSE.
5. Only statements 1 and 2 are correct, so option (b) is right.

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3. 2019 · Q4 Association & Contingency · Numerical

QUESTION

If for two attributes A and B: $(A) = 50$, $(B) = 60$, $(AB) = 40$, $N = 100$, then

OPTIONS

(a) A and B are independent

(b) A and B are positively associated

(c) A and B are negatively associated

(d) nothing definite can be said about the association between A and B without further information

ANSWER (AI-derived — no official key exists for this paper)

(b) A and B are positively associated

EXAM SHORTCUT (AI-derived explanation)

Under independence the expected (AB) is (A)(B)/N = 50 x 60/100 = 30. The observed 40 exceeds 30, so A and B are positively associated.

TIPS & TRICKS (AI-derived explanation)

- The whole test is one comparison: observed (AB) against (A)(B)/N. Greater means positive association, smaller means negative, equal means independence.
- Compute (A)(B)/N first and keep it as a single number - here 30 - then a one-glance comparison with 40 finishes the item.
- Yule's Q would confirm the sign: the 2x2 table is (AB)=40, (A beta)=10, (alpha B)=20, (alpha beta)=30, so ad - bc = 1200 - 200 > 0, again positive.
- Option (d) is a trap for candidates who think a 2x2 table is insufficient - it is entirely sufficient, since the four cell frequencies follow from (A), (B), (AB) and N.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For two attributes A and B, independence means (AB) = (A)(B)/N.
2. Here (A) = 50, (B) = 60 and N = 100, so the expected frequency under independence is (50)(60)/100 = 30.
3. The observed frequency is (AB) = 40, which is greater than 30.
4. Since the observed joint frequency exceeds the frequency expected under independence, the attributes are positively associated.
5. As a check, the full table is (AB) = 40, (A beta) = 10, (alpha B) = 20, (alpha beta) = 30, giving Yule's Q = (1200 - 200)/(1200 + 200) = 5/7 > 0.
6. Hence option (b) is correct.

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4. 2019 · Q26 Association & Contingency · Numerical

QUESTION

For $\chi^2 = 2.38$ and N=200, the coefficient of contingency C for the given data is

OPTIONS

(a) 0.01

(b) 0.1084

(c) 0.1096

(d) 0.1840

ANSWER (AI-derived — no official key exists for this paper)

(b) 0.1084

EXAM SHORTCUT (AI-derived explanation)

C = sqrt(chi^2/(chi^2 + N)) = sqrt(2.38/202.38) = sqrt(0.01176) = 0.1084.

TIPS & TRICKS (AI-derived explanation)

- Pearson's coefficient of contingency is C = sqrt(chi^2/(chi^2 + N)). The N in the denominator is the TOTAL frequency, added to chi-square, not multiplied.
- C always lies in [0, 1) and can never reach 1 - a useful check that your value is small when chi-square is small relative to N.
- Estimate before computing: 2.38/202 is about 0.0118, whose square root is about 0.109. Options (b) and (c) differ only in the fourth decimal, so carry one extra digit: 0.011760 gives 0.10844, i.e. 0.1084.
- Do not confuse C with Cramer's V = sqrt(chi^2/(N min(r-1, c-1))), which would give a different value here.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Pearson's coefficient of contingency is C = sqrt(chi^2/(chi^2 + N)).
2. Substituting chi^2 = 2.38 and N = 200: chi^2 + N = 2.38 + 200 = 202.38.
3. The ratio is 2.38/202.38 = 0.011760.
4. Taking the square root: C = sqrt(0.011760) = 0.10844.
5. To four decimal places this is 0.1084.
6. Hence option (b) is correct.

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5. 2020 · Q2 Association & Contingency · Numerical

QUESTION

Contingency table (Material vs. Leakage):

	A	B	C	D	E	Total
Leaked	36	22	X_1	X_2	42	200
Not leaked	90	Y	46	78	168	400

Column totals for materials C and D are equal. What are X_1, X_2, Y?

OPTIONS

(a) $66,34,18$

(b) 34,66,18

(c) 70,30,20

(d) 30,70,20

ANSWER **(AI-derived — no official key exists for this paper)**

(a) 66,34,18

EXAM SHORTCUT **(AI-derived explanation)**

Row sums give X1 + X2 = 100 and Y = 18. Equal column totals for C and D give X1 + 46 = X2 + 78, i.e. X1 - X2 = 32. Solving, X1 = 66 and X2 = 34.

TIPS & TRICKS **(AI-derived explanation)**

- Extract every equation the table offers BEFORE solving: each row total is one equation and the stated condition is another. Three unknowns need three equations.
- The 'not leaked' row is fully determined except for Y, so Y = 400 - (90 + 46 + 78 + 168) = 18 comes out immediately and independently.
- Sum and difference: X1 + X2 = 100 and X1 - X2 = 32 give X1 = 66 and X2 = 34 by inspection - add and halve.
- Options (a) and (b) differ only by swapping X1 and X2, so pin down which column is C: the larger leaked count goes with the SMALLER not-leaked count (46) to keep the totals equal.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. From the 'leaked' row: 36 + 22 + X1 + X2 + 42 = 200, so X1 + X2 = 200 - 100 = 100.
2. From the 'not leaked' row: 90 + Y + 46 + 78 + 168 = 400, so Y = 400 - 382 = 18.
3. The column total for material C is X1 + 46 and for material D is X2 + 78; these are stated to be equal.
4. Hence X1 + 46 = X2 + 78, giving X1 - X2 = 32.
5. Solving X1 + X2 = 100 with X1 - X2 = 32: X1 = 66 and X2 = 34.
6. Check: column C total = 66 + 46 = 112 and column D total = 34 + 78 = 112, equal as required. Option (a) is correct.

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6. **2020 · Q12** Association & Contingency · Theoretical

QUESTION

For attributes, the relations: 1. $(AB)>(A)$ 2. $(AB)<(A)+(B)-N$ 3. $(ABC)>(AB)+(AC)+(BC)-(A)-(B)-(C)+N$ Which are correct for **inconsistency** of data?

OPTIONS

(a) 1 and 2 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

ANSWER **(AI-derived — no official key exists for this paper)**

(d) 1, 2 and 3

EXAM SHORTCUT (AI-derived explanation)

Consistency requires every ultimate class frequency to be non-negative. Each of the three relations makes one such class negative - (AB) > (A) makes (A beta) < 0, (AB) < (A)+(B)-N makes (alpha beta) < 0, and the third makes (alpha beta gamma) < 0. All three signal inconsistency.

TIPS & TRICKS (AI-derived explanation)

- Test for consistency by computing the ULTIMATE class frequencies; the data are inconsistent as soon as any of them is negative.
- Two attributes give the standard bounds max(0, (A)+(B)-N) <= (AB) <= min((A),(B)). Statements 1 and 2 are simply violations of the upper and lower bound respectively.
- For three attributes, inclusion-exclusion gives (alpha beta gamma) = N - (A) - (B) - (C) + (AB) + (AC) + (BC) - (ABC). Requiring this to be non-negative gives (ABC) <= (AB)+(AC)+(BC)-(A)-(B)-(C)+N; statement 3 is the violation of that bound.
- When all listed relations are constructed as violations of the same non-negativity principle, expect 'all of them' - but derive at least the three-attribute one, since its sign is easy to get backwards.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A set of attribute frequencies is CONSISTENT only if every ultimate class frequency is non-negative.
2. Statement 1: since the class AB is contained in A, we always have (AB) <= (A); indeed (A beta) = (A) - (AB) >= 0. So (AB) > (A) forces (A beta) < 0 and the data are inconsistent. CORRECT indicator.
3. Statement 2: (alpha beta) = N - (A) - (B) + (AB) must be non-negative, which gives (AB) >= (A) + (B) - N. So (AB) < (A) + (B) - N makes (alpha beta) < 0 and the data are inconsistent. CORRECT indicator.
4. Statement 3: by inclusion-exclusion, (alpha beta gamma) = N - (A) - (B) - (C) + (AB) + (AC) + (BC) - (ABC).
5. Non-negativity of (alpha beta gamma) requires (ABC) <= (AB) + (AC) + (BC) - (A) - (B) - (C) + N. Hence the strict inequality in statement 3 makes (alpha beta gamma) < 0 and signals inconsistency. CORRECT indicator.
6. All three relations indicate inconsistency, option (d).

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7. 2021 · Q12 Association & Contingency · Theoretical

QUESTION

Consider the following conditions on attributes A and B: 1. $(AB) \leq (A)$ 2. $(AB) \leq (B)$ 3. $(AB) \geq (A)+(B)-N$ Which are required for consistency of the data on A and B?

OPTIONS

(a) 1 and 2 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

ANSWER (AI-derived — no official key exists for this paper)

(d) 1, 2 and 3

EXAM SHORTCUT (AI-derived explanation)

The class frequencies (A beta) = (A) - (AB), (alpha B) = (B) - (AB) and (alpha beta) = N - (A) - (B) + (AB) must all be non-negative, which gives exactly the three stated conditions.

TIPS & TRICKS (AI-derived explanation)

- Consistency = every ultimate class frequency is non-negative. Write the four cell frequencies in terms of (A), (B), (AB) and N, then impose non-negativity.
- Conditions 1 and 2 come from (A beta) >= 0 and (alpha B) >= 0; condition 3 comes from (alpha beta) >= 0. One condition per empty-looking cell.
- Combined form worth memorising: max(0, (A) + (B) - N) <= (AB) <= min((A), (B)) - the Frechet bounds for a 2x2 table.
- When each listed condition corresponds to a distinct cell of the table, all of them are needed - 'all three' is the natural key.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For two attributes the four ultimate class frequencies are (AB), (A beta) = (A) - (AB), (alpha B) = (B) - (AB) and (alpha beta) = N - (A) - (B) + (AB).
2. The data are consistent precisely when all four are non-negative.
3. (A beta) >= 0 gives (AB) <= (A) - condition 1.
4. (alpha B) >= 0 gives (AB) <= (B) - condition 2.
5. (alpha beta) >= 0 gives (AB) >= (A) + (B) - N - condition 3.
6. All three conditions are therefore required, option (d).

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8. 2022 · Q56 Association & Contingency · Numerical

QUESTION

Among 1482 persons exposed to COVID-19, 368 were attacked with the virus. Among the 1482, 343 were vaccinated, and among them only 35 were attacked. Consider: 1. Vaccination and attack are positively associated. 2. Vaccination may be a preventive measure.

OPTIONS

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

ANSWER (AI-derived — no official key exists for this paper)

(b) 2 only

EXAM SHORTCUT (AI-derived explanation)

Expected attacks among the vaccinated under independence = 343 x 368/1482 = 85.2, but only 35 were observed. Far fewer than expected means NEGATIVE association, so vaccination looks preventive - statement 2 only.

TIPS & TRICKS (AI-derived explanation)

- Independence test for attributes: compare the observed (AB) with (A)(B)/N. Observed above means positive association, below means negative.
- Rates make it vivid: 35/343 = 10.2% among the vaccinated against 333/1139 = 29.2% among the unvaccinated - a large protective difference.

- Statement 1 and statement 2 point in OPPOSITE directions, so exactly one can be right; deciding the sign of the association settles both.
- Yule's Q would confirm it: the diagonal product 35(806) is far below the off-diagonal 308(333), giving a strongly negative Q.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let V denote vaccination and A denote being attacked. Given N = 1482, (A) = 368, (V) = 343 and (AV) = 35.
2. Under independence the expected number both vaccinated and attacked would be (V)(A)/N = 343 x 368/1482 = 85.2.
3. The observed value 35 is far BELOW this, so vaccination and attack are negatively associated. Statement 1, which claims positive association, is FALSE.
4. Equivalently, the attack rate among the vaccinated is 35/343 = 10.2%, whereas among the 1139 unvaccinated it is (368 - 35)/1139 = 333/1139 = 29.2%.
5. A substantially lower attack rate among the vaccinated is exactly the evidence that vaccination may act as a preventive measure. Statement 2 is TRUE.
6. Only statement 2 is correct, option (b).

---

9. 2023 · Q24 Association & Contingency · Numerical

QUESTION

Consider the contingency table presenting the number of skilled and unskilled labour against their gender. What is the value of χ^2 for testing the association between gender and the nature of work?

	Skilled	Unskilled
Male	30	30
Female	20	20

OPTIONS

(a) 50

(b) 16.66

(c) 12.07

(d) 0

ANSWER (AI-derived — no official key exists for this paper)

(d) 0

EXAM SHORTCUT (AI-derived explanation)

Row totals 60 and 40, column totals 50 and 50, N = 100, so every expected frequency equals the observed one (30, 30, 20, 20). Hence chi-square = 0.

TIPS & TRICKS (AI-derived explanation)

- Compute one expected value first: 60 x 50/100 = 30, which already equals the observed 30 - a strong hint that the whole table is exactly independent.
- Independence test by cross products: ad - bc = 30(20) - 30(20) = 0, so the attributes are exactly independent and chi-square must be 0.

- In this table each row splits 50:50 between skilled and unskilled, so gender carries NO information about the nature of work.
- chi-square is non-negative and equals 0 only under exact independence - recognising that here saves all four cell computations.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The observed table is Male: 30 skilled, 30 unskilled; Female: 20 skilled, 20 unskilled.
2. Row totals are 60 and 40; column totals are 50 and 50; the grand total is N = 100.
3. Expected frequencies under independence: Male-Skilled = 60 x 50/100 = 30, Male-Unskilled = 30, Female-Skilled = 40 x 50/100 = 20, Female-Unskilled = 20.
4. Every expected frequency equals the corresponding observed frequency, so each term $(O - E)^2/E$ is zero.
5. Equivalently, ad - bc = 30(20) - 30(20) = 0, the condition for exact independence.
6. Hence chi-square = 0, option (d).

---

10. 2025 · Q20 Association & Contingency · Numerical

QUESTION

The male population in a state is 25 lakh. Among them, literates number 2 lakh, criminals number 26000, and literate criminals number 2000. What is the association between literacy and criminality?

OPTIONS

(a) No association

(b) Negatively associated

(c) Positively associated

(d) Association cannot be established

ANSWER (AI-derived — no official key exists for this paper)

(b) Negatively associated

EXAM SHORTCUT (AI-derived explanation)

Expected literate criminals under independence = (200000)(26000)/2500000 = 2080. The observed 2000 is lower, so literacy and criminality are negatively associated.

TIPS & TRICKS (AI-derived explanation)

- Independence test for attributes: compare the observed (AB) with (A)(B)/N. Below means negative association, above means positive.
- Simplify before multiplying: 200000/2500000 = 0.08, so the expected count is 0.08 x 26000 = 2080 - one multiplication instead of a large product.
- Rate comparison makes it vivid: 2000/200000 = 1.0% of literates are criminals against 24000/2300000 = 1.04% of illiterates - a small but genuine difference.
- The margin is narrow (2000 against 2080), so compute exactly rather than estimating; a rough calculation could easily suggest 'no association'.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let A denote literacy and B criminality, with N = 2,500,000, (A) = 200,000, (B) = 26,000 and (AB) = 2,000.
2. Under independence the expected number of literate criminals would be (A)(B)/N = 200,000 x 26,000/2,500,000 = 0.08 x 26,000 = 2,080.

3. The observed value 2,000 is LESS than the expected 2,080.
4. Since the observed joint frequency falls short of the frequency expected under independence, the attributes are negatively associated - literacy is associated with a slightly lower rate of criminality.
5. Hence they are negatively associated, option (b).

---

11. 2025 · Q53 Association & Contingency · Theoretical

QUESTION

For three distinct attributes A, B, C with $(A)=(\alpha)=(B)=(\beta)=(C)=(\gamma)=N/2$ such that $(ABC)=(\alpha\beta\gamma)$: I. $(AB)=(\alpha\beta)$, $(A\beta)=(\alpha B)$ II. $2(ABC) = (AB)+(AC)+(BC)-N/2$ Which is/are correct?

OPTIONS

(a) I only

(b) II only

(c) Both I and II

(d) Neither I nor II

ANSWER (AI-derived — no official key exists for this paper)

(c) Both I and II

EXAM SHORTCUT (AI-derived explanation)

With every attribute and its negation at N/2, the identity (alpha beta) = N - (A) - (B) + (AB) collapses to (AB), giving statement I. Inclusion-exclusion with (ABC) = (alpha beta gamma) then yields 2(ABC) = (AB)+(AC)+(BC) - N/2.

TIPS & TRICKS (AI-derived explanation)

- Two-attribute identity: (alpha beta) = N - (A) - (B) + (AB). With (A) = (B) = N/2 the constants cancel and (alpha beta) = (AB) exactly.
- The pair (A beta) = (alpha B) follows from the row and column totals: (AB) + (A beta) = (A) = N/2 and (AB) + (alpha B) = (B) = N/2.
- Three-attribute inclusion-exclusion: (alpha beta gamma) = N - (A)-(B)-(C) + (AB)+(AC)+(BC) - (ABC). Substitute (A)=(B)=(C)=N/2 to get -N/2 + sums - (ABC).
- Imposing (ABC) = (alpha beta gamma) then gives 2(ABC) = (AB)+(AC)+(BC) - N/2 - exactly statement II.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement I: from the 2x2 table, (AB) + (A beta) = (A) = N/2 and (AB) + (alpha B) = (B) = N/2, so subtracting gives (A beta) = (alpha B).
2. Also (alpha beta) = N - (A) - (B) + (AB) = N - N/2 - N/2 + (AB) = (AB). Both parts of statement I hold. TRUE.
3. Statement II: by inclusion-exclusion for three attributes, (alpha beta gamma) = N - (A) - (B) - (C) + (AB) + (AC) + (BC) - (ABC).
4. With (A) = (B) = (C) = N/2 this becomes (alpha beta gamma) = N - 3N/2 + (AB)+(AC)+(BC) - (ABC) = -N/2 + (AB)+(AC)+(BC) - (ABC).
5. Imposing the given condition (ABC) = (alpha beta gamma): (ABC) = -N/2 + (AB)+(AC)+(BC) - (ABC).
6. Rearranging, 2(ABC) = (AB) + (AC) + (BC) - N/2. TRUE. Both statements hold, option (c).

---

12. 2025 · Q60 Association & Contingency · Theoretical

QUESTION

For a 2×2 table of attributes A, B:

	B	not B
A	a	b
not A	c	d

Q = Yule's Coefficient of Association, Y = Coefficient of Colligation. I. Q=Y=1 if bc=0 II. Q=-1 if ad=0 III. Q=0 if ad=bc Which are correct?

OPTIONS

(a) I and II only

(b) II and III only

(c) I and III only

(d) I, II and III

ANSWER (AI-derived — no official key exists for this paper)

(d) I, II and III

EXAM SHORTCUT (AI-derived explanation)

Q = (ad - bc)/(ad + bc) and Y = (sqrt(ad) - sqrt(bc))/(sqrt(ad) + sqrt(bc)). If bc = 0 both equal 1; if ad = 0 then Q = -1; if ad = bc then Q = 0. All three statements hold.

TIPS & TRICKS (AI-derived explanation)

- Yule's Q and the coefficient of colligation Y always share the same SIGN and reach +-1 together - so complete association is detected identically by both.
- bc = 0 means one off-diagonal cell is empty (complete association), giving Q = ad/ad = 1 and Y = sqrt(ad)/sqrt(ad) = 1.
- ad = 0 means one diagonal cell is empty (complete disassociation), giving Q = -bc/bc = -1.
- ad = bc is precisely the independence condition, and it makes the numerator of Q vanish - so Q = 0 marks independence.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Yule's coefficient of association is Q = (ad - bc)/(ad + bc), and the coefficient of colligation is Y = (sqrt(ad) - sqrt(bc))/(sqrt(ad) + sqrt(bc)).
2. Statement I: if bc = 0 then Q = ad/ad = 1 and Y = sqrt(ad)/sqrt(ad) = 1, so both equal 1 - complete positive association. TRUE.
3. Statement II: if ad = 0 then Q = (0 - bc)/(0 + bc) = -1, indicating complete negative association. TRUE.
4. Statement III: if ad = bc the numerator of Q vanishes, so Q = 0. This is exactly the condition for the two attributes to be independent. TRUE.
5. All three statements are correct.
6. Hence option (d) is correct.

S5 · STATISTICAL METHODS

Curve Fitting & Orthogonal Polynomials

3 questions · 2018: 0 2019: 0 2020: 0 2021: 1 2022: 0 2023: 1 2024: 1 2025: 0 2026: 0

Weight. 3 questions (1.8% of Statistical Methods) · appeared in 3 of 9 papers · peak year 2021 (1)

Examiner's pattern. Thin (3 items) but never zero for long. Orthogonality is the recurring idea — the degree-3 orthogonal polynomial for a standard normal (Hermite), orthogonality checks on pairs of quadratic forms, and picking between competing least-squares models.

Must-know. Orthogonality: sum over the design of p_i(x) p_j(x) = 0 for i != j. Hermite: H3(x) = x^3 - 3x for the standard normal weight.

1. 2021 · Q17 Curve Fitting & Orthogonal Polynomials · Theoretical

QUESTION

If x is a standard normal variate, the third-degree orthogonal polynomial is

OPTIONS

(a) x^3+x^2-1

(b) x^3+3x+4

(c) x^3-3x

(d) x^3+x^2-6

ANSWER (AI-derived — no official key exists for this paper)

(c) x^3-3x

EXAM SHORTCUT (AI-derived explanation)

The orthogonal polynomials for the standard normal weight are the Hermite polynomials. The third is He_3(x) = x^3 - 3x.

TIPS & TRICKS (AI-derived explanation)

- Match weight to family: normal weight gives Hermite, e^(-x) on (0, infinity) gives Laguerre, constant weight on (-1,1) gives Legendre, and (1-x^2)^(-1/2) gives Chebyshev.
- First few probabilists' Hermite polynomials: He_0 = 1, He_1 = x, He_2 = x^2 - 1, He_3 = x^3 - 3x, He_4 = x^4 - 6x^2 + 3.
- Parity argument alone settles it: the normal weight is even, so an odd-degree orthogonal polynomial must be an ODD function. Options (a) and (d) contain even terms and are eliminated at once.
- Orthogonality to lower degrees is easy to verify: E(He_3(X)) = E(X^3) - 3E(X) = 0 and E(X . He_3(X)) = E(X^4) - 3E(X^2) = 3 - 3 = 0.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Orthogonal polynomials with respect to the standard normal density are the (probabilists') Hermite polynomials He_n(x).
2. They satisfy E[He_m(X) He_n(X)] = 0 for m not equal to n, where X ~ N(0,1).
3. The first few are He_0 = 1, He_1 = x, He_2 = x^2 - 1 and He_3 = x^3 - 3x.
4. Check orthogonality of x^3 - 3x to the lower degrees, using E(X) = 0, E(X^2) = 1, E(X^3) = 0, E(X^4) = 3: E(He_3) = 0 - 0 = 0, and E(X He_3) = E(X^4) - 3E(X^2) = 3 - 3 = 0.

5. The normal weight is symmetric, so an odd-degree orthogonal polynomial must be odd, which x^3 - 3x is.
6. Hence the third-degree orthogonal polynomial is x^3 - 3x, option (c).

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2. 2023 · Q37 Bivariate & Multivariate Data · Numerical

QUESTION

For a bivariate data, using the principles of least-squares, the following models were constructed: E_1: Y=3X+5 with $\sum(Y_i$-3X_i-5$)^2$=15.48 E_2: Y=-X^2+8X+2 with $\sum(Y_i$+X_i^2-8X_i-2$)^2$=13.36 What is the estimated value of Y for X=6?

OPTIONS

(a) 4.5

(b) 14

(c) 18.5

(d) 23

ANSWER **(AI-derived — no official key exists for this paper)**

(b) 14

EXAM SHORTCUT **(AI-derived explanation)**

Choose the model with the smaller residual sum of squares: E2 has 13.36 against E1's 15.48. Then Y = -36 + 48 + 2 = 14 at X = 6.

TIPS & TRICKS **(AI-derived explanation)**

- Least squares selects the model with the SMALLER sum of squared residuals - that single comparison decides which equation to substitute into.
- Compute both predictions to see the trap: E1 gives 3(6) + 5 = 23 and E2 gives 14. Both appear as options, so the model choice is the whole question.
- Substitute carefully into the quadratic: -(6^2) + 8(6) + 2 = -36 + 48 + 2 = 14. The leading minus sign applies to the square.
- The quadratic's advantage here is modest (13.36 against 15.48), a reminder that a better fit is not automatically a better model - but the question asks only for the least-squares choice.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. Two least-squares models are offered, with their residual sums of squares stated.
2. Model E1: Y = 3X + 5 with sum of squared residuals 15.48.
3. Model E2: Y = -X^2 + 8X + 2 with sum of squared residuals 13.36.
4. The principle of least squares prefers the model with the smaller residual sum of squares, so E2 is selected since 13.36 < 15.48.
5. Substituting X = 6 into E2: Y = -(36) + 8(6) + 2 = -36 + 48 + 2 = 14, option (b). (Model E1 would have given 23, which is offered as option (d).)

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3. 2024 · Q11 Order Statistics - Minimum, Maximum & Median · Numerical

QUESTION

Consider the following pairs: I. $x_1^2+2x_1x_2+x_2^2$; $y_1^2+2y_1y_2+y_2^2$ II. $x_1^2-2x_1x_2-x_2^2$; $2y_1^2+y_1y_2-y_2^2$ III. $x_1^2+2x_1x_2-2x_2^2$; $2y_1^2+y_1y_2+2y_2^2$ IV. $x_1^2+x_1x_2-x_2^2$; $y_1^2+y_1y_2-y_2^2$ How many of the above are non-orthogonal pairs?

OPTIONS

(a) Only one pair

(b) Only two pairs

(c) Only three pairs

(d) All four pairs

ANSWER (AI-derived — no official key exists for this paper)

(c) Only three pairs

EXAM SHORTCUT (AI-derived explanation)

Treat each quadratic form as its coefficient triple and take the dot product. Pairs I (6), II (1) and IV (3) give non-zero values, while pair III gives 2 + 2 - 4 = 0. So three pairs are non-orthogonal.

TIPS & TRICKS (AI-derived explanation)

- Reduce each form to the coefficient vector of (square, cross, square): e.g. $x1^2$ + 2x1x2 + $x2^2$ becomes (1, 2, 1). Orthogonality is then a simple dot product.
- Work through the four dot products systematically: I gives 1+4+1 = 6; II gives 2-2+1 = 1; III gives 2+2-4 = 0; IV gives 1+1+1 = 3.
- Only ONE pair is orthogonal, so THREE are non-orthogonal - read the question direction carefully, since it asks for the non-orthogonal count.
- Sign care in pair II: the coefficients are (1, -2, -1) and (2, 1, -1), so the middle product is -2 and the last is +1.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Represent each quadratic form by the vector of its coefficients in the order (first square, cross term, second square); two forms are orthogonal when the dot product of their coefficient vectors is zero.
2. Pair I: (1, 2, 1) and (1, 2, 1) give 1 + 4 + 1 = 6, not zero. NON-ORTHOGONAL.
3. Pair II: (1, -2, -1) and (2, 1, -1) give 2 - 2 + 1 = 1, not zero. NON-ORTHOGONAL.
4. Pair III: (1, 2, -2) and (2, 1, 2) give 2 + 2 - 4 = 0. ORTHOGONAL.
5. Pair IV: (1, 1, -1) and (1, 1, -1) give 1 + 1 + 1 = 3, not zero. NON-ORTHOGONAL.
6. Three of the four pairs are non-orthogonal, option (c).

S6 · STATISTICAL METHODS

Bivariate Normal Distribution

25 questions · 2018: 1 2019: 1 2020: 1 2021: 5 2022: 3 2023: 8 2024: 2 2025: 2 2026: 2

Weight. 25 questions (15.2% of Statistical Methods) · appeared in 9 of 9 papers · peak year 2023 (8)

Examiner's pattern. Very high yield (26 items) and highly formulaic. The recurring asks are the regression coefficient b_yx = rhosigma_y/sigma_x, the conditional law Y|X=x, independence of X+Y and X-Y, E[max(X,Y)], E(e^(X+Y)) via the MGF, and the characterisation 'every linear combination is normal'. 2021 and 2023 both ran multi-question BVN clusters.

Must-know. Y|X=x ~ N(mu_y + rho(sigma_y/sigma_x)(x - mu_x), sigma_y^2 (1 - rho^2)); for BVN(0,0,1,1,rho), E[max] = sqrt((1-rho)/pi); MGF = exp(t'mu + 0.5 t'St).

1. 2018 · Q27 Distribution of the Correlation Coefficient · Theoretical

QUESTION

Let $(X, Y) \sim BVN(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$. 1. X, Y are independent iff $\rho = 0$. 2. Every linear combination of X, Y is a normal variate. 3. The regressions of Y on X and X on Y are linear and homoscedastic.

OPTIONS

(a) 1 and 2 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

ANSWER (AI-derived — no official key exists for this paper)

(d) 1, 2 and 3

EXAM SHORTCUT (AI-derived explanation)

All three are standard properties of the bivariate normal: zero correlation gives independence, every linear combination is normal, and both regression lines are linear with constant (homoscedastic) conditional variance. Answer: all three.

TIPS & TRICKS (AI-derived explanation)

- 'rho = 0 implies independence' is TRUE for the bivariate normal and FALSE in general - the examiner exploits this asymmetry constantly. Check whether normality is stated.
- The conditional variance of Y given X is sigma_2^2(1 - rho^2), which does not depend on x. That constancy is exactly what homoscedasticity means.
- The conditional mean is E(Y|X=x) = mu_2 + rho (sigma_2/sigma_1)(x - mu_1) - linear in x, which gives the linear regression property.
- In a 1/2/3 item where every statement is a bookwork property of the same distribution, (d) 'all of them' is very often the key - but confirm the one you are least sure of, here statement 3.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement 1: for the bivariate normal, the joint density factorises into the product of the two marginal normal densities precisely when rho = 0. So independence holds if and only if rho = 0. TRUE. (Note this equivalence is special to the normal; in general rho = 0 does not imply independence.)

2. Statement 2: if (X,Y) is bivariate normal then aX + bY is univariate normal for all constants a, b - this is a defining property of the multivariate normal family. TRUE.
3. Statement 3: the conditional distribution of Y given X = x is N(mu_2 + rho (sigma_2/sigma_1)(x - mu_1), sigma_2^2(1 - rho^2)).
4. The conditional mean is linear in x, so the regression of Y on X is linear; the conditional variance sigma_2^2(1 - rho^2) is free of x, so the regression is homoscedastic. By symmetry the same holds for X on Y. TRUE.
5. All three statements are correct, so option (d) is right.

---

2. 2019 · Q7 Distribution of the Correlation Coefficient · Numerical

QUESTION

Let X, Y, Z be independently and identically distributed standard normal variates, and $(X+\theta Z, Y+\theta Z)$ follows a bivariate normal distribution with correlation coefficient 0.25. Then the absolute value of θ is

OPTIONS

(a) $\frac{1}{\sqrt{2}}$

(b) $\frac{1}{\sqrt{3}}$

(c) $\frac{1}{\sqrt{2}+1}$

(d) $\frac{1}{\sqrt{3}+1}$

ANSWER (AI-derived — no official key exists for this paper)

(b) $\frac{1}{\sqrt{3}}$

EXAM SHORTCUT (AI-derived explanation)

Cov(X+tZ, Y+tZ) = t^2 and each variance is 1 + t^2, so rho = t^2/(1+t^2) = 0.25. Cross-multiplying, 4t^2 = 1 + t^2 gives t^2 = 1/3 and |t| = 1/sqrt3.

TIPS & TRICKS (AI-derived explanation)

- The shared component Z is what creates the correlation: only the Z-terms survive in the covariance because X, Y, Z are independent.
- The form rho = t^2/(1 + t^2) is the standard 'common factor' correlation; note that it is always in [0,1) and increases in |t|, so a small rho means a small |t|.
- Invert quickly: rho = t^2/(1+t^2) rearranges to t^2 = rho/(1 - rho) = 0.25/0.75 = 1/3. Memorise that inversion - it saves the whole cross-multiplication.
- Only |theta| is asked because rho depends on theta^2; both +1/sqrt3 and -1/sqrt3 give correlation 0.25.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let U = X + theta Z and V = Y + theta Z, with X, Y, Z independent standard normal.
2. Var(U) = Var(X) + theta^2 Var(Z) = 1 + theta^2, and similarly Var(V) = 1 + theta^2.
3. Cov(U, V) = Cov(X, Y) + theta Cov(X, Z) + theta Cov(Z, Y) + theta^2 Var(Z) = 0 + 0 + 0 + theta^2 = theta^2.
4. Therefore rho = theta^2/(1 + theta^2).
5. Setting theta^2/(1 + theta^2) = 0.25 gives 4 theta^2 = 1 + theta^2, so 3 theta^2 = 1 and theta^2 = 1/3.

6. Hence |theta| = 1/sqrt(3), option (b).

---

3. 2020 · Q31 Bivariate Normal Distribution · Conceptual

QUESTION

A bivariate normal random vector (X, Y) with $V(X)=2,\ V(Y)=3,\ \mathrm{Cov}(X, Y)=\alpha$ does **not** possess a density function. Then

OPTIONS

(a) $\alpha=0$

(b) $\alpha=\pm\sqrt{6}$

(c) $|\alpha|<\sqrt{6}$

(d) $|\alpha|>6$

ANSWER (AI-derived — no official key exists for this paper)

(b) $\alpha=\pm\sqrt{6}$

EXAM SHORTCUT (AI-derived explanation)

A bivariate normal has no density exactly when its covariance matrix is singular, i.e. |rho| = 1, i.e. Cov^2 = V(X)V(Y) = 6. So alpha = +-sqrt(6).

TIPS & TRICKS (AI-derived explanation)

- Singular covariance matrix means the determinant V(X)V(Y) - Cov^2 = 0, which is the same as rho^2 = 1. Compute the determinant, do not reason about rho separately.
- A singular bivariate normal is concentrated on a LINE in the plane, which has zero area, so no two-dimensional density can exist.
- The Cauchy-Schwarz inequality guarantees |Cov| <= sqrt(V(X)V(Y)) = sqrt6, so |alpha| > 6 in option (d) is impossible for any distribution at all.
- Option (c) |alpha| < sqrt6 is precisely the case where the density DOES exist - the exact opposite of what is asked.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The bivariate normal density exists if and only if its covariance matrix is positive definite, i.e. non-singular.
2. The covariance matrix here is [[2, alpha], [alpha, 3]], with determinant 2(3) - alpha^2 = 6 - alpha^2.
3. The density fails to exist exactly when this determinant is zero, that is alpha^2 = 6.
4. Hence alpha = +sqrt(6) or -sqrt(6), which corresponds to correlation rho = +-1.
5. In that case the distribution is degenerate, concentrated on a straight line in the plane, so no two-dimensional density function exists.
6. Option (b) is correct.

---

4. 2021 · Q14 Bivariate Normal Distribution · Theoretical

QUESTION

Consider the following statements: 1. (X, Y) possesses a bivariate normal distribution iff every linear combination of X and Y is a normal variate. 2. If the marginals of X and Y are normal, then (X, Y) is always bivariate normal. Which is/are correct?

OPTIONS

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

ANSWER (AI-derived — no official key exists for this paper)

(a) 1 only

EXAM SHORTCUT (AI-derived explanation)

Statement 1 is the Cramer-Wold characterisation of the bivariate normal and is true. Statement 2 is false - normal marginals do not force joint normality.

TIPS & TRICKS (AI-derived explanation)

- Cramer-Wold device: a random vector is multivariate normal if and only if EVERY linear combination of its components is univariate normal. This is the standard definition, so statement 1 is true by definition.
- Standard counterexample for statement 2: take Z ~ N(0,1) and S = +-1 independently with equal probability; then Z and SZ are each N(0,1) but the pair is not bivariate normal (it is concentrated on two lines).
- The direction of implication matters: joint normality implies normal marginals, but not conversely - the classic one-way street of this chapter.
- The word 'always' in statement 2 is the giveaway; universal claims in this area are usually the false ones.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement 1: by the Cramer-Wold device, a random vector (X,Y) has a bivariate normal distribution if and only if aX + bY is univariate normal for every pair of constants a, b. This is the standard characterisation. TRUE.
2. Statement 2: joint normality implies normal marginals, but the converse fails.
3. A standard counterexample: let Z ~ N(0,1) and let S take the values +1 and -1 with probability 1/2 each, independently of Z. Put X = Z and Y = SZ.
4. Then both X and Y are N(0,1), but the pair (X,Y) is concentrated on the two lines y = x and y = -x and has no bivariate normal density.
5. So normal marginals do not imply bivariate normality. Statement 2 is FALSE.
6. Only statement 1 is correct, option (a).

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5. **2021 · Q15** Distribution of the Correlation Coefficient · Numerical

QUESTION

If $(X,Y) \sim BVN(0,0,1,1,\rho)$, what is $E[\max(X,Y)]$?

OPTIONS

(a) $\sqrt{\pi\rho}$

(b) $\sqrt{\frac{\rho}{\pi}}$

(c) $\sqrt{\pi(1-\rho)}$

(d) $\sqrt{\frac{1-\rho}{\pi}}$

ANSWER (AI-derived — no official key exists for this paper)

(d) $\sqrt{\frac{1-\rho}{\pi}}$

EXAM SHORTCUT (AI-derived explanation)

max(X,Y) = [(X+Y) + |X-Y|]/2. Here X - Y ~ N(0, 2(1-rho)), so E|X-Y| = sqrt(2(1-rho)) sqrt(2/pi) = 2 sqrt((1-rho)/pi), and half of that is sqrt((1-rho)/pi).

TIPS & TRICKS (AI-derived explanation)

- Same identity as the independent case, only the variance of X - Y changes: Var(X - Y) = 1 + 1 - 2rho = 2(1 - rho).
- For W ~ N(0, tau^2), E|W| = tau sqrt(2/pi). The two factors of sqrt(2) cancel neatly, leaving the tidy answer.
- Check rho = 0: the formula gives 1/sqrt(pi), matching the independent-case result of 2019 Q36 - a strong consistency test.
- Check rho = 1: X = Y so the maximum is X itself with mean 0, and the formula duly gives 0. Only option (d) satisfies both limits.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For a standard bivariate normal pair, use max(X, Y) = [(X + Y) + |X - Y|]/2.
2. E(X + Y) = 0 since both means are zero.
3. X - Y is normal with mean 0 and variance Var(X) + Var(Y) - 2 Cov(X,Y) = 1 + 1 - 2rho = 2(1 - rho).
4. With tau = sqrt(2(1-rho)) and the half-normal mean formula E|W| = tau sqrt(2/pi), we get E|X - Y| = sqrt(2(1-rho)) sqrt(2/pi) = 2 sqrt((1-rho)/pi).
5. Therefore E[max(X,Y)] = [0 + 2 sqrt((1-rho)/pi)]/2 = sqrt((1 - rho)/pi).
6. Checks: rho = 0 gives 1/sqrt(pi) and rho = 1 gives 0, both correct. Option (d).

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6. **2021 · Q18** Bivariate Normal Distribution · Numerical

COMMON DATA FOR THIS GROUP

Q18–Q20 refer to the following: Let (X, Y) follow a Bivariate Normal distribution with joint p.d.f.

$$f(x,y)=k\exp[-\frac{8}{27}\{(x-7)^2-2(x-7)(y+5)+4(y+5)^2\}],\ (x,y)\in \mathrm{R}^2$$

QUESTION

What are $V(X)$ and $V(Y)$ respectively?

OPTIONS

(a) 9/4 and 9/16

(b) 9/16 and 9/32

(c) 9/4 and 9/32

(d) 9/16 and 9/16

ANSWER (AI-derived — no official key exists for this paper)

(a) 9/4 and 9/16

EXAM SHORTCUT (AI-derived explanation)

Matching the quadratic form: the ratio of the u^2 and v^2 coefficients is 1 : 4, so $sigma\_2^2$ = $sigma\_1^2/4$; the cross term then forces rho = 1/2, whence $1/(2(1-rho^2))$ = 2/3 and $sigma\_1^2$ = (2/3)(27/8) = 9/4, $sigma\_2^2$ = 9/16.

TIPS & TRICKS (AI-derived explanation)

- Write u = x - 7 and v = y + 5 first; the means are read off directly as mu_1 = 7 and mu_2 = -5 before any algebra.
- Compare the printed form Q = $A[u^2 - 2uv + 4v^2]$ with the standard $[u^2/s1^2 - 2\ rho\ uv/(s1\ s2) + v^2/s2^2]$ divided by $2(1-rho^2)$. Three coefficient equations determine rho, s1 and s2.
- Shortcut for rho: dividing the cross-term equation by the u^2 equation gives rho = s2/s1, so the ratio of the standard deviations IS the correlation here - a neat structural fact of this particular form.
- Sanity check: V(X) = 9/4 > V(Y) = 9/16, consistent with the smaller coefficient on u^2 (a larger variance means a flatter quadratic in that direction).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Put u = x - 7 and v = y + 5, so the exponent is $-(8/27)[u^2 - 2uv + 4v^2]$.
2. The bivariate normal exponent is $-(1/(2(1-rho^2)))[u^2/s1^2 - 2\ rho\ u\ v/(s1\ s2) + v^2/s2^2]$. Write $A = 1/(2(1-rho^2))$.
3. Matching coefficients: $A/s1^2$ = 8/27, A rho/(s1 s2) = 8/27, and $A/s2^2$ = 4(8/27) = 32/27.
4. Dividing the first by the third: $s2^2/s1^2$ = (8/27)/(32/27) = 1/4, so s1 = 2 s2. Dividing the second by the first: rho $s1^2$/(s1 s2) = 1, i.e. rho s1/s2 = 1, so rho = s2/s1 = 1/2.
5. Then A = 1/(2(1 - 1/4)) = 2/3, and from $A/s1^2$ = 8/27 we get $s1^2$ = (2/3)(27/8) = 9/4.
6. Hence V(X) = 9/4 and V(Y) = $s1^2/4$ = 9/16, option (a).

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7. 2021 · Q19 Bivariate Normal Distribution · Numerical

COMMON DATA FOR THIS GROUP

Q18–Q20 refer to the following: Let (X, Y) follow a Bivariate Normal distribution with joint p.d.f.

$$f(x,y)=k\exp[-\frac{8}{27}\{(x-7)^2-2(x-7)(y+5)+4(y+5)^2\}],\ (x,y)\in \mathrm{R}^2$$

QUESTION

What is the approximate value of k?

OPTIONS

(a) $0.31/\pi$

(b) $0.41/\pi$

(c) $0.51/\pi$

(d) $0.61/\pi$

ANSWER (AI-derived — no official key exists for this paper)

(c) $0.51/\pi$

EXAM SHORTCUT (AI-derived explanation)

k = 1/(2 pi s1 s2 sqrt(1-rho^2)) with s1 = 3/2, s2 = 3/4, rho = 1/2. The product s1 s2 sqrt(3)/2 = (9/8)(0.866) = 0.9743, so k = 1/(2 pi x 0.9743) = 0.51/pi.

TIPS & TRICKS (AI-derived explanation)

- The bivariate normal normalising constant is 1/(2 pi s1 s2 sqrt(1 - rho^2)) - memorise it as '2 pi times the two SDs times the root of one minus rho squared'.
- Take square roots of the variances first: s1 = 3/2 and s2 = 3/4 from V(X) = 9/4 and V(Y) = 9/16. Using the variances directly is the standard error here.
- The options are all of the form c/pi, so factor out 1/pi early: k = 1/(2 pi (0.9743)) = (1/1.9486)/pi = 0.513/pi.
- sqrt(1 - 1/4) = sqrt(3)/2 = 0.866 - keep four significant figures, since the options differ only in the second decimal of c.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. From the previous part, s1^2 = 9/4 and s2^2 = 9/16, so s1 = 3/2 and s2 = 3/4, with rho = 1/2.
2. The bivariate normal density has normalising constant k = 1/[2 pi s1 s2 sqrt(1 - rho^2)].
3. Compute sqrt(1 - rho^2) = sqrt(1 - 1/4) = sqrt(3)/2 = 0.8660.
4. s1 s2 = (3/2)(3/4) = 9/8 = 1.125, so s1 s2 sqrt(1-rho^2) = 1.125 x 0.8660 = 0.9743.
5. Hence k = 1/(2 pi x 0.9743) = 1/(1.9486 pi) = 0.5132/pi.
6. To two decimals this is 0.51/pi, option (c).

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8. **2021 · Q20** Bivariate Normal Distribution · Numerical

COMMON DATA FOR THIS GROUP

Q18–Q20 refer to the following: Let (X, Y) follow a Bivariate Normal distribution with joint p.d.f.

$$f(x,y)=k\exp[-\frac{8}{27}\{(x-7)^2-2(x-7)(y+5)+4(y+5)^2\}],\ (x,y)\in \mathrm{R}^2$$

QUESTION

What is the line of regression of Y on X?

OPTIONS

(a) $Y=6.75X-0.25$

(b) $Y=0.25X-6.75$

(c) $Y=6.25X+0.45$

(d) $Y=6.25X+6.75$

ANSWER (AI-derived — no official key exists for this paper)

(b) $Y=0.25X-6.75$

EXAM SHORTCUT (AI-derived explanation)

The regression of Y on X is E(Y|X=x) = mu_2 + rho(s2/s1)(x - mu_1) = -5 + (1/2)(1/2)(x - 7) = 0.25x - 6.75.

TIPS & TRICKS (AI-derived explanation)

- Regression coefficient b_YX = rho s2/s1 = (1/2)(3/4)/(3/2) = 1/4. Compute this single number first and the line follows.
- The slope of a regression line can never exceed |rho| times the SD ratio, so a slope of 6.75 (options (a), (c), (d)) is impossible here - the answer is identifiable almost by inspection.
- Constant term: -5 - 0.25(7) = -5 - 1.75 = -6.75. Keep the sign of mu_2 = -5, which comes from the (y + 5) in the exponent.
- Check that the line passes through the means: at x = 7, Y = 0.25(7) - 6.75 = -5 = mu_2, as every regression line must.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. From the exponent, mu_1 = 7 and mu_2 = -5; from the earlier parts, s1 = 3/2, s2 = 3/4 and rho = 1/2.
2. The regression of Y on X for a bivariate normal is E(Y | X = x) = mu_2 + rho (s2/s1)(x - mu_1).
3. The regression coefficient is b_YX = rho s2/s1 = (1/2)(3/4)/(3/2) = (1/2)(1/2) = 1/4 = 0.25.
4. Substituting: Y = -5 + 0.25(x - 7) = -5 + 0.25x - 1.75.
5. So Y = 0.25x - 6.75.
6. Check: at x = 7 the line gives -5, the mean of Y, as required. Option (b) is correct.

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9. 2022 · Q13 Bivariate Normal Distribution · Numerical

QUESTION

If $(X,Y) \sim BVN(10,20,100,100,0.5)$, what is $E(e^{X+Y})$?

OPTIONS

(a) e^{150}

(b) e^{180}

(c) e^{100}

(d) e^{220}

ANSWER (AI-derived — no official key exists for this paper)

(b) e^{180}

EXAM SHORTCUT (AI-derived explanation)

X + Y is normal with mean 30 and variance 100 + 100 + 2(0.5)(100) = 300. Then E(e^W) = exp(mean + variance/2) = exp(30 + 150) = e^{180}.

TIPS & TRICKS (AI-derived explanation)

- The parameters in BVN(mu1, mu2, sigma1^2, sigma2^2, rho) are VARIANCES here, both 100, so each standard deviation is 10.
- Var(X+Y) = sigma1^2 + sigma2^2 + 2 rho sigma1 sigma2 = 100 + 100 + 2(0.5)(10)(10) = 300. Forgetting the covariance term gives 200 and the wrong answer.

- E(e^W) for W ~ N(m, v) is exp(m + v/2) - the MGF at t = 1. The halving of the variance is the step most often dropped.
- Arithmetic: 30 + 300/2 = 30 + 150 = 180. Option (a) e^{150} comes from forgetting the mean; option (d) e^{220} from using v rather than v/2 with the wrong variance.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. (X,Y) ~ BVN(10, 20, 100, 100, 0.5), so E(X) = 10, E(Y) = 20, Var(X) = Var(Y) = 100 and rho = 0.5, giving sigma_X = sigma_Y = 10.
2. Any linear combination of a bivariate normal pair is normal, so W = X + Y is normal.
3. E(W) = 10 + 20 = 30.
4. Var(W) = Var(X) + Var(Y) + 2 Cov(X,Y) = 100 + 100 + 2(0.5)(10)(10) = 200 + 100 = 300.
5. For W ~ N(m, v), E(e^W) = exp(m + v/2) = exp(30 + 150).
6. Hence E(e^(X+Y)) = e^{180}, option (b).

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10. 2022 · Q15 Distribution of the Correlation Coefficient · Theoretical

QUESTION

Let (X_1,X_2) follow bivariate normal with $E(X_1)=E(X_2)=0$, $V(X_1)=V(X_2)=1$, $\mathrm{Corr}(X_1,X_2)=\rho$. What is $E(X_1^{\,2}X_2^{\,2})$?

OPTIONS

(a) $\rho^2+\frac{1}{2}$

(b) $\frac{\rho^2}{2}+\frac{2}{3}$

(c) $3\rho^2+4$

(d) $2\rho^2+1$

ANSWER (AI-derived — no official key exists for this paper)

(d) $2\rho^2+1$

EXAM SHORTCUT (AI-derived explanation)

For a standard bivariate normal pair, $E(X1^2\ X2^2) = 1 + 2\ rho^2$ - a standard fourth-moment result.

TIPS & TRICKS (AI-derived explanation)

- Isserlis (Wick) theorem for zero-mean joint normals: E(X1 X2 X3 X4) = E(X1X2)E(X3X4) + E(X1X3)E(X2X4) + E(X1X4)E(X2X3). With the pairs (X1,X1,X2,X2) this gives 1 + rho^2 + rho^2.
- Check rho = 0: independence gives $E(X1^2)E(X2^2)$ = 1, and the formula returns 1. Only options (a) and (d) even approach this, and (a) gives 1/2.
- Check rho = 1: X1 = X2 so the answer is $E(X^4)$ = 3, and 2(1) + 1 = 3. Option (d) passes both limits.
- Conditional route: $E(X1^2\ X2^2) = E[X1^2\ E(X2^2|X1)]$ with $E(X2^2|X1) = (1 - rho^2) + rho^2\ X1^2$, giving $(1-rho^2) + 3\ rho^2 = 1 + 2\ rho^2$.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. (X1, X2) is standard bivariate normal with correlation rho, so E(X1) = E(X2) = 0, Var = 1 each and E(X1 X2) = rho.
2. Conditionally, X2 given X1 = x is N(rho x, 1 - rho^2), so $E(X2^2 \mid X1 = x) = (1 - rho^2) + rho^2\ x^2$.

3. Hence E(X1^2 X2^2) = E[X1^2 ((1 - rho^2) + rho^2 X1^2)] = (1 - rho^2) E(X1^2) + rho^2 E(X1^4).
4. For a standard normal, E(X1^2) = 1 and E(X1^4) = 3.
5. So E(X1^2 X2^2) = (1 - rho^2) + 3 rho^2 = 1 + 2 rho^2.
6. Checks: rho = 0 gives 1 and rho = 1 gives 3, both correct. Option (d).

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11. 2022 · Q18 Distribution of the Correlation Coefficient · Theoretical

QUESTION

Let $(X,Y)\sim\ BVN(0,0,1,1,\rho)$. Consider the statements: 1. $(X+Y)$ and $(X\text{-}Y)$ are independently distributed. 2. $(X+Y)$ follows normal distribution with mean 0 and variance $2(1+\rho)$. 3. $(X\text{-}Y)$ follows normal distribution with mean 0 and variance $2(1\text{-}\rho)$. Which of the above is/are correct?

OPTIONS

(a) 1 only

(b) 1 and 2 only

(c) 2 and 3 only

(d) 1, 2 and 3

ANSWER (AI-derived — no official key exists for this paper)

(d) 1, 2 and 3

EXAM SHORTCUT (AI-derived explanation)

Cov(X+Y, X-Y) = Var(X) - Var(Y) = 0, and joint normality upgrades that to independence. The variances 1 + 1 +- 2rho give 2(1+rho) and 2(1-rho). All three statements hold.

TIPS & TRICKS (AI-derived explanation)

- For jointly normal variables with EQUAL variances, the sum and difference are always independent - the covariance is Var(X) - Var(Y) = 0 and normality does the rest.
- Var(X +- Y) = Var(X) + Var(Y) +- 2 Cov(X,Y) = 2 +- 2rho = 2(1 +- rho). Both statements 2 and 3 come from this one line.
- Zero covariance implies independence ONLY under joint normality - here it is given, which is what makes statement 1 true.
- Sanity check at rho = 1: X = Y, so X - Y is degenerate with variance 2(1-1) = 0, exactly as statement 3 predicts.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. (X,Y) ~ BVN(0,0,1,1,rho), so Var(X) = Var(Y) = 1 and Cov(X,Y) = rho.
2. Statement 2: X + Y is a linear combination of jointly normal variables, hence normal, with mean 0 and Var(X+Y) = 1 + 1 + 2rho = 2(1 + rho). TRUE.
3. Statement 3: similarly X - Y is normal with mean 0 and Var(X-Y) = 1 + 1 - 2rho = 2(1 - rho). TRUE.
4. Statement 1: Cov(X+Y, X-Y) = Var(X) - Var(Y) + Cov(Y,X) - Cov(X,Y) = 1 - 1 = 0.
5. Since (X+Y, X-Y) is a linear transformation of a bivariate normal pair it is itself bivariate normal, and for jointly normal variables zero covariance implies independence. TRUE.
6. All three statements are correct, option (d).

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12. 2023 · Q26 Bivariate Normal Distribution · Numerical

COMMON DATA FOR THIS GROUP

Questions 26–30 are based on the following: the pdf of bivariate normal distribution is given by

$$f(x,y) = \lambda\, e^{-(2x^2+3xy+4y^2-7x-11y)}, \quad -\infty<x<\infty,\ -\infty<y<\infty$$

where λ *is a constant.*

QUESTION

What is the value of $E(X)+E(Y)+3$?

OPTIONS

(a) 0

(b) 1

(c) 4

(d) 5

ANSWER (AI-derived — no official key exists for this paper)

(d) 5

EXAM SHORTCUT (AI-derived explanation)

The means are where the quadratic form is stationary: 4x + 3y = 7 and 3x + 8y = 11 give x = y = 1. So E(X) + E(Y) + 3 = 5.

TIPS & TRICKS (AI-derived explanation)

- The mean vector of a normal density is the MINIMISER of the exponent's quadratic form - set both partial derivatives to zero and solve the 2x2 system.
- Partial derivatives: d/dx of ($2x^2$ + 3xy - 7x) is 4x + 3y - 7; d/dy of (3xy + $4y^2$ - 11y) is 3x + 8y - 11. Watch the factor 2 on the squared terms.
- Elimination: 8(4x + 3y) - 3(3x + 8y) = 32x - 9x = 23x, and 8(7) - 3(11) = 23, so x = 1 immediately - the determinant 23 reappears throughout this set.
- The offset '+3' in the question is pure decoration; find the two means and add.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Write the exponent as -Q(x,y) with Q = $2x^2$ + 3xy + $4y^2$ - 7x - 11y.
2. For a bivariate normal density the mean vector is the point at which the quadratic form Q attains its minimum, i.e. where both partial derivatives vanish.
3. dQ/dx = 4x + 3y - 7 = 0 and dQ/dy = 3x + 8y - 11 = 0.
4. Multiply the first by 8 and the second by 3: 32x + 24y = 56 and 9x + 24y = 33. Subtracting gives 23x = 23, so x = 1.
5. Substituting back, 4(1) + 3y = 7 gives y = 1. Hence E(X) = 1 and E(Y) = 1.
6. Therefore E(X) + E(Y) + 3 = 1 + 1 + 3 = 5, option (d).

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13. 2023 · Q27 Bivariate Normal Distribution · Numerical

COMMON DATA FOR THIS GROUP

Questions 26–30 are based on the following: the pdf of bivariate normal distribution is given by

$$f(x,y) = \lambda\, e^{-(2x^2+3xy+4y^2-7x-11y)}, \quad -\infty < x < \infty,\ -\infty < y < \infty$$

where λ is a constant.

QUESTION

What is the value of $23(\mathrm{Var}(X)+\mathrm{Var}(Y))$?

OPTIONS

(a) 4

(b) 8

(c) 12

(d) 17

ANSWER (AI-derived — no official key exists for this paper)

(c) 12

EXAM SHORTCUT (AI-derived explanation)

The precision matrix is A = [[4,3],[3,8]] with determinant 23, so the covariance matrix is (1/23)[[8,-3],[-3,4]]. Then 23(Var X + Var Y) = 8 + 4 = 12.

TIPS & TRICKS (AI-derived explanation)

- Reading the precision matrix: exp(-Q) with Q = a x^2 + b xy + c y^2 means Q = (1/2)z'Az, so A11 = 2a, A22 = 2c and A12 = b. Here A = [[4,3],[3,8]].
- Covariance = inverse of the precision matrix: for a 2x2 matrix, swap the diagonal, negate the off-diagonal and divide by the determinant 4(8) - 9 = 23.
- Var(X) = 8/23 and Var(Y) = 4/23 - note the SWAP: the larger variance goes with the smaller coefficient in the quadratic form.
- The factor 23 in the question is exactly the determinant, placed there to clear the denominators - a strong hint that this route is intended.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The exponent is -Q with Q = $2x^2$ + 3xy + $4y^2$ - 7x - 11y; the quadratic part is $2x^2$ + 3xy + $4y^2$.
2. Writing the quadratic part as (1/2) z' A z with z = (x - mu1, y - mu2)', we read A11 = 2(2) = 4, A22 = 2(4) = 8 and A12 = A21 = 3.
3. So the precision matrix is A = [[4, 3], [3, 8]], with determinant 4(8) - 3(3) = 32 - 9 = 23.
4. The covariance matrix is the inverse: Sigma = (1/23)[[8, -3], [-3, 4]].
5. Hence Var(X) = 8/23 and Var(Y) = 4/23.
6. Therefore 23(Var(X) + Var(Y)) = 8 + 4 = 12, option (c).

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14. **2023 · Q28** Distribution of the Correlation Coefficient · Numerical

COMMON DATA FOR THIS GROUP

Questions 26–30 are based on the following: the pdf of bivariate normal distribution is given by

$$f(x,y) = \lambda\, e^{-(2x^2+3xy+4y^2-7x-11y)}, \quad -\infty < x < \infty,\ -\infty < y < \infty$$

where λ is a constant.

QUESTION

If r is the correlation coefficient between X and Y, then what is the value of $\sqrt{2}\ r$?

OPTIONS

(a) $-\frac{3}{4}$

(b) $-\frac{1}{4}$

(c) $-\frac{1}{\sqrt{2}}$

(d) $-\frac{1}{2}$

ANSWER **(AI-derived — no official key exists for this paper)**

(a) $-\frac{3}{4}$

EXAM SHORTCUT **(AI-derived explanation)**

From Sigma = (1/23)[[8,-3],[-3,4]], r = (-3/23)/sqrt((8/23)(4/23)) = -3/sqrt(32) = -3/(4 sqrt2), so sqrt(2) r = -3/4.

TIPS & TRICKS **(AI-derived explanation)**

- The 23s cancel in the correlation, leaving r = -3/sqrt(8 x 4) = -3/sqrt(32) - so the determinant never needs to be carried through.
- Sign rule: the correlation takes the OPPOSITE sign to the off-diagonal entry of the precision matrix. Here that entry is +3, so r is negative - all four options duly are.
- sqrt(32) = 4 sqrt(2), so multiplying by sqrt(2) clears the surd exactly and gives the clean -3/4. The question's sqrt(2) factor is the hint that this will happen.
- Validity check: |r| = 3/(4 sqrt2) = 0.53 < 1, so the covariance matrix is positive definite and the density exists.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. From the previous part, the covariance matrix is Sigma = (1/23)[[8, -3], [-3, 4]].
2. So Var(X) = 8/23, Var(Y) = 4/23 and Cov(X,Y) = -3/23.
3. The correlation is r = Cov/sqrt(Var(X) Var(Y)) = (-3/23)/sqrt((8/23)(4/23)) = (-3/23)/(sqrt(32)/23).
4. The factors of 23 cancel, giving r = -3/sqrt(32) = -3/(4 sqrt(2)).
5. Multiplying by sqrt(2): sqrt(2) r = -3 sqrt(2)/(4 sqrt(2)) = -3/4.
6. Option (a) is correct.

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15. 2023 · Q29 Bivariate Normal Distribution · Numerical

COMMON DATA FOR THIS GROUP

Questions 26–30 are based on the following: the pdf of bivariate normal distribution is given by

$$f(x,y) = \lambda\ e^{-(2x^2+3xy+4y^2-7x-11y)}, \quad -\infty < x < \infty,\ -\infty < y < \infty$$

where λ is a constant.

QUESTION

What is the distribution of $Y \mid X = x$ equal to?

OPTIONS

(a) $N(\frac{11-3x}{8}, \frac{1}{8})$

(b) $N(1, \frac{1}{8})$

(c) $N(\frac{1+x}{2}, \frac{1}{8})$

(d) $N(\frac{4-3x}{8}, \frac{7}{8})$

ANSWER (AI-derived — no official key exists for this paper)

(a) $N(\frac{11-3x}{8}, \frac{1}{8})$

EXAM SHORTCUT (AI-derived explanation)

Collect the y-terms of the exponent: $4y^2 + (3x - 11)y$. Completing the square gives a normal with mean (11 - 3x)/8 and variance 1/(2 x 4) = 1/8.

TIPS & TRICKS (AI-derived explanation)

- For a conditional distribution, treat x as FIXED and complete the square in y only - the x-only terms are absorbed into the normalising constant.
- Coefficient reading: $\exp(-c\, y^2 + \ldots)$ corresponds to variance 1/(2c). Here c = 4, so the conditional variance is 1/8 regardless of x.
- The conditional mean is -(coefficient of y)/(2c) = -(3x - 11)/8 = (11 - 3x)/8. Getting the sign right is the crux.
- Cross-check at x = 1 (the mean of X): the conditional mean is (11-3)/8 = 1 = E(Y), exactly as the regression line must pass through the means.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The joint density is proportional to $\exp(-(2x^2 + 3xy + 4y^2 - 7x - 11y))$.
2. For fixed x, keep only the terms involving y: $-(4y^2 + 3xy - 11y) = -(4y^2 + (3x - 11)y)$.
3. Complete the square: $4y^2 + (3x-11)y = 4[y + (3x-11)/8]^2$ - constant, where the constant depends only on x.
4. So the conditional density of Y given X = x is proportional to $\exp(-4[y - (11 - 3x)/8]^2)$.
5. Comparing with $\exp(-(y - m)^2/(2\ \text{sigma}^2))$ gives 2 sigma^2 = 1/4, i.e. sigma^2 = 1/8, and m = (11 - 3x)/8.
6. Hence Y | X = x ~ N((11 - 3x)/8, 1/8). At x = 1 the mean is 1 = E(Y), as required. Option (a) is correct.

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16. 2023 · Q30 Bivariate Normal Distribution · Numerical

COMMON DATA FOR THIS GROUP

Questions 26–30 are based on the following: the pdf of bivariate normal distribution is given by

$$f(x,y) = \lambda\, e^{-(2x^2+3xy+4y^2-7x-11y)}, \quad -\infty < x < \infty,\ -\infty < y < \infty$$

where λ *is a constant.*

QUESTION

If $E(X)=\mu_1$ and $E(Y)=\mu_2$, then what is the value of $2\pi\, f(\mu_1,\mu_2)$?

OPTIONS

(a) $\sqrt{23}$

(b) $\sqrt{6}$

(c) $4\sqrt{2}$

(d) $16\sqrt{2}$

ANSWER **(AI-derived — no official key exists for this paper)**

(a) $\sqrt{23}$

EXAM SHORTCUT **(AI-derived explanation)**

At the mean vector the bivariate normal density attains its maximum 1/(2 pi sqrt(det Sigma)). Here det Sigma = 1/23, so 2 pi f(mu1, mu2) = sqrt(23).

TIPS & TRICKS **(AI-derived explanation)**

- Peak value of a bivariate normal: f(mu) = 1/(2 pi sqrt(det Sigma)) = sqrt(det A)/(2 pi), where A is the precision matrix. Multiplying by 2 pi leaves sqrt(det A).
- det A = 23 was already computed for the variances, so this part is a one-line reuse - the 2 pi in the question exists precisely to cancel.
- Equivalently f(mu) = 1/(2 pi sigma1 sigma2 sqrt(1 - r^2)); substituting sigma1^2 = 8/23, sigma2^2 = 4/23 and r^2 = 9/32 gives the same sqrt(23)/(2 pi).
- The exponential factor is exp(0) = 1 at the mean, which is why only the normalising constant survives.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. For a bivariate normal density, the maximum occurs at the mean vector and equals the normalising constant 1/(2 pi sqrt(det Sigma)).
2. From the earlier parts, the precision matrix is A = [[4,3],[3,8]] with det A = 23, so Sigma = A^(-1) has det Sigma = 1/23.
3. Hence f(mu1, mu2) = 1/(2 pi sqrt(1/23)) = sqrt(23)/(2 pi).
4. Multiplying by 2 pi: 2 pi f(mu1, mu2) = sqrt(23).
5. Equivalently, using f(mu) = 1/(2 pi sigma1 sigma2 sqrt(1-r^2)) with sigma1 sigma2 = sqrt(32)/23 and 1 - r^2 = 23/32 gives the same value.
6. Option (a) is correct.

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17. **2023 · Q31** Distribution of the Correlation Coefficient · Numerical

QUESTION

If $(X,Y) \sim BVN(0,0,1,1;\rho)$, then what is the value of $\dfrac{E((Y-\rho X)^4 \mid X)}{(1-\rho^2)^2}$?

OPTIONS

(a) 1

(b) 3

(c) 9

(d) 81

ANSWER (AI-derived — no official key exists for this paper)

(b) 3

EXAM SHORTCUT (AI-derived explanation)

Given X, the residual Y - rho X is N(0, 1 - rho^2). The fourth moment of a centred normal is 3 sigma^4 = 3(1-rho^2)^2, so the ratio is 3.

TIPS & TRICKS (AI-derived explanation)

- Conditional law for a standard bivariate normal: Y | X = x ~ N(rho x, 1 - rho^2). Subtracting rho X centres it at zero.
- Fourth central moment of N(0, sigma^2) is 3 sigma^4 - memorise the sequence 1, 0, sigma^2, 0, 3 sigma^4 for the first four moments.
- The denominator (1 - rho^2)^2 is placed exactly to cancel sigma^4, leaving the pure constant 3 - free of both x and rho.
- Note the answer does not depend on X at all, so the conditional expectation is a constant and equals the unconditional one - a useful structural observation.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For (X,Y) ~ BVN(0,0,1,1,rho), the conditional distribution is Y | X = x ~ N(rho x, 1 - rho^2).
2. Hence the residual Y - rho X, conditionally on X, is normal with mean 0 and variance sigma^2 = 1 - rho^2.
3. For a centred normal variable with variance sigma^2, the fourth moment is E(W^4) = 3 sigma^4.
4. So E[(Y - rho X)^4 | X] = 3(1 - rho^2)^2.
5. Dividing by (1 - rho^2)^2 gives 3, independently of x and of rho.
6. Option (b) is correct.

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18. 2023 · Q39 Distribution of the Correlation Coefficient · Numerical

COMMON DATA FOR THIS GROUP

Questions 39–40 are based on: $(X,Y)\sim BVN(0,0,1,1;\rho)$

QUESTION

What is the value of $\mathrm{Cov}(Y\text{-}\rho\ X,\ X)$?

OPTIONS

(a) 0

(b) 0.5

(c) 0.75

(d) 1

ANSWER (AI-derived — no official key exists for this paper)

(a) 0

EXAM SHORTCUT (AI-derived explanation)

Cov(Y - rho X, X) = Cov(Y,X) - rho Var(X) = rho - rho(1) = 0.

TIPS & TRICKS (AI-derived explanation)

- Use bilinearity of covariance: Cov(Y - rho X, X) = Cov(Y,X) - rho Cov(X,X). No integration is needed.
- For the standard bivariate normal, Cov(X,Y) = rho and Var(X) = 1, so the two terms cancel exactly.
- Interpretation: Y - rho X is the RESIDUAL from the regression of Y on X, and a least-squares residual is always uncorrelated with the predictor.
- Under joint normality this zero covariance upgrades to full independence - which is exactly what the next part exploits.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For (X,Y) ~ BVN(0,0,1,1,rho) we have Var(X) = Var(Y) = 1 and Cov(X,Y) = rho.
2. By the bilinearity of covariance, Cov(Y - rho X, X) = Cov(Y, X) - rho Cov(X, X).
3. Cov(X,X) = Var(X) = 1.
4. So the expression is rho - rho(1) = 0.
5. This reflects the general fact that the residual Y - rho X from the regression of Y on X is uncorrelated with X, and under joint normality it is in fact independent of X.
6. Option (a) is correct.

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19. 2023 · Q40 Distribution of the Correlation Coefficient · Numerical

COMMON DATA FOR THIS GROUP

Questions 39–40 are based on: $(X,Y) \sim BVN(0,0,1,1;\rho)$

QUESTION

What is the moment generating function of $X^2+Y^2-2\rho XY$?

OPTIONS

(a) $(1-2t)^{-1}$

(b) $(1-2\rho t)^{-1}$

(c) $(1-2t+2\rho^2 t)^{-1}$

(d) $(1-2t+2\rho t)^{-1}$

ANSWER (AI-derived — no official key exists for this paper)

(c) $(1-2t+2\rho^2 t)^{-1}$

EXAM SHORTCUT (AI-derived explanation)

The quadratic form (X^2 - 2 rho XY + Y^2)/(1 - rho^2) is chi-square with 2 d.f., so W = (1-rho^2) chi-square(2) and its MGF is [1 - 2(1-rho^2)t]^(-1) = (1 - 2t + 2 rho^2 t)^(-1).

TIPS & TRICKS (AI-derived explanation)

- Standard result: for a bivariate normal vector Z with covariance Sigma, Z' Sigma^(-1) Z ~ chi-square(2). Here Sigma^(-1) = (1/(1-rho^2))[[1,-rho],[-rho,1]], which produces exactly this quadratic form.

- MGF of chi-square with v d.f. is (1 - 2t)^(-v/2); with v = 2 that is (1 - 2t)^(-1), and scaling by c replaces t by ct.
- Expand the bracket at the end: 1 - 2(1-rho^2)t = 1 - 2t + 2 rho^2 t - the sign of the rho^2 term separates options (c) and (d).
- Check rho = 0: X^2 + Y^2 ~ chi-square(2) with MGF (1-2t)^(-1), and option (c) duly reduces to that.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For (X,Y) ~ BVN(0,0,1,1,rho), the covariance matrix is Sigma = [[1, rho],[rho, 1]] with inverse Sigma^(-1) = (1/(1-rho^2))[[1, -rho],[-rho, 1]].
2. The standard quadratic-form result gives Z' Sigma^(-1) Z ~ chi-square with 2 degrees of freedom, where Z = (X, Y)'.
3. Expanding, Z' Sigma^(-1) Z = (X^2 - 2 rho XY + Y^2)/(1 - rho^2).
4. Hence W = X^2 + Y^2 - 2 rho XY = (1 - rho^2) times a chi-square(2) variable.
5. The MGF of chi-square(2) is (1 - 2t)^(-1), so for a constant multiple c the MGF is (1 - 2ct)^(-1) with c = 1 - rho^2.
6. Therefore M(t) = [1 - 2(1 - rho^2)t]^(-1) = (1 - 2t + 2 rho^2 t)^(-1), option (c).

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20. 2024 · Q12 Measures of Location · Numerical

QUESTION

Suppose X and Y are zero-mean jointly normally distributed random variables such that $\sigma_X^2=4$, $\sigma_Y^2=\frac{17}{9}$, $E(XY)=2$. If $Z=2X-3Y$, then what is the pdf of Z?

OPTIONS

(a) $\frac{1}{\sqrt{12\pi}}e^{-\frac{z^2}{12}}$

(b) $\frac{1}{\sqrt{18\pi}}e^{-\frac{z^2}{18}}$

(c) $\frac{1}{\sqrt{6\pi}}e^{-\frac{z^2}{9}}$

(d) $\frac{1}{\sqrt{8\pi}}e^{-\frac{z^2}{8}}$

ANSWER (AI-derived — no official key exists for this paper)

(b) $\frac{1}{\sqrt{18\pi}}e^{-\frac{z^2}{18}}$

EXAM SHORTCUT (AI-derived explanation)

Var(Z) = 4(4) + 9(17/9) - 12 Cov = 16 + 17 - 24 = 9, so Z ~ N(0,9) with density (1/(3 sqrt(2 pi))) e^(-z^2/18) = (1/sqrt(18 pi)) e^(-z^2/18).

TIPS & TRICKS (AI-derived explanation)

- Var(aX + bY) = a^2 Var X + b^2 Var Y + 2ab Cov. With a = 2 and b = -3 the cross term is 2(2)(-3)(2) = -24 - the minus sign is essential.
- E(XY) = Cov(X,Y) here because both means are zero; the question gives the raw product moment deliberately.
- Convert the density: 1/(sigma sqrt(2 pi)) with sigma = 3 is 1/(3 sqrt(2 pi)) = 1/sqrt(9 x 2 pi) = 1/sqrt(18 pi), and the exponent is -z^2/(2 x 9) = -z^2/18.

- Cross-check the two parts of the option: the constant sqrt(18 pi) and the exponent denominator 18 must agree, which only option (b) achieves.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. X and Y are jointly normal with zero means, Var(X) = 4, Var(Y) = 17/9 and Cov(X,Y) = E(XY) = 2 (the means being zero).
2. Z = 2X - 3Y is a linear combination of jointly normal variables, so Z is normal with mean 0.
3. Var(Z) = 2^2 Var(X) + $(-3)^2$ Var(Y) + 2(2)(-3) Cov(X,Y) = 4(4) + 9(17/9) - 12(2) = 16 + 17 - 24 = 9, so sigma_Z = 3.
4. The N(0,9) density is f(z) = (1/(3 sqrt(2 pi))) e^(-z^2/18), and 3 sqrt(2 pi) = sqrt(9 x 2 pi) = sqrt(18 pi).
5. Hence f(z) = (1/sqrt(18 pi)) e^(-z^2/18), option (b).

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21. 2024 · Q36 Bivariate Normal Distribution · Numerical

QUESTION

Let X and Y be jointly distributed as $BN(1, 0, 36, 16, 1/2)$. Consider the following statements: I. $E(Y| X=7)=2$ II. $\mathrm{Cov}(X+Y, Y\text{-}X)$=-20 III. $V(X| Y=1)=27$ Which of the statements given above are correct?

OPTIONS

(a) I and II only

(b) II and III only

(c) I and III only

(d) I, II and III

ANSWER (AI-derived — no official key exists for this paper)

(d) I, II and III

EXAM SHORTCUT (AI-derived explanation)

With s1 = 6, s2 = 4 and rho = 1/2: E(Y|X=7) = 0 + (1/2)(4/6)(6) = 2; Cov(X+Y, Y-X) = Var Y - Var X = 16 - 36 = -20; V(X|Y=1) = 36(1 - 1/4) = 27. All three hold.

TIPS & TRICKS (AI-derived explanation)

- Take square roots of the variances first: 36 and 16 give s1 = 6 and s2 = 4, so the regression coefficient rho s2/s1 = (1/2)(2/3) = 1/3.
- Cov(X+Y, Y-X) expands to Cov(X,Y) - Var(X) + Var(Y) - Cov(Y,X) = Var(Y) - Var(X); the covariance terms cancel, so rho never enters.
- Conditional variance is sigma^2(1 - rho^2) and does NOT depend on the conditioning value - the '1' in V(X|Y=1) is a distractor.
- Check statement I: x = 7 is one standard deviation... in fact 6 units above mu1 = 1, and (1/3)(6) = 2, matching the stated value.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The parameters are mu1 = 1, mu2 = 0, sigma1^2 = 36, sigma2^2 = 16 and rho = 1/2, so sigma1 = 6, sigma2 = 4 and Cov(X,Y) = rho sigma1 sigma2 = (1/2)(6)(4) = 12.
2. Statement I: E(Y | X = x) = mu2 + rho (sigma2/sigma1)(x - mu1) = 0 + (1/2)(4/6)(7 - 1) = (1/3)(6) = 2. TRUE.

3. Statement II: Cov(X+Y, Y-X) = Cov(X,Y) - Cov(X,X) + Cov(Y,Y) - Cov(Y,X) = Var(Y) - Var(X) = 16 - 36 = -20. TRUE.
4. Statement III: V(X | Y = y) = sigma1^2 (1 - rho^2) = 36(1 - 1/4) = 36(3/4) = 27, independent of the value of y. TRUE.
5. All three statements are correct.
6. Option (d) is correct.

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22. 2025 · Q13 Bivariate Normal Distribution · Numerical

QUESTION

Let (X, Y) follow a bivariate normal distribution with $E(X)=E(Y)=2,\ V(X)=100,\ V(Y)=144,\ \mathrm{Corr}(X,Y)=0.5.$ Which statement(s) is/are correct? I. $E(Y \mid X=x)$ = $3.2+0.6x$ II. $V(Y \mid X=x)$ = 108

OPTIONS

(a) I only

(b) II only

(c) Both I and II

(d) Neither I nor II

ANSWER (AI-derived — no official key exists for this paper)

(b) II only

EXAM SHORTCUT (AI-derived explanation)

With s1 = 10, s2 = 12 and rho = 0.5, E(Y|X=x) = 2 + 0.6(x - 2) = 0.8 + 0.6x, not 3.2 + 0.6x, so statement I fails. V(Y|X) = 144(1 - 0.25) = 108, so statement II holds.

TIPS & TRICKS (AI-derived explanation)

- Regression coefficient: rho s2/s1 = 0.5(12/10) = 0.6. The SLOPE in statement I is right; only the intercept is wrong.
- Intercept = mu2 - slope x mu1 = 2 - 0.6(2) = 0.8. Statement I quotes 3.2, which would need mu2 = 4.4 - a straightforward misprint to detect.
- Check that the line passes through the means: at x = 2 the correct line gives 2 = E(Y), whereas 3.2 + 0.6(2) = 4.4 does not.
- Conditional variance s2^2(1 - rho^2) = 144(0.75) = 108 does not depend on x - a reliable and quick confirmation of statement II.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The parameters are E(X) = E(Y) = 2, V(X) = 100, V(Y) = 144 and rho = 0.5, so s1 = 10 and s2 = 12.
2. Statement I: E(Y | X = x) = mu2 + rho (s2/s1)(x - mu1) = 2 + 0.5(12/10)(x - 2) = 2 + 0.6(x - 2) = 0.6x + 0.8, not 3.2 + 0.6x. (Check: at x = 2 the correct line gives 2 = E(Y), whereas the stated line gives 4.4.) FALSE.
3. Statement II: V(Y | X = x) = s2^2 (1 - rho^2) = 144(1 - 0.25) = 144(0.75) = 108, independent of x. TRUE.
4. Only statement II is correct, option (b).

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23. 2025 · Q18 Distribution of the Correlation Coefficient · Factual

QUESTION

If X and Y have a bivariate normal distribution with parameters $\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho$, the marginal distribution of X is:

OPTIONS

(a) $N(\mu_1, \sigma_1^2)$

(b) $N(\mu_1 + \rho\frac{\sigma_1}{\sigma_2}(y - \mu_2),\ \sigma_1^2)$

(c) $N(\mu_1,\ \sigma_1^2(1-\rho^2))$

(d) $N(\mu_1 + \rho\frac{\sigma_1}{\sigma_2}(y - \mu_2),\ \sigma_1^2(1-\rho^2))$

ANSWER (AI-derived — no official key exists for this paper)

(a) $N(\mu_1, \sigma_1^2)$

EXAM SHORTCUT (AI-derived explanation)

The marginal distribution of X in a bivariate normal is simply N(mu1, sigma1^2) - the correlation affects only the CONDITIONAL law.

TIPS & TRICKS (AI-derived explanation)

- Marginal versus conditional: the marginal of X is N(mu1, sigma1^2); the conditional X | Y = y is N(mu1 + rho(s1/s2)(y - mu2), s1^2(1 - rho^2)).
- Any option containing y is a CONDITIONAL distribution, so options (b) and (d) are eliminated at once - a marginal cannot depend on the other variable.
- The factor (1 - rho^2) belongs to the conditional variance only, which rules out option (c).
- Structural point: integrating out Y removes all information about the dependence, which is why rho disappears from the marginal.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For a bivariate normal distribution with parameters mu1, mu2, sigma1^2, sigma2^2 and rho, integrating the joint density over y gives the marginal density of X.
2. The result is the univariate normal density with mean mu1 and variance sigma1^2: X ~ N(mu1, sigma1^2).
3. The correlation rho does not appear, because integrating out Y discards all information about the dependence between the variables.
4. By contrast, the CONDITIONAL distribution X | Y = y is N(mu1 + rho(sigma1/sigma2)(y - mu2), sigma1^2(1 - rho^2)) - option (d) - and options (b) and (c) are partial versions of that.
5. A marginal distribution cannot depend on y, which immediately rules out options (b) and (d).
6. Hence the marginal of X is N(mu1, sigma1^2), option (a).

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24. 2026 · Q13 Distribution of the Correlation Coefficient · Numerical

QUESTION

(X, Y) has a bivariate normal distribution with $\mu_x = 5,\ \mu_y = 8,\ \sigma_x^2 = 16,\ \sigma_y^2 = 9,\ \rho_{xy} = 0.6$. What is the regression coefficient of Y on X?

OPTIONS

(a) 7/16

(b) 9/20

(c) 1/2

(d) 3/8

ANSWER **(AI-derived — no official key exists for this paper)**

(b) 9/20

EXAM SHORTCUT **(AI-derived explanation)**

b(Y on X) = rho *sigma_y/sigma_x = 0.6* (3/4) = 0.45 = 9/20. The means 5 and 8 are irrelevant.

TIPS & TRICKS **(AI-derived explanation)**

- Take square roots first: sigma_x = 4 and sigma_y = 3 from the variances 16 and 9. Using the variances directly is the standard trap and would give 0.6*(9/16) = 27/80.
- Y on X puts sigma_y on top: b_yx = rho sigma_y/sigma_x. X on Y flips it: b_xy = rho sigma_x/sigma_y = 0.6(4/3) = 0.8.
- Check: b_yx *b_xy = rho^2, so 0.45* 0.8 = 0.36 = 0.6^2. A five-second confirmation.
- The means mu_x and mu_y affect only the intercept of the regression line, never the slope - ignore them here.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. For a bivariate normal distribution, the regression of Y on X is linear: E(Y | X = x) = mu_y + rho (sigma_y/sigma_x)(x - mu_x).
2. The regression coefficient of Y on X is therefore b_yx = rho * sigma_y / sigma_x.
3. Given sigma_x^2 = 16 so sigma_x = 4, and sigma_y^2 = 9 so sigma_y = 3, with rho = 0.6.
4. b_yx = 0.6 * (3/4) = 0.45 = 9/20.
5. Check with the reverse coefficient: b_xy = 0.6 *(4/3) = 0.8, and b_yx* b_xy = 0.45 * 0.8 = 0.36 = rho^2, as required.
6. Hence option (b) is correct.

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25. **2026 · Q29** Distribution of the Correlation Coefficient · Conceptual

QUESTION

$(X,Y) \sim BVN(\mu_1,\mu_2,\sigma_1^2,\sigma_2^2,\rho)$. If $\sigma_1^2=2,\ \sigma_2^2=3,\ \rho=0$, consider: I. $E(X)=E(Y)$ II. $E(XY)=2$ III. X and Y are independent Which is/are correct?

OPTIONS

(a) I only

(b) II only

(c) III

(d) I and II

ANSWER **(AI-derived — no official key exists for this paper)**

(c) III

EXAM SHORTCUT **(AI-derived explanation)**

The means mu_1 and mu_2 are never specified, so neither E(X) = E(Y) nor E(XY) = 2 can be asserted. But for the bivariate normal, rho = 0 does imply independence. Only III. Answer (c).

TIPS & TRICKS **(AI-derived explanation)**

- Zero correlation implies independence ONLY for the joint normal - this is the single most-tested exception in the whole topic.
- Ask what the data actually determine: the question gives variances and rho but leaves mu_1, mu_2 as free symbols, so any statement about means is unverifiable.
- With rho = 0, E(XY) = E(X)E(Y) = mu_1 mu_2, not sigma_1^2 = 2. Statement II confuses the covariance with a variance.
- Beware the counterexample direction too: uncorrelated normal MARGINALS need not be independent unless the JOINT distribution is bivariate normal, which is given here.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. The distribution is bivariate normal with parameters mu_1, mu_2, sigma_1^2 = 2, sigma_2^2 = 3 and rho = 0. Note that mu_1 and mu_2 are not specified numerically.
2. Statement I: E(X) = mu_1 and E(Y) = mu_2. Nothing in the data forces mu_1 = mu_2, so this cannot be asserted. Incorrect.
3. Statement II: Cov(X,Y) = rho sigma_1 sigma_2 = 0, so E(XY) = E(X)E(Y) = mu_1 mu_2, which is unknown. There is no basis for the value 2 (that is sigma_1^2, a different quantity). Incorrect.
4. Statement III: for a bivariate normal density, setting rho = 0 makes the exponent separate into a function of x alone plus a function of y alone, so f(x,y) = f_X(x) f_Y(y). Hence X and Y are independent. Correct.
5. Only statement III holds.
6. Hence option (c) is correct.

S7 · STATISTICAL METHODS

Regression - Linear & Polynomial

8 questions · 2018: 1 2019: 2 2020: 0 2021: 1 2022: 2 2023: 0 2024: 1 2025: 0 2026: 1

Weight. 8 questions (4.9% of Statistical Methods) · appeared in 6 of 9 papers · peak year 2019 (2)

Examiner's pattern. The two-regression-lines problem dominates: you are handed two equations and must decide which is which, then extract r, and often solve for the means. The validity check is |r| <= 1 — the wrong assignment gives |r| > 1. Regression derived from a joint pdf also appears.

Must-know. b_yx b_xy = r^2; both lines pass through (xbar, ybar); r takes the sign of the b's; b_yx = rsigma_y/sigma_x.

1. 2018 · Q26 Linear Regression · Numerical

QUESTION

If the joint p.d.f. of (X,Y) is $f(x,y)=\frac{1}{3}x^2e^{-y(1+x)}$, $x\geq 0, y\geq 0$, the regression equation of Y on X is

OPTIONS

(a) $y=\frac{1}{1+x}$

(b) $y=\frac{x^2}{1+x}$

(c) $y=1+x$

(d) $y=\frac{x^2}{2(1+x)^2}$

ANSWER (AI-derived — no official key exists for this paper)

(a) $y=\frac{1}{1+x}$

EXAM SHORTCUT (AI-derived explanation)

Fix x and look at the y-part: f is proportional to e^(-y(1+x)), which is an Exponential with RATE (1+x). Its mean is the reciprocal, so E(Y|X=x) = 1/(1+x).

TIPS & TRICKS (AI-derived explanation)

- The regression of Y on X IS the conditional mean E(Y|X=x). Whenever the joint density factorises as g(x) e^(-y h(x)), the conditional law of Y is Exponential(h(x)) and the answer is 1/h(x) - one line, no integration.
- Any multiplicative function of x alone (here $x^2/3$) is absorbed into the normalising constant of the conditional density and cannot appear in E(Y|X). That rules out options (b) and (d) at a glance.
- Option (c) y = 1 + x inverts the relationship: it is the RATE, not the mean. Reciprocal errors are the standard trap in exponential conditionals.
- Note that the printed joint density is not normalisable over the stated range (the x-integral of $x^2/(1+x)$ diverges), but the conditional distribution of Y given X - and hence the answer - is unaffected.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The joint density is f(x,y) = (1/3) x^2 e^(-y(1+x)) for x >= 0, y >= 0.
2. For a fixed value X = x, the conditional density of Y is f(y|x) = f(x,y)/f_X(x), which is proportional to e^(-y(1+x)) for y >= 0.
3. Normalising, f(y|x) = (1 + x) e^(-y(1+x)), which is the Exponential density with rate parameter (1 + x).
4. The mean of an Exponential distribution with rate lambda is 1/lambda, so E(Y | X = x) = 1/(1 + x).

5. The regression of Y on X is the curve y = E(Y|X = x) = 1/(1 + x).
6. Hence option (a) is correct.

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2. 2019 · Q5 Linear Regression · Numerical

QUESTION

If $x = 4y+5$ and $y = kx+4$ are the lines of regression of x on y and of y on x respectively, which one of the following is true?

OPTIONS

(a) $k \geq 1$

(b) $k \leq -1$

(c) $-1 \leq k \leq 0$

(d) $0 \leq k \leq \frac{1}{4}$

ANSWER (AI-derived — no official key exists for this paper)

(d) $0 \leq k \leq \frac{1}{4}$

EXAM SHORTCUT (AI-derived explanation)

b_xy = 4 from the first line and b_yx = k from the second. Since r^2 = b_xy b_yx must lie in [0,1], we need 0 <= 4k <= 1, i.e. 0 <= k <= 1/4.

TIPS & TRICKS (AI-derived explanation)

- Read the regression coefficients straight off: in x = 4y + 5 the coefficient of y is b_xy = 4; in y = kx + 4 the coefficient of x is b_yx = k.
- The master constraint is r^2 = b_xy . b_yx with 0 <= r^2 <= 1. Every 'which values are possible' regression item reduces to this inequality.
- Both regression coefficients must share the sign of r, so with b_xy = 4 > 0 we need k > 0 - that alone eliminates options (b) and (c).
- The geometric mean of the two regression coefficients is |r|; here sqrt(4k) <= 1 gives the same bound in one step.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The line of regression of x on y is x = b_xy . y + constant, so from x = 4y + 5 we read b_xy = 4.
2. The line of regression of y on x is y = b_yx . x + constant, so from y = kx + 4 we read b_yx = k.
3. The fundamental relation is r^2 = b_xy . b_yx = 4k.
4. Since r is a correlation coefficient, 0 <= r^2 <= 1, so 0 <= 4k <= 1.
5. Dividing by 4: 0 <= k <= 1/4.
6. Hence option (d) is correct.

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3. 2019 · Q6 Distribution of the Correlation Coefficient · Theoretical

QUESTION

The regression lines of Y on X and of X on Y are respectively $Y = aX+b$ and $X = cY+d$. If r_{XY} is the correlation coefficient between X and Y, and $\frac{s_X}{s_Y}$ is the ratio of standard deviation of X to that of Y, the pair $(r_{XY}, \frac{s_X}{s_Y})$ would assume the value

OPTIONS

(a) $(\sqrt{ac}, \sqrt{\frac{c}{a}})$

(b) $(\sqrt{\frac{c}{a}}, \sqrt{ac})$

(c) $(\frac{bd}{\sqrt{ac}}, \sqrt{ac})$

(d) $(\sqrt{ac}, \sqrt{\frac{a}{c}})$

ANSWER (AI-derived — no official key exists for this paper)

(a) $(\sqrt{ac}, \sqrt{\frac{c}{a}})$

EXAM SHORTCUT (AI-derived explanation)

a = r s_Y/s_X and c = r s_X/s_Y. Multiplying gives ac = r^2, so r = sqrt(ac); dividing gives c/a = $(s\_X/s\_Y)^2$, so s_X/s_Y = sqrt(c/a).

TIPS & TRICKS (AI-derived explanation)

- Two moves only: MULTIPLY the regression coefficients to isolate r, DIVIDE them to isolate the ratio of standard deviations. Every question of this type uses that pair.
- Keep the subscripts straight: b_YX = a = r s_Y/s_X carries s_Y on top, while b_XY = c = r s_X/s_Y carries s_X on top. Getting these the wrong way round turns (a) into (d).
- The intercepts b and d carry no information about r or the standard deviations - they only locate the means. Option (c), which uses bd, is therefore impossible.
- Since c/a = s_X^2/s_Y^2, the ratio s_X/s_Y is sqrt(c/a) - note it is c over a, not a over c, which is the sole difference between options (a) and (d).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The regression of Y on X is Y = aX + b, so its slope is the regression coefficient b_YX = a = r (s_Y/s_X).
2. The regression of X on Y is X = cY + d, so its slope is b_XY = c = r (s_X/s_Y).
3. Multiplying: ac = r^2 (s_Y/s_X)(s_X/s_Y) = r^2, hence r = sqrt(ac) (taking the sign common to a and c).
4. Dividing: c/a = [r s_X/s_Y] / [r s_Y/s_X] = $(s\_X/s\_Y)^2$.
5. Therefore s_X/s_Y = sqrt(c/a).
6. The pair is (sqrt(ac), sqrt(c/a)), option (a).

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4. 2021 · Q3 Linear Regression · Numerical

COMMON DATA FOR THIS GROUP

Q1–Q3 refer to the following: Let the joint p.d.f. of X and Y be

$$f(x,y)=\begin{cases} 6(1-x-y), & x>0,\ y>0,\ x+y<1 \\ 0, & \text{otherwise} \end{cases}$$

QUESTION

What is the equation of the line of regression of X on Y?

OPTIONS

(a) $X=\frac{3-5Y}{15}$

(b) $X=\frac{1-Y}{3}$

(c) $X=\frac{5-3Y}{15}$

(d) $X=\frac{1-Y}{5}$

ANSWER (AI-derived — no official key exists for this paper)

(b) $X=\frac{1-Y}{3}$

EXAM SHORTCUT (AI-derived explanation)

E(X^2) = 6 Gamma(3)Gamma(1)Gamma(2)/Gamma(6) = 1/10, so Var(X) = 1/10 - 1/16 = 3/80 and by symmetry Var(Y) = 3/80. Then b_XY = (-1/80)/(3/80) = -1/3, giving X = 1/4 - (1/3)(Y - 1/4) = (1 - Y)/3.

TIPS & TRICKS (AI-derived explanation)

- Regression of X on Y: X = E(X) + [Cov/Var(Y)](Y - E(Y)). Write the template first and slot in the three numbers.
- Both variances equal 3/80 by symmetry, so the regression coefficient is simply Cov/Var = -1/3 - a clean number that signals the working is right.
- Simplify at the end: 1/4 + 1/12 = 4/12 = 1/3, so the line collapses to (1 - Y)/3 with no stray fractions.
- Sign check: the covariance is negative, so the slope must be negative. Options (b) and (d) both have negative slopes; the constant term then separates them.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. From the previous parts, E(X) = E(Y) = 1/4 and Cov(X,Y) = -1/80.
2. E(X^2) = 6 times the Dirichlet integral with a = 3, b = 1, c = 2: 6 Gamma(3)Gamma(1)Gamma(2)/Gamma(6) = 6(2)/120 = 1/10.
3. Var(X) = 1/10 - $(1/4)^2$ = 1/10 - 1/16 = 8/80 - 5/80 = 3/80, and by symmetry Var(Y) = 3/80.
4. Regression coefficient of X on Y: b_XY = Cov(X,Y)/Var(Y) = (-1/80)/(3/80) = -1/3.
5. The line is X - E(X) = b_XY (Y - E(Y)), i.e. X = 1/4 - (1/3)(Y - 1/4) = 1/4 - Y/3 + 1/12.
6. Since 1/4 + 1/12 = 1/3, this is X = 1/3 - Y/3 = (1 - Y)/3. Option (b) is correct.

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5. 2022 · Q10 Linear Regression · Numerical

COMMON DATA FOR THIS GROUP

Q9–Q10 share this setup. Let the joint density function of X and Y be

$$f(x,y) = \begin{cases} \frac{y}{(1+x)^4}\exp\left(-\frac{y}{1+x}\right), & x\geq 0,\ y\geq 0 \\ 0, & \text{otherwise} \end{cases}$$

QUESTION

What is the equation of regression of Y on X?

OPTIONS

(a) $Y=2X+2$

(b) $Y=3X+5$

(c) $Y=\frac{X}{2}+5$

(d) $Y=3X+\frac{5}{2}$

ANSWER (AI-derived — no official key exists for this paper)

(a) $Y=2X+2$

EXAM SHORTCUT (AI-derived explanation)

The regression of Y on X is the conditional mean, which from the previous part is E(Y|X=x) = 2(1+x) = 2x + 2.

TIPS & TRICKS (AI-derived explanation)

- Regression of Y on X IS the conditional expectation E(Y|X = x) - no least-squares computation is needed once the conditional law is known.
- The line here happens to be exactly linear, which is why the answer is a clean straight line rather than a curve.
- Reuse the previous part: shared-stem items are designed so that the conditional mean found there is the regression asked for here.
- Check against the earlier value: at x = 10 the line gives 22, matching E(Y|X=10) exactly.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The regression curve of Y on X is by definition the graph of E(Y | X = x) against x.
2. From the joint density, the conditional distribution of Y given X = x is Gamma with shape 2 and scale (1 + x).
3. Hence E(Y | X = x) = 2(1 + x).
4. Expanding: E(Y | X = x) = 2x + 2.
5. The regression is therefore the straight line Y = 2X + 2, consistent with the value 22 obtained at x = 10.
6. Option (a) is correct.

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6. 2022 · Q14 Linear Regression · Numerical

QUESTION

Given $n=4$, $\sum x^2=164$, $\sum y^2=572$, $\sum x=24$, $\sum y=44$, $\sum xy=300$. What is the estimated value of X when $Y=10$?

OPTIONS

(a) 2.59

(b) 3.59

(c) 4.59

(d) 5.59

ANSWER (AI-derived — no official key exists for this paper)

(d) 5.59

EXAM SHORTCUT (AI-derived explanation)

xbar = 6, ybar = 11, Sxy = 300 - 264 = 36, Syy = 572 - 484 = 88. So b_XY = 36/88 = 0.4091 and X = 6 + 0.4091(10 - 11) = 5.59.

TIPS & TRICKS (AI-derived explanation)

- Predicting X from Y needs the regression of X ON Y, whose coefficient is Sxy/Syy - dividing by Sxx instead is the classic reversal error.
- Corrected sums: Sxy = sum xy - n xbar ybar and Syy = sum y^2 - n $ybar^2$. Compute the means first (6 and 11) and the rest is arithmetic.
- Y = 10 lies just BELOW ybar = 11 and the relationship is positive, so the predicted X must lie just below xbar = 6 - which immediately identifies 5.59.
- sum x^2 = 164 is not needed for this prediction; it would be required only for the regression of Y on X or for r.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. With n = 4: xbar = 24/4 = 6 and ybar = 44/4 = 11.
2. Corrected sum of products: Sxy = sum xy - n xbar ybar = 300 - 4(6)(11) = 300 - 264 = 36.
3. Corrected sum of squares for Y: Syy = sum y^2 - n $ybar^2$ = 572 - 4(121) = 572 - 484 = 88.
4. Regression coefficient of X on Y: b_XY = Sxy/Syy = 36/88 = 0.40909.
5. The regression line is X = xbar + b_XY (Y - ybar) = 6 + 0.40909(Y - 11).
6. At Y = 10: X = 6 + 0.40909(-1) = 5.591, i.e. 5.59, option (d).

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7. 2024 · Q15 Distribution of the Correlation Coefficient · Numerical

QUESTION

It is known that $8X$-$10Y$+66=0 and $40X$-$18Y$=214 are two regression equations. Consider the following statements: I. The regression coefficient of X on Y is 0.8. II. The correlation coefficient is 0.6. III. The most probable value of Y, when X=30, is 30.6. Which of the statements given above are correct?

OPTIONS

(a) I and II only

(b) II and III only

(c) I and III only

(d) I, II and III

ANSWER (AI-derived — no official key exists for this paper)

(b) II and III only

EXAM SHORTCUT (AI-derived explanation)

Assigning 8X - 10Y + 66 = 0 as Y on X gives b_YX = 0.8, and 40X - 18Y = 214 as X on Y gives b_XY = 0.45, whose product 0.36 is admissible so r = 0.6. Statement I mislabels 0.8 as b_XY, so only II and III hold.

TIPS & TRICKS (AI-derived explanation)

- Assign the lines by TESTING: the product of the two regression coefficients must lie in [0,1]. The other assignment gives 1.25 x 2.222 = 2.78 > 1 and is impossible.
- b_YX is the coefficient of X when Y is the subject; b_XY is the coefficient of Y when X is the subject. Statement I attaches 0.8 to the wrong one.
- r = sqrt(b_YX b_XY) = sqrt(0.36) = 0.6, positive because both coefficients are positive.
- For prediction of Y from X use the Y on X line: Y = 0.8(30) + 6.6 = 30.6, confirming statement III.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Try 8X - 10Y + 66 = 0 as the regression of Y on X: 10Y = 8X + 66, so Y = 0.8X + 6.6 and b_YX = 0.8.
2. Then 40X - 18Y = 214 must be the regression of X on Y: 40X = 18Y + 214, so X = 0.45Y + 5.35 and b_XY = 0.45.
3. Check: b_YX b_XY = 0.8 x 0.45 = 0.36, which lies in [0,1], so this assignment is admissible. (The opposite assignment gives 1.25 x 2.222 = 2.78 > 1 and is impossible.)
4. Statement I: the regression coefficient of X on Y is b_XY = 0.45, not 0.8 (0.8 is b_YX). FALSE.
5. Statement II: r = sqrt(0.36) = 0.6, positive since both coefficients are positive. TRUE.
6. Statement III: predicting Y from X uses the Y on X line, giving Y = 0.8(30) + 6.6 = 30.6. TRUE. Only II and III hold, option (b).

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8. 2026 · Q30 Linear Regression · Conceptual

QUESTION

The joint pdf of X and Y is $f(x,y)=\frac{y}{(1+x)^4}e^{\frac{-y}{1+x}};\ x,y\geq 0$. The regression of Y on X is:

OPTIONS

(a) exponential

(b) linear

(c) polynomial of degree 2

(d) polynomial of degree 3

ANSWER (AI-derived — no official key exists for this paper)

(b) linear

EXAM SHORTCUT (AI-derived explanation)

Given X = x, the density in y is proportional to y e^{-y/(1+x)}, a Gamma with shape 2 and scale (1+x), whose mean is 2(1+x) = 2 + 2x. That is linear in x.

TIPS & TRICKS (AI-derived explanation)

- The regression of Y on X is the conditional mean E(Y | X = x) viewed as a function of x - identify the conditional family first, then quote its mean.
- Recognise the kernel: y^(k-1) e^{-y/theta} is Gamma(shape k, scale theta) with mean k*theta. Here k = 2 and theta = 1 + x.
- You do not even need the marginal of X: the y-dependence alone identifies the conditional family, and the normalising factor is forced.
- For completeness f_X(x) = $1/(1+x)^2$ on x >= 0 - a Pareto-type density; the exponential shape in y must not be confused with an exponential regression curve.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Write theta = 1 + x. The joint density is f(x, y) = y e^{-y/theta}/theta^4 for x, y >= 0.
2. Marginal of X: f_X(x) = integral from 0 to infinity of y e^{-y/theta} dy / theta^4. Using integral from 0 to infinity of y e^{-y/theta} dy = theta^2 (Gamma function with shape 2), we get f_X(x) = theta^2/theta^4 = 1/theta^2 = 1/(1+x)^2.
3. Check: integral from 0 to infinity of dx/(1+x)^2 = 1, so f_X is a proper density.
4. Conditional density: f(y | x) = f(x,y)/f_X(x) = [y e^{-y/theta}/theta^4] * theta^2 = y e^{-y/theta}/theta^2.
5. This is the Gamma density with shape 2 and scale theta, whose mean is 2 theta.
6. Therefore E(Y | X = x) = 2(1 + x) = 2 + 2x, which is a straight line in x.
7. Hence the regression of Y on X is linear, and option (b) is correct.

S8 · STATISTICAL METHODS

Correlation: Distribution of r, Partial, Multiple, Intraclass, Correlation Ratio

33 questions · 2018: 6 2019: 5 2020: 6 2021: 2 2022: 4 2023: 2 2024: 0 2025: 4 2026: 4

Weight. 33 questions (20.1% of Statistical Methods) · appeared in 8 of 9 papers · peak year 2018 (6)

Examiner's pattern. The largest topic in Statistical Methods (32 items). The equicorrelated trivariate setup — all pairwise r equal — is the most repeated single template, asked for both partial and multiple correlation. Also constant: invariance of r under linear transformation (only the SIGN can change), intraclass correlation range, correlation ratio properties, and testing the significance of r.

Must-know. r12.3 = (r12 - r13r23)/sqrt((1-r13^2)(1-r23^2)); R^2_1.23 = (r12^2 + r13^2 - 2 r12 r13 r23)/(1 - r23^2); equicorrelated: R^2_1.23 = 2r^2/(1+r); intraclass r lies in [-1/(k-1), 1]; eta^2 >= r^2.

1. 2018 · Q22 Distribution of the Correlation Coefficient · Theoretical

QUESTION

The minimum value of $\mathrm{Var}(Y\text{-}aX)$ over all a is

OPTIONS

(a) $\rho^2 \dfrac{\mathrm{Var}(Y)}{\mathrm{Var}(X)}$

(b) $\rho^2 \dfrac{\mathrm{Var}(X)}{\mathrm{Var}(Y)}$

(c) $\rho^2 \mathrm{Var}(Y)$

(d) $(1\text{-}\rho^2)\mathrm{Var}(Y)$

ANSWER (AI-derived — no official key exists for this paper)

(d) $(1\text{-}\rho^2)\mathrm{Var}(Y)$

EXAM SHORTCUT (AI-derived explanation)

Var(Y - aX) is a quadratic in a; minimising it is exactly least squares. The minimum is the residual variance of the regression of Y on X, which is Var(Y)(1 - rho^2).

TIPS & TRICKS (AI-derived explanation)

- Remember the identity: minimum residual variance = (1 - rho^2) x Var(Y). The factor (1 - rho^2) is the proportion of variance NOT explained; rho^2 is the proportion explained.
- Dimensional check kills two options instantly: the answer must have the units of Var(Y). Options (a) and (b) are ratios of variances and are dimensionally wrong.
- The minimising value is a = Cov(X,Y)/Var(X), which is precisely the regression coefficient b_(YX) - the question is least squares in disguise.
- If rho = 0 the minimum must equal Var(Y) (X is useless); if rho = +-1 it must be 0. Only option (d) satisfies both limits.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Write g(a) = Var(Y - aX) = Var(Y) - 2a Cov(X,Y) + a^2 Var(X). This is an upward parabola in a.
2. Setting g'(a) = -2 Cov(X,Y) + 2a Var(X) = 0 gives the minimising value a* = Cov(X,Y)/Var(X).
3. Substituting back: g(a*) = Var(Y) - [Cov(X,Y)]^2/Var(X).

4. Since rho = Cov(X,Y)/(sd(X) sd(Y)), we have $[\text{Cov}(X,Y)]^2$ = rho^2 Var(X) Var(Y), so $[\text{Cov}(X,Y)]^2$/Var(X) = rho^2 Var(Y).
5. Therefore the minimum value is Var(Y) - rho^2 Var(Y) = (1 - rho^2) Var(Y).
6. Hence option (d) is correct.

---

2. 2018 · Q28 Distribution of the Correlation Coefficient · Conceptual

QUESTION

The correlation between Maths and Statistics marks for 60 students is 0.65. After Maths marks are all increased by 3, the correlation between original Statistics marks and revised Maths marks is

OPTIONS

(a) 0.60

(b) 0.65

(c) 0.70

(d) Almost 1

ANSWER (AI-derived — no official key exists for this paper)

(b) 0.65

EXAM SHORTCUT (AI-derived explanation)

Correlation is invariant under a change of origin. Adding 3 to every Maths mark shifts the origin only, so r is unchanged at 0.65.

TIPS & TRICKS (AI-derived explanation)

- r is invariant under any change of origin and under a change of scale by a POSITIVE constant; multiplying by a negative constant flips only the sign, never the magnitude.
- The sample size 60 is deliberately irrelevant here - the examiner adds it purely as a distraction.
- Formally, Cov(X + c, Y) = Cov(X, Y) and Var(X + c) = Var(X), so every ingredient of r is unchanged.
- Option (d) 'almost 1' punishes the belief that a systematic adjustment somehow strengthens the relationship - the scatter pattern is simply translated.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let X be Maths marks and Y be Statistics marks, with r(X,Y) = 0.65.
2. The revised Maths marks are X' = X + 3, a pure change of origin.
3. Covariance is unaffected by a shift: Cov(X + 3, Y) = Cov(X, Y).
4. Variance is also unaffected: Var(X + 3) = Var(X), so sd(X') = sd(X).
5. Therefore r(X', Y) = Cov(X',Y)/(sd(X') sd(Y)) = Cov(X,Y)/(sd(X) sd(Y)) = r(X,Y) = 0.65.
6. Hence option (b) is correct.

---

3. 2018 · Q29 Multiple Correlation · Numerical

QUESTION

If the zero-order correlation coefficients among 3 variates all equal ρ, the multiple correlation $R^2_{1.23}$ equals

OPTIONS

(a) ρ

(b) $\frac{2\rho^2}{1+\rho}$

(c) $\frac{\rho^2}{1+\rho}$

(d) $+1$

ANSWER (AI-derived — no official key exists for this paper)

(b) $\frac{2\rho^2}{1+\rho}$

EXAM SHORTCUT (AI-derived explanation)

Put r12 = r13 = r23 = rho in R^2 = ($r12^2$ + $r13^2$ - 2 r12 r13 r23)/(1 - $r23^2$). Numerator = $2rho^2$ - $2rho^3$ = $2rho^2$(1-rho); denominator = (1-rho)(1+rho). Cancel (1-rho) to get $2rho^2$/(1+rho).

TIPS & TRICKS (AI-derived explanation)

- Memorise R^2_(1.23) = ($r12^2$ + $r13^2$ - 2 r12 r13 r23)/(1 - $r23^2$). Every multiple-correlation numerical in this paper reduces to it.
- The cancellation of (1 - rho) is the whole trick; factor 1 - rho^2 as (1-rho)(1+rho) before simplifying rather than after.
- Check the limits: rho = 0 must give R^2 = 0 (option (b) gives 0, correct) and rho = 1 must give R^2 = 1 (option (b) gives 2/2 = 1, correct). Option (c) fails the rho = 1 test, giving 1/2.
- R^2 is a squared quantity, so a bare rho as in option (a) cannot be right on dimensional grounds.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The square of the multiple correlation of X1 on X2, X3 is R^2_(1.23) = ($r12^2$ + $r13^2$ - 2 r12 r13 r23)/(1 - $r23^2$).
2. All three zero-order correlations equal rho, so r12 = r13 = r23 = rho.
3. Numerator: rho^2 + rho^2 - 2 rho . rho . rho = $2rho^2$ - $2rho^3$ = $2rho^2$ (1 - rho).
4. Denominator: 1 - rho^2 = (1 - rho)(1 + rho).
5. Cancelling the common factor (1 - rho) gives R^2_(1.23) = $2rho^2$/(1 + rho).
6. Hence option (b) is correct.

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4. **2018 · Q30** Distribution of the Correlation Coefficient · Theoretical

QUESTION

If r is the correlation coefficient of a sample of N independent observations from a bivariate normal population with true correlation zero, then $E(1\text{-}r^2)^{-1}$ is

OPTIONS

(a) $\frac{N\text{-}3}{N\text{-}4}$

(b) $\frac{2(N\text{-}3)}{N\text{-}4}$

(c) 0

(d) 1

ANSWER (AI-derived — no official key exists for this paper)

(a) $\frac{N\text{-}3}{N\text{-}4}$

EXAM SHORTCUT (AI-derived explanation)

Under rho = 0 the density of r is proportional to (1 - r^2)^((N-4)/2). Multiplying by (1-r^2)^(-1) lowers the exponent by one, so the expectation is a ratio of Beta functions that collapses to (N-3)/(N-4).

TIPS & TRICKS (AI-derived explanation)

- Null density to memorise: f(r) = (1 - r^2)^((N-4)/2) / B(1/2, (N-2)/2) on (-1, 1). Almost every 'distribution of r when rho = 0' item is solved by shifting its exponent.
- The standard integral is: integral over (-1,1) of (1 - r^2)^(a-1) dr = B(1/2, a). Recognising the integrand as a Beta kernel converts the whole problem into Gamma-function algebra.
- Use Gamma(z+1) = z Gamma(z) and Gamma(z + 3/2) = (z + 1/2) Gamma(z + 1/2) - after those two substitutions everything cancels except (N-3)/(N-4).
- Sanity check with large N: the answer must tend to 1 because r tends to 0 and (1-r^2)^(-1) tends to 1. Only options (a) and (d) do that, and (d) ignores N entirely.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. When the population correlation is zero, the density of the sample correlation coefficient based on N observations is f(r) = (1 - r^2)^((N-4)/2) / B(1/2, (N-2)/2), for -1 < r < 1.
2. Hence E[(1 - r^2)^(-1)] = [1/B(1/2, (N-2)/2)] times the integral over (-1,1) of (1 - r^2)^((N-6)/2) dr.
3. Using the standard result integral over (-1,1) of (1 - r^2)^(a-1) dr = B(1/2, a), with a - 1 = (N-6)/2 so that a = (N-4)/2, the integral equals B(1/2, (N-4)/2).
4. Therefore E[(1-r^2)^(-1)] = B(1/2, (N-4)/2) / B(1/2, (N-2)/2).
5. Writing B(1/2, a) = Gamma(1/2)Gamma(a)/Gamma(a + 1/2) and putting a1 = (N-4)/2, a2 = a1 + 1, the ratio becomes [Gamma(a1) Gamma(a1 + 3/2)] / [Gamma(a1 + 1/2) Gamma(a1 + 1)].
6. Using Gamma(a1 + 1) = a1 Gamma(a1) and Gamma(a1 + 3/2) = (a1 + 1/2) Gamma(a1 + 1/2), this equals (a1 + 1/2)/a1 = [(N-3)/2]/[(N-4)/2] = (N-3)/(N-4), option (a).

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5. **2018 · Q31** Partial Correlation · Theoretical

QUESTION

If $ax_1+bx_2+cx_3=0$ holds identically for all x_1,x_2,x_3, the partial correlation coefficient $r_{12.3}$ is

OPTIONS

(a) 0

(b) 1

(c) -1

(d) 0.5

ANSWER (AI-derived — no official key exists for this paper)

(c) -1

EXAM SHORTCUT (AI-derived explanation)

An exact linear relation makes the two residuals proportional. From a x1 + b x2 + c x3 = 0 we get x1 = -(b/a)x2 - (c/a)x3, so after removing x3 the residual of x1 is a NEGATIVE multiple of that of x2 (for a, b of the same sign), giving r12.3 = -1.

TIPS & TRICKS (AI-derived explanation)

- Exact linear dependence always forces a partial correlation of magnitude 1 - so the real question is only the sign, and options (a) and (d) can be struck out at once.
- Sign rule: r12.3 = -sign(b/a). With the usual textbook convention that a, b, c are positive constants (as in the classic x1 + x2 + x3 = 0), the value is -1.
- The formula r12.3 = (r12 - r13 r23)/sqrt((1 - r13^2)(1 - r23^2)) also gives +-1 here, but the residual argument reaches the sign far faster.
- Partial correlation removes the linear effect of x3 from BOTH variables; what remains is the pure x1-x2 link, which is deterministic here.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The relation a x1 + b x2 + c x3 = 0 holds identically, so the three variables are exactly linearly dependent.
2. Solve for x1: x1 = -(b/a) x2 - (c/a) x3.
3. Regress each of x1 and x2 on x3 and take residuals. Because the relation above is exact, the residual of x1 after removing x3 equals -(b/a) times the residual of x2 after removing x3.
4. A residual pair related by an exact multiplicative constant has correlation +1 if the constant is positive and -1 if it is negative.
5. Here the constant is -(b/a), which is negative when a and b have the same sign - the standard convention in this classical result (e.g. the case x1 + x2 + x3 = 0).
6. Hence r12.3 = -1, option (c). (If a and b had opposite signs the value would be +1; the magnitude is always 1.)

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6. 2018 · Q32 Distribution of the Correlation Coefficient · Numerical

QUESTION

X_1, X_2, X_3, X_4 are four uncorrelated random variables, each with variance σ^2. What is the correlation between $U = X_1 + X_2 + X_3$ and $V = X_1 + X_2 + X_4$?

OPTIONS

(a) 0

(b) 1

(c) $\frac{1}{3}$

(d) $\frac{2}{3}$

ANSWER (AI-derived — no official key exists for this paper)

(d) $\frac{2}{3}$

EXAM SHORTCUT (AI-derived explanation)

U and V share X1 and X2. Cov(U,V) = Var(X1) + Var(X2) = 2 sigma^2, while Var(U) = Var(V) = 3 sigma^2. So r = 2 sigma^2/(3 sigma^2) = 2/3.

TIPS & TRICKS (AI-derived explanation)

- For sums of uncorrelated variables with a common variance, the correlation is simply (number of SHARED terms)/sqrt(count in U x count in V). Here 2/sqrt(3 x 3) = 2/3.
- Uncorrelated cross terms vanish, so the covariance counts only the overlap - never expand all nine products.
- sigma^2 cancels completely, which is why the answer is a pure number; if your working leaves a sigma you have slipped.
- Extreme checks: no shared terms would give 0 and complete overlap would give 1, so a two-out-of-three overlap must land strictly between - eliminating (a) and (b).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. X1, X2, X3, X4 are uncorrelated, each with variance sigma^2, so all cross-covariances are 0.
2. Var(U) = Var(X1 + X2 + X3) = 3 sigma^2, and similarly Var(V) = Var(X1 + X2 + X4) = 3 sigma^2.
3. Cov(U, V) = Cov(X1 + X2 + X3, X1 + X2 + X4). Every cross term between distinct variables is 0, so only the shared variables contribute.
4. Cov(U, V) = Var(X1) + Var(X2) = sigma^2 + sigma^2 = 2 sigma^2.
5. Therefore r(U, V) = 2 sigma^2 / sqrt(3 sigma^2 . 3 sigma^2) = 2 sigma^2/(3 sigma^2) = 2/3.
6. Hence option (d) is correct.

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7. 2019 · Q8 Distribution of the Correlation Coefficient · Factual

QUESTION

If r is the observed correlation coefficient in a sample of n pairs of observations from a correlated bivariate normal population, then the statistic $\frac{1}{2}\log_e(\frac{1+r}{1\text{-}r})$ is approximately normal with variance

OPTIONS

(a) $\frac{1}{n\text{-}1}$

(b) $\frac{1}{n}$

(c) $\frac{1}{n\text{-}2}$

(d) $\frac{1}{n\text{-}3}$

ANSWER (AI-derived — no official key exists for this paper)

(d) $\frac{1}{n\text{-}3}$

EXAM SHORTCUT (AI-derived explanation)

The expression is Fisher's z-transformation, z = (1/2) log[(1+r)/(1-r)] = arctanh(r). Its asymptotic variance is 1/(n-3).

TIPS & TRICKS (AI-derived explanation)

- Fisher's z: mean approximately (1/2)log[(1+rho)/(1-rho)] and variance exactly 1/(n-3). The variance is FREE of rho - that is the entire point of the transformation.
- Do not confuse it with the null-case test statistic r sqrt(n-2)/sqrt(1-r^2) ~ t with n-2 d.f., which is where the tempting 1/(n-2) in option (c) comes from.
- Use z when testing rho = rho0 for a non-zero rho0, or when comparing two independent correlations; use the t-statistic only for rho = 0.

- Remember the pairing 'z goes with n-3' as a fixed unit; the confidence interval for rho is built by transforming, adding +-1.96/sqrt(n-3), and transforming back.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The statistic (1/2) log_e[(1+r)/(1-r)] is Fisher's z-transformation of the sample correlation coefficient, equal to arctanh(r).
2. For samples from a bivariate normal population with correlation rho, z is approximately normally distributed even for moderate n.
3. Its asymptotic mean is (1/2) log_e[(1+rho)/(1-rho)] = arctanh(rho).
4. Its asymptotic variance is 1/(n - 3), which notably does not depend on rho - this variance-stabilising property is the reason for the transformation.
5. Hence the variance is 1/(n-3), option (d).

---

8. 2019 · Q9 Distribution of the Correlation Coefficient · Numerical

QUESTION

Consider the pairs of observations on X and Y: $(-1,-2)$, $(0,-1)$, $(1,0)$, $(2,1)$. If the correlation coefficient between X and Y is r, and that between X^2 and Y^2 is s, which one of the following is correct?

OPTIONS

(a) $r>0$ but $s<0$

(b) $r>0$ and $s>0$

(c) $r<0$ but $s>0$

(d) $r<0$ and $s<0$

ANSWER (AI-derived — no official key exists for this paper)

(a) $r>0$ but $s<0$

EXAM SHORTCUT (AI-derived explanation)

The four points satisfy Y = X - 1 exactly, so r = +1 > 0. But X^2 = 1, 0, 1, 4 against Y^2 = 4, 1, 0, 1 - as X^2 rises Y^2 broadly falls, and the covariance works out to -0.25 per point, so s < 0.

TIPS & TRICKS (AI-derived explanation)

- Check for an exact linear relation first: Y = X - 1 here, which forces r = +1 with no arithmetic at all.
- Squaring destroys linearity when the data straddle zero. Both X and Y take negative values, so the squares reverse the order of some pairs - that is why the sign can flip.
- Compute the covariance of X^2 and Y^2 by deviations: both have mean 1.5, giving deviations (-0.5,-1.5,-0.5,2.5) and (2.5,-0.5,-1.5,-0.5) whose products sum to -1. Negative sum, negative s.
- Moral: r measures LINEAR association only; a perfect linear relation between X and Y implies nothing about the correlation between their squares.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The observations are (-1,-2), (0,-1), (1,0), (2,1). Each satisfies Y = X - 1 exactly.
2. A perfect increasing linear relation gives r = +1, so r > 0.
3. Now form the squares: X^2 = 1, 0, 1, 4 and Y^2 = 4, 1, 0, 1.

4. Both sets have mean (1+0+1+4)/4 = 1.5. Deviations of X^2: -0.5, -1.5, -0.5, 2.5. Deviations of Y^2: 2.5, -0.5, -1.5, -0.5.
5. Sum of products of deviations: (-0.5)(2.5) + (-1.5)(-0.5) + (-0.5)(-1.5) + (2.5)(-0.5) = -1.25 + 0.75 + 0.75 - 1.25 = -1.0, so the covariance is negative and s < 0.
6. Therefore r > 0 but s < 0, option (a).

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9. 2019 · Q10 Distribution of the Correlation Coefficient · Numerical

QUESTION

The coefficient of correlation between X and Y is 0.6. Their covariance is 4.8. If the variance of X is 9, then the standard deviation of Y is

OPTIONS

(a) 7.21

(b) 5.23

(c) 3.22

(d) 2.67

ANSWER (AI-derived — no official key exists for this paper)

(d) 2.67

EXAM SHORTCUT (AI-derived explanation)

r = Cov/(s_X s_Y) gives 0.6 = 4.8/(3 s_Y), so 3 s_Y = 8 and s_Y = 8/3 = 2.67.

TIPS & TRICKS (AI-derived explanation)

- Rearranged form worth memorising: s_Y = Cov/(r . s_X). One substitution and you are done.
- Take the square root of the variance first: Var(X) = 9 gives s_X = 3, not 9. Using 9 directly yields 0.89 and no matching option.
- The arithmetic is clean: 4.8/0.6 = 8, then 8/3 = 2.67. Look for such cancellations before reaching for a calculator.
- Consistency check: with s_Y = 8/3, Cov = r s_X s_Y = 0.6 x 3 x 2.667 = 4.8, matching the datum exactly.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The correlation coefficient is defined by r = Cov(X,Y)/(s_X s_Y).
2. Given Var(X) = 9, the standard deviation is s_X = 3.
3. Substituting the known values: 0.6 = 4.8/(3 s_Y).
4. Therefore 3 s_Y = 4.8/0.6 = 8.
5. So s_Y = 8/3 = 2.67.
6. Hence option (d) is correct.

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10. 2019 · Q19 Distribution of the Correlation Coefficient · Theoretical

QUESTION

r is the correlation coefficient between X and Y. If $Z = aY + b$, where $a(\neq 0)$ and b are real, the correlation coefficient between X and Z is

OPTIONS

(a) r

(b) $r|a|$

(c) $(\frac{a}{|a|})|r|$

(d) $(\frac{a}{|a|})r$

ANSWER (AI-derived — no official key exists for this paper)

(d) $(\frac{a}{|a|})r$

EXAM SHORTCUT (AI-derived explanation)

A linear change of scale multiplies the covariance by a and the standard deviation by |a|, so the correlation is multiplied by a/|a| - the SIGN of a. Hence r(X,Z) = (a/|a|) r.

TIPS & TRICKS (AI-derived explanation)

- Rule: correlation is unchanged by a shift, and under scaling it is multiplied by the sign of the multiplier. The magnitude |r| never changes.
- Option (a) is right only when a > 0; the question allows any non-zero a, so the sign factor must appear explicitly.
- Option (b) r|a| is dimensionally impossible - a correlation is bounded by 1, but r|a| could exceed 1 for large a.
- Option (c) uses |r| and so loses the original sign of r; only option (d) preserves r's own sign while applying the sign of a.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let Z = aY + b with a not equal to 0.
2. Covariance: Cov(X, Z) = Cov(X, aY + b) = a Cov(X, Y).
3. Standard deviation: sd(Z) = |a| sd(Y), since the constant b shifts but does not scale.
4. Therefore r(X, Z) = a Cov(X,Y) / [sd(X) . |a| sd(Y)] = (a/|a|) . Cov(X,Y)/(sd(X) sd(Y)) = (a/|a|) r.
5. So the correlation keeps its magnitude |r| and takes the sign of a times the sign of r.
6. Hence option (d) is correct.

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11. 2019 · Q21 Distribution of the Correlation Coefficient · Theoretical

QUESTION

Consider the following statements: 1. Correlation coefficient is a measure of proportion or percentage in a relationship. 2. Statistic Z^2 is an equivalent measure of the statistic χ^2. 3. t-distribution becomes mesokurtic when degrees of freedom approach a very large value. 4. (1-α) significant points of $F(n_2,n_1)$ distribution are the same as reciprocal of (1-α) significant points of $F(n_1,n_2)$ distribution. Which of the above statements are correct?

OPTIONS

(a) 2 and 3 only

(b) 2 and 4 only

(c) 3 and 4 only

(d) 1, 2, 3 and 4

ANSWER (AI-derived — no official key exists for this paper)

(a) 2 and 3 only

EXAM SHORTCUT (AI-derived explanation)

Statement 1 confuses r with r^2 (only the coefficient of DETERMINATION is a proportion), and statement 4 mis-states the F reciprocal relation, which pairs alpha with 1-alpha, not 1-alpha with 1-alpha. Statements 2 and 3 are standard.

TIPS & TRICKS (AI-derived explanation)

- r is not a percentage; r^2 is. A correlation of 0.6 explains 36% of the variance, not 60% - the single most common misreading in this chapter.
- Z^2 ~ chi-square(1) is the exact link behind statement 2, and it is why a two-sided Z-test and a 1-d.f. chi-square test always agree.
- As d.f. increase, t tends to N(0,1): its kurtosis falls from leptokurtic towards the mesokurtic value 3. Statement 3 is therefore true.
- The correct F identity is F_(1-alpha)(n1, n2) = 1/F_alpha(n2, n1) - the tail areas must be COMPLEMENTARY. Statement 4 keeps 1-alpha on both sides and is false.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement 1: the correlation coefficient measures the strength and direction of a LINEAR relationship; it is the coefficient of determination r^2 that gives the proportion of variance explained. FALSE.
2. Statement 2: if Z ~ N(0,1) then Z^2 ~ chi-square with 1 degree of freedom, so the two statistics carry equivalent information for a two-sided test. TRUE.
3. Statement 3: the t-distribution with n d.f. has kurtosis 3 + 6/(n-4), which tends to 3 as n grows; a kurtosis of 3 is exactly mesokurtic (the normal case). TRUE.
4. Statement 4: the correct relation between the tabulated points of the two F distributions is F_(1-alpha)(n1, n2) = 1/F_alpha(n2, n1), where the tail areas are complementary. The statement uses (1-alpha) on both sides, which holds only for alpha = 1/2. FALSE.
5. Only statements 2 and 3 are correct, option (a).

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12. 2020 · Q8 Partial Correlation · Numerical

QUESTION

Let r_{ij} denote the correlation between X_i, X_j. If $r_{12} = \sin^2\theta$, $r_{13} = \cos\theta$, $r_{23} = \sin\theta$ $(-\pi \le \theta \le \pi)$, the partial correlation $r_{12.3}$ is

OPTIONS

(a) $2\csc 2\theta$

(b) $-2\csc 2\theta$

(c) $\tan\theta - 1$

(d) $2\cot 2\theta$

ANSWER (AI-derived — no official key exists for this paper)

(c) $\tan\theta - 1$

EXAM SHORTCUT (AI-derived explanation)

Substitute into r12.3 = (r12 - r13 r23)/sqrt((1-r13^2)(1-r23^2)). The numerator is $\sin^2$ t - sin t cos t and the denominator is |sin t cos t|, so the ratio is tan t - 1.

TIPS & TRICKS (AI-derived explanation)

- The denominator simplifies first: (1 - $\cos^2$ t)(1 - $\sin^2$ t) = $\sin^2$ t $\cos^2$ t, whose square root is |sin t cos t| - spotting this collapses the whole expression.
- Factor the numerator as sin t (sin t - cos t) so that the common sin t cancels against the denominator, leaving (sin t - cos t)/cos t = tan t - 1.
- Memorise the partial correlation formula r12.3 = (r12 - r13 r23)/sqrt((1 - r13^2)(1 - r23^2)); every item in this sub-topic is one substitution away from it.
- Options (a), (b) and (d) all involve 2 theta, which never arises from this cancellation - a structural cue that (c) is the intended form.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The partial correlation formula is r12.3 = (r12 - r13 r23)/sqrt[(1 - r13^2)(1 - r23^2)].
2. Substituting r12 = $\sin^2$ t, r13 = cos t, r23 = sin t, the numerator is $\sin^2$ t - (cos t)(sin t) = sin t (sin t - cos t).
3. For the denominator, 1 - r13^2 = 1 - $\cos^2$ t = $\sin^2$ t and 1 - r23^2 = 1 - $\sin^2$ t = $\cos^2$ t.
4. So the denominator is sqrt($\sin^2$ t $\cos^2$ t) = |sin t cos t|.
5. Taking the principal branch where sin t cos t > 0, r12.3 = sin t (sin t - cos t)/(sin t cos t) = (sin t - cos t)/cos t.
6. This equals tan t - 1, option (c).

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13. 2020 · Q9 Non-parametric Tests · Numerical

QUESTION

When two judges rank two individuals only, Spearman's rank correlation coefficient can assume the values

OPTIONS

(a) -1 and 0 only

(b) -1 and 1 only

(c) 0 and 1 only

(d) -1,0 and 1

ANSWER (AI-derived — no official key exists for this paper)

(b) -1 and 1 only

EXAM SHORTCUT (AI-derived explanation)

With only two individuals the second judge either agrees (d = 0, rho = 1) or reverses (sum d^2 = 2, rho = 1 - 12/6 = -1). No other outcome exists.

TIPS & TRICKS (AI-derived explanation)

- Spearman's rho = 1 - 6 sum(d^2)/(n(n^2 - 1)). With n = 2 the denominator is 2(3) = 6, so rho = 1 - sum(d^2).
- Only two rankings of two items are possible, so only two values of rho can occur - enumerate them rather than reasoning abstractly.

- 0 is impossible because sum(d^2) would have to be 1, but with n = 2 the differences are either both 0 or both +-1, giving sum(d^2) equal to 0 or 2.
- General principle: for small n the set of attainable rank correlations is discrete, and the gaps between attainable values are large - a reason rank tests need adequate n.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Spearman's rank correlation is rho = 1 - 6 sum(d_i^2)/(n(n^2 - 1)), where d_i are the differences of ranks.
2. With n = 2, n(n^2 - 1) = 2(4 - 1) = 6, so rho = 1 - sum(d_i^2).
3. There are only two possible rankings by the second judge. If the judges agree, the ranks are (1,2) and (1,2), so d = (0,0) and sum(d^2) = 0, giving rho = 1.
4. If the judges disagree, the ranks are (1,2) and (2,1), so d = (-1, +1) and sum(d^2) = 2, giving rho = 1 - 2 = -1.
5. No other configuration exists, so rho can only be -1 or +1; the value 0 is unattainable.
6. Option (b) is correct.

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14. 2020 · Q13 Distribution of the Correlation Coefficient · Conceptual

QUESTION

If $D=X\text{-}Y$ and $S_D^{\ 2}=S_X^{\ 2}+S_Y^{\ 2}$, the correlation coefficient r between X,Y is

OPTIONS

(a) greater than zero but not equal to one

(b) less than zero

(c) equal to zero

(d) equal to one

ANSWER (AI-derived — no official key exists for this paper)

(c) equal to zero

EXAM SHORTCUT (AI-derived explanation)

In general Var(X - Y) = S_X^2 + S_Y^2 - 2 r S_X S_Y. The given identity removes the cross term, so 2 r S_X S_Y = 0 and hence r = 0.

TIPS & TRICKS (AI-derived explanation)

- The variance of a difference carries a MINUS 2rS_X S_Y term; the plain additive formula holds only when that term vanishes.
- Since standard deviations are strictly positive for non-degenerate variables, 2 r S_X S_Y = 0 forces r = 0 exactly - not merely 'small'.
- Note this gives UNCORRELATED, not independent: the same identity would hold for dependent but uncorrelated variables.
- Companion identity: Var(X + Y) = S_X^2 + S_Y^2 + 2 r S_X S_Y, which likewise reduces to the additive form precisely when r = 0.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For D = X - Y, the general variance formula is S_D^2 = S_X^2 + S_Y^2 - 2 r S_X S_Y, where r is the correlation between X and Y.

2. The question states that $S\_D^2 = S\_X^2 + S\_Y^2$.
3. Equating the two expressions gives $S\_X^2 + S\_Y^2 - 2 r S\_X S\_Y = S\_X^2 + S\_Y^2$.
4. Hence 2 r S_X S_Y = 0.
5. Since S_X and S_Y are positive for non-degenerate variables, r = 0.
6. Hence the correlation is exactly zero, option (c).

---

15. 2020 · Q14 Distribution of the Correlation Coefficient · Conceptual

QUESTION

X has finite variance σ^2. If Y=20-X, the correlation between X and $(X+Y)X$ is

OPTIONS

(a) -1

(b) 0

(c) 1/20

(d) 1

ANSWER (AI-derived — no official key exists for this paper)

(d) 1

EXAM SHORTCUT (AI-derived explanation)

Y = 20 - X makes X + Y = 20 identically, so (X + Y)X = 20X. The correlation of X with 20X is +1.

TIPS & TRICKS (AI-derived explanation)

- Simplify the expression BEFORE computing anything: the sum X + Y collapses to the constant 20, which is the entire content of the question.
- Correlation between a variable and a positive multiple of itself is exactly +1; a negative multiple would give -1.
- The variance sigma^2 is stated only to guarantee that X is non-degenerate so that the correlation is defined - it never enters the answer.
- Option (a) -1 is the correlation between X and Y themselves, the natural trap if you stop at Y = 20 - X without simplifying the product.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Since Y = 20 - X, the sum X + Y = X + (20 - X) = 20 is a constant.
2. Therefore the second variable is (X + Y)X = 20X.
3. The correlation between X and 20X is Cov(X, 20X)/(sd(X) . sd(20X)) = 20 Var(X)/(sd(X) . 20 sd(X)) = 1.
4. In general, corr(X, aX) equals +1 for a > 0, and here a = 20 > 0.
5. The finite variance $\text{sigma}^2 > 0$ ensures the correlation is well defined.
6. Hence the correlation is 1, option (d).

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16. 2020 · Q15 Distribution of the Correlation Coefficient · Conceptual

QUESTION

The correlation coefficient in a bivariate set-up is 0.7. 1. The percentage of variation in the dependent variable explained by the independent variable is 49. 2. Both regression coefficients are positive. 3. Both regression coefficients are negative.

OPTIONS

(a) 1 only

(b) 1 and 2

(c) 1 and 3

(d) 2 only

ANSWER (AI-derived — no official key exists for this paper)

(b) 1 and 2

EXAM SHORTCUT (AI-derived explanation)

$r^2 = 0.49$ gives 49% explained variation, so statement 1 holds. Since $r = +0.7 > 0$, both regression coefficients share that positive sign, so statement 2 holds and statement 3 fails.

TIPS & TRICKS (AI-derived explanation)

- Percentage of variation explained is r^2 x 100, not r x 100. Here $0.7^2 = 0.49$, i.e. 49%.
- Sign rule: r, b_yx and b_xy always carry the SAME sign, because r^2 = b_yx b_xy and r = sign(b) sqrt(b_yx b_xy).
- Statements 2 and 3 are direct contradictions, so exactly one of them must be true - decide by the sign of r and both are settled at once.
- Remember 'r^2 explains, r describes': r^2 is the coefficient of determination and r merely indicates strength and direction.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The correlation coefficient is r = 0.7.
2. Statement 1: the proportion of variation in the dependent variable explained by the independent variable is the coefficient of determination $r^2 = 0.49$, i.e. 49%. TRUE.
3. The two regression coefficients satisfy b_yx b_xy = r^2 and both carry the same sign as r.
4. Statement 2: since r = +0.7 is positive, both regression coefficients are positive. TRUE.
5. Statement 3: this contradicts statement 2 and is therefore FALSE.
6. Statements 1 and 2 are correct, option (b).

---

17. 2020 · Q16 Multiple Correlation · Numerical

QUESTION

In a trivariate distribution, if $r_{12}=r_{23}=r_{31}=\rho\neq1$, the value of $R_{1.23}$ is

OPTIONS

(a) $\frac{\rho}{\sqrt{1+\rho}}$

(b) $\frac{1}{\sqrt{1+\rho}}$

(c) $\frac{1}{1+\rho}$

(d) $\frac{\sqrt{2}\ \rho}{\sqrt{1+\rho}}$

ANSWER **(AI-derived — no official key exists for this paper)**

(d) $\frac{\sqrt{2}\ \rho}{\sqrt{1+\rho}}$

EXAM SHORTCUT **(AI-derived explanation)**

R^2 = (r12^2 + r13^2 - 2 r12 r13 r23)/(1 - r23^2) = 2rho^2(1-rho)/[(1-rho)(1+rho)] = 2rho^2/(1+rho). Taking the square root gives sqrt(2) rho/sqrt(1+rho).

TIPS & TRICKS **(AI-derived explanation)**

- Compute R SQUARED first and take the square root only at the very end - working with R directly invites algebraic slips.
- The cancellation of (1 - rho) between numerator and denominator is the whole trick; factor 1 - rho^2 as (1-rho)(1+rho) before simplifying.
- Check the limit rho -> 1: R^2 -> 2/2 = 1, so R -> 1 as the variables become perfectly correlated. Option (d) satisfies this; option (a) gives 1/sqrt2 and fails.
- This is the square root of the 2018 Q29 result - the same identity appears in both papers, once as R^2 and once as R.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. For a trivariate distribution, R^2_(1.23) = (r12^2 + r13^2 - 2 r12 r13 r23)/(1 - r23^2).
2. With r12 = r13 = r23 = rho, the numerator is rho^2 + rho^2 - 2rho^3 = 2rho^2(1 - rho).
3. The denominator is 1 - rho^2 = (1 - rho)(1 + rho).
4. Cancelling (1 - rho), which is non-zero since rho is not 1, gives R^2_(1.23) = 2rho^2/(1 + rho).
5. Taking the positive square root: R_(1.23) = sqrt(2) rho/sqrt(1 + rho).
6. Hence option (d) is correct.

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18. 2021 · Q32 Partial Correlation · Numerical

QUESTION

For three variables x_1, x_2, x_3 with every pairwise correlation equal to r, what is the partial correlation $r_{12.3}$?

OPTIONS

(a) r

(b) $\frac{1}{r+1}$

(c) $\frac{r}{r+1}$

(d) $\frac{1}{1-r}$

ANSWER **(AI-derived — no official key exists for this paper)**

(c) $\frac{r}{r+1}$

EXAM SHORTCUT **(AI-derived explanation)**

r12.3 = (r - r^2)/sqrt((1-r^2)(1-r^2)) = r(1-r)/(1-r^2) = r/(1+r) after cancelling (1 - r).

TIPS & TRICKS **(AI-derived explanation)**

- Substitute all three correlations as r and the formula collapses immediately - no need to keep the subscripts separate.
- Factor 1 - r^2 as (1-r)(1+r) BEFORE cancelling; the common factor (1 - r) is what produces the clean answer.
- Check r = 0: independent variables must give partial correlation 0, and r/(1+r) = 0. Check r = 1: the formula gives 1/2, and indeed perfect mutual correlation is a degenerate boundary case.
- Note the contrast with 2018 Q31: an exact linear relation gives partial correlation +-1, whereas equal pairwise correlations give the milder r/(1+r).

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. The partial correlation formula is r12.3 = (r12 - r13 r23)/sqrt[(1 - $r13^2$)(1 - $r23^2$)].
2. With r12 = r13 = r23 = r, the numerator is r - r . r = r - r^2 = r(1 - r).
3. The denominator is sqrt[(1 - r^2)(1 - r^2)] = 1 - r^2 = (1 - r)(1 + r).
4. Cancelling the common factor (1 - r), valid since r is not 1, gives r12.3 = r/(1 + r).
5. Check: r = 0 gives 0, as it must for uncorrelated variables.
6. Option (c) is correct.

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19. **2021 · Q34** Distribution of the Correlation Coefficient · Numerical

QUESTION

The correlation coefficient of a random sample of size 11 from a bivariate normal population is 0.6. The t-statistic for testing significance of the population correlation is

OPTIONS

(a) 2.25

(b) 2.8

(c) 6.75

(d) 8.44

ANSWER **(AI-derived — no official key exists for this paper)**

(a) 2.25

EXAM SHORTCUT **(AI-derived explanation)**

t = r sqrt(n-2)/sqrt(1 - r^2) = 0.6 sqrt(9)/sqrt(0.64) = 1.8/0.8 = 2.25.

TIPS & TRICKS **(AI-derived explanation)**

- Test statistic for H0: rho = 0 is t = r sqrt(n-2)/sqrt(1-r^2) with n - 2 degrees of freedom. Memorise it as one block.

- The numbers are chosen to be clean: n - 2 = 9 gives sqrt = 3, and 1 - 0.36 = 0.64 gives sqrt = 0.8. Spot these before reaching for a calculator.
- Do not use Fisher's z here - that is for testing a NON-ZERO rho or comparing two correlations. For rho = 0 the exact t-test applies.
- With 9 d.f. the 5% two-tailed critical value is 2.262, so t = 2.25 is just short of significance - a nice illustration of how borderline a correlation of 0.6 is at n = 11.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. To test H0: rho = 0 against an alternative, the statistic is t = r sqrt(n - 2)/sqrt(1 - r^2), which follows Student's t with n - 2 degrees of freedom.
2. Here r = 0.6 and n = 11, so n - 2 = 9 and sqrt(9) = 3.
3. 1 - r^2 = 1 - 0.36 = 0.64, so sqrt(1 - r^2) = 0.8.
4. t = 0.6 x 3/0.8 = 1.8/0.8 = 2.25.
5. The statistic has 9 degrees of freedom.
6. Hence t = 2.25, option (a).

---

20. 2022 · Q11 Distribution of the Correlation Coefficient · Numerical

QUESTION

Consider the bivariate data:

X	1	2	3	4	5
Y	4	1	0	1	4

What is the correlation coefficient?

OPTIONS

(a) 0

(b) $-\frac{1}{2}$

(c) -1

(d) 1

ANSWER (AI-derived — no official key exists for this paper)

(a) 0

EXAM SHORTCUT (AI-derived explanation)

The data satisfy Y = $(X-3)^2$ exactly - a symmetric parabola about x = 3. The deviation products cancel in pairs, so the covariance and hence r are 0.

TIPS & TRICKS (AI-derived explanation)

- Recognise the pattern first: 4, 1, 0, 1, 4 is $(X-3)^2$, symmetric about the mean of X. Symmetry forces zero LINEAR correlation.
- Deviations of X are -2, -1, 0, 1, 2 and of Y are 2, -1, -2, -1, 2; the products -4, 1, 0, -1, 4 sum to 0.
- This is the standard illustration that r = 0 does NOT mean no relationship - here the relationship is perfect but quadratic.

- Option (d) 1 punishes candidates who see an exact functional relationship and assume perfect correlation; r measures linearity only.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The data are (1,4), (2,1), (3,0), (4,1), (5,4). Note that Y = $(X - 3)^2$ exactly.
2. Mean of X: (1+2+3+4+5)/5 = 3. Mean of Y: (4+1+0+1+4)/5 = 2.
3. Deviations of X: -2, -1, 0, 1, 2. Deviations of Y: 2, -1, -2, -1, 2.
4. Products of deviations: (-2)(2) = -4, (-1)(-1) = 1, (0)(-2) = 0, (1)(-1) = -1, (2)(2) = 4.
5. Their sum is -4 + 1 + 0 - 1 + 4 = 0, so the covariance is 0 and hence r = 0.
6. The variables are perfectly related but the relationship is quadratic, not linear. Option (a) is correct.

---

21. 2022 · Q16 Multiple Correlation · Numerical

QUESTION

For a trivariate distribution, $r_{12}=r_{13}=r_{23}=r$ where r_{ij} is the correlation coefficient between X_i and X_j. If $R^2_{1.23}$ is the multiple correlation coefficient, the expression for (1-$R^2_{1.23}$) is:

OPTIONS

(a) $(1-r)(1-2r)(1+r)^{-1}$

(b) $(1-r)(1+2r)(1+r)^{-1}$

(c) $(1-r)(1-2r)(1-r)^{-1}$

(d) $(1-r)(1+2r)(1-r)^{-1}$

ANSWER (AI-derived — no official key exists for this paper)

(b) $(1-r)(1+2r)(1+r)^{-1}$

EXAM SHORTCUT (AI-derived explanation)

$R^2 = 2r^2/(1+r)$, so $1 - R^2 = (1 + r - 2r^2)/(1+r)$. Factorising the numerator as (1-r)(1+2r) gives (1-r)(1+2r)/(1+r).

TIPS & TRICKS (AI-derived explanation)

- Start from the known result R^2_(1.23) = $2r^2/(1 + r)$ for equal pairwise correlations, then subtract from 1 over the common denominator.
- Factorise $-2r^2 + r + 1$ carefully: it equals -(2r + 1)(r - 1) = (1 - r)(1 + 2r). Getting this factorisation right is the whole question.
- Options (c) and (d) have (1 - r) in the denominator, which would cancel with the numerator's (1 - r) and leave no r-dependence of the right form - a structural giveaway.
- Check r = 0: 1 - R^2 must be 1, and (1)(1)/(1) = 1. Check r = 1: R^2 = 1 so 1 - R^2 = 0, and (0)(3)/(2) = 0. Only option (b) passes both.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For a trivariate distribution with all pairwise correlations equal to r, R^2_(1.23) = $(r^2 + r^2 - 2r^3)/(1 - r^2) = 2r^2(1-r)/[(1-r)(1+r)] = 2r^2/(1 + r)$.
2. Therefore 1 - R^2_(1.23) = $1 - 2r^2/(1 + r) = [(1 + r) - 2r^2]/(1 + r)$.
3. Factorise the numerator: $-2r^2 + r + 1 = -(2r^2 - r - 1) = -(2r + 1)(r - 1) = (1 - r)(1 + 2r)$.

4. Hence 1 - R^2_(1.23) = (1 - r)(1 + 2r)/(1 + r).
5. Checks: at r = 0 the value is 1 and at r = 1 it is 0, both as required.
6. Option (b) is correct.

---

22. 2022 · Q19 Distribution of the Correlation Coefficient · Theoretical

QUESTION

If r is a sample correlation coefficient in a random sample of size n from a bivariate normal population with $\rho=0$, the distribution of r^2 is:

OPTIONS

(a) $\beta(1,n\text{-}2)$

(b) $\beta(\frac{1}{2},\frac{n\text{-}2}{2})$

(c) $\beta(2,n\text{-}3)$

(d) $\beta(\frac{1}{3},\frac{n\text{-}3}{3})$

ANSWER (AI-derived — no official key exists for this paper)

(b) $\beta(\frac{1}{2},\frac{n\text{-}2}{2})$

EXAM SHORTCUT (AI-derived explanation)

Under rho = 0 the density of r is proportional to (1-r^2)^((n-4)/2). Substituting t = r^2 gives the Beta(1/2, (n-2)/2) density.

TIPS & TRICKS (AI-derived explanation)

- Null density of r: f(r) = (1 - r^2)^((n-4)/2)/B(1/2, (n-2)/2) on (-1,1). Recognising the Beta constant already reveals the parameters of r^2.
- The change of variable t = r^2 contributes a factor t^(-1/2) from dr/dt, which is what produces the first Beta parameter 1/2.
- Consistency check with the F link: r^2/(1-r^2) x (n-2) ~ F(1, n-2), and a Beta(1/2, (n-2)/2) variable transformed that way is exactly F(1, n-2).
- The two Beta parameters must sum to (n-1)/2 here; option (b) gives 1/2 + (n-2)/2 = (n-1)/2, while the others do not fit the standard theory.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. When the population correlation is zero, the sample correlation coefficient based on n observations has density f(r) = (1 - r^2)^((n-4)/2)/B(1/2, (n-2)/2) on (-1, 1).
2. Let T = r^2. For 0 < t < 1, the two values r = +-sqrt(t) both map to t, and |dr/dt| = 1/(2 sqrt t) for each.
3. Hence the density of T is g(t) = 2 . f(sqrt t) . 1/(2 sqrt t) = t^(-1/2) (1 - t)^((n-4)/2)/B(1/2, (n-2)/2).
4. Comparing with the Beta density t^(a-1)(1-t)^(b-1)/B(a,b): a - 1 = -1/2 gives a = 1/2, and b - 1 = (n-4)/2 gives b = (n-2)/2.
5. So r^2 ~ Beta(1/2, (n-2)/2). This is consistent with the standard result that (n-2) r^2/(1 - r^2) ~ F(1, n-2).
6. Option (b) is correct.

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23. 2022 · Q20 Partial Correlation · Theoretical

QUESTION

If $R_{1.23}$=0, which of the following statements is/are correct? 1. X_1 is uncorrelated with X_2 and X_3. 2. X_2 is uncorrelated with X_3. 3. $(1\text{-}r_{12}^{\ 2})(1\text{-}r_{13.2}^{\ 2})=1$.

OPTIONS

(a) 1 only

(b) 1 and 2

(c) 1 and 3

(d) 2 and 3

ANSWER (AI-derived — no official key exists for this paper)

(c) 1 and 3

EXAM SHORTCUT (AI-derived explanation)

R_(1.23) = 0 forces r12 = r13 = 0 (the numerator is a positive definite form), so statement 1 holds. Statement 3 is the identity 1 - R^2 = (1-r12^2)(1-r13.2^2) with R = 0. Statement 2 imposes nothing on r23 and is false.

TIPS & TRICKS (AI-derived explanation)

- The numerator of R^2 is r12^2 + r13^2 - 2 r12 r13 r23, a positive definite quadratic form in (r12, r13) whenever |r23| < 1, so it vanishes only when both are zero.
- Key identity to memorise: 1 - R^2_(1.23) = (1 - r12^2)(1 - r13.2^2). Setting R = 0 gives statement 3 immediately.
- R_(1.23) says nothing about the relationship BETWEEN the predictors, so r23 is entirely unconstrained - statement 2 must be false.
- Interpretation: R_(1.23) = 0 means X2 and X3 together explain none of the variation in X1, which requires X1 to be uncorrelated with each of them separately.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. R^2_(1.23) = (r12^2 + r13^2 - 2 r12 r13 r23)/(1 - r23^2).
2. For |r23| < 1 the numerator is a positive definite quadratic form in r12 and r13 (its discriminant 4r23^2 - 4 is negative), so it vanishes only when r12 = r13 = 0.
3. Statement 1: R_(1.23) = 0 therefore forces X1 to be uncorrelated with both X2 and X3. TRUE.
4. Statement 2: nothing in the condition restricts r23, the correlation between the two predictors, which may take any admissible value. FALSE.
5. Statement 3: the standard identity is 1 - R^2_(1.23) = (1 - r12^2)(1 - r13.2^2). With R_(1.23) = 0 the left side is 1, so the product equals 1. TRUE. (Directly: r12 = 0 and r13.2 = 0, so the product is 1 x 1.)
6. Statements 1 and 3 are correct, option (c).

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24. 2023 · Q25 Distribution of the Correlation Coefficient · Numerical

QUESTION

If $\mathrm{Var}(X)=4,\ \mathrm{Cov}(X,Y)=32$ and correlation coefficient $r(X,Y)=0.8$ for some bivariate data, then what is $V(Y)$ equal to?

OPTIONS

(a) 400

(b) 200

(c) $200\sqrt{2}$

(d) $400\sqrt{2}$

ANSWER (AI-derived — no official key exists for this paper)

(a) 400

EXAM SHORTCUT (AI-derived explanation)

r = Cov/(sd_X sd_Y) gives 0.8 = 32/(2 sd_Y), so sd_Y = 20 and V(Y) = 400.

TIPS & TRICKS (AI-derived explanation)

- Take the square root of Var(X) = 4 to get sd_X = 2 BEFORE substituting - using 4 directly is the standard error here.
- Rearranged form: sd_Y = Cov/(r sd_X) = 32/(0.8 x 2) = 20. One division and one multiplication.
- Remember to SQUARE the standard deviation at the end: the question asks for the variance, so 20 becomes 400.
- Consistency check: Cov = r sd_X sd_Y = 0.8 x 2 x 20 = 32, matching the datum exactly.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The correlation coefficient is r = Cov(X,Y)/(sd(X) sd(Y)).
2. Given Var(X) = 4, the standard deviation is sd(X) = 2.
3. Substituting the known values: 0.8 = 32/(2 sd(Y)).
4. So 2 sd(Y) = 32/0.8 = 40, giving sd(Y) = 20.
5. Therefore V(Y) = 20^2 = 400.
6. Check: r sd(X) sd(Y) = 0.8(2)(20) = 32 = Cov(X,Y). Option (a) is correct.

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25. 2023 · Q32 Intraclass Correlation · Theoretical

QUESTION

For h families containing equal number k of members in each family, the range of intra-class correlation coefficient ρ is:

OPTIONS

(a) 0 to 1

(b) −1 to 1

(c) $-\frac{1}{k-1}$ to 0

(d) $-\frac{1}{k-1}$ to 1

ANSWER (AI-derived — no official key exists for this paper)

(d) $-\frac{1}{k-1}$ to 1

EXAM SHORTCUT (AI-derived explanation)

The intra-class correlation lies between -1/(k-1) and +1, the lower bound arising because the k members' deviations within a family must sum to zero.

TIPS & TRICKS (AI-derived explanation)

- Range to memorise: -1/(k-1) <= rho <= 1, where k is the family (group) size. The lower bound is NOT -1 - that is the key discriminator.
- Reason for the asymmetry: within a group the deviations from the group mean sum to zero, so the members cannot all be strongly negatively correlated with one another.
- As k grows the lower bound tends to 0, so large groups permit almost no negative intra-class correlation - a useful sanity anchor.
- For k = 2 the bound is -1, which is why the pairwise case looks like an ordinary correlation and can mislead you into choosing option (b).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The intra-class correlation coefficient measures the similarity of members within the same group, here families of k members each.
2. It can be written as rho = (MS_between - MS_within)/(MS_between + (k-1) MS_within) in the analysis-of-variance formulation.
3. Its maximum is +1, attained when all members of a family are identical so that the within-family variation vanishes.
4. Its minimum is not -1: because the k deviations from the family mean must sum to zero, the members cannot be arbitrarily negatively correlated. Setting the between-family mean square to zero gives the smallest value -1/(k-1).
5. Hence the range is -1/(k - 1) to 1.
6. Option (d) is correct.

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26. 2025 · Q14 Distribution of the Correlation Coefficient · Numerical

QUESTION

In a trivariate distribution, r_{12}=0.8, r_{13}=-0.4, r_{23}=0. The correlation coefficient between X_1 and X_3 after eliminating the linear effect of X_2 from each is:

OPTIONS

(a) $\frac{2}{3}$

(b) $\frac{1}{3}$

(c) $-\frac{1}{3}$

(d) $-\frac{2}{3}$

ANSWER (AI-derived — no official key exists for this paper)

(d) $-\frac{2}{3}$

EXAM SHORTCUT (AI-derived explanation)

r13.2 = (r13 - r12 r23)/sqrt[(1-r12^2)(1-r23^2)] = (-0.4 - 0)/sqrt(0.36 x 1) = -0.4/0.6 = -2/3.

TIPS & TRICKS (AI-derived explanation)

- 'Eliminating the linear effect of X2' is the definition of the PARTIAL correlation r13.2 - identify the notation before substituting.
- r23 = 0 simplifies everything: the numerator reduces to r13 and the second bracket to 1.
- 1 - r12^2 = 1 - 0.64 = 0.36, whose square root is exactly 0.6 - the numbers are chosen to be clean.
- Sign check: r13 is negative and the denominator is positive, so the partial correlation must be negative - options (a) and (b) are eliminated at once.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Removing the linear effect of X2 from both X1 and X3 gives the partial correlation r13.2.
2. The formula is r13.2 = (r13 - r12 r23)/sqrt[(1 - r12^2)(1 - r23^2)].
3. Substituting r12 = 0.8, r13 = -0.4 and r23 = 0: the numerator is -0.4 - (0.8)(0) = -0.4.
4. The denominator is sqrt[(1 - 0.64)(1 - 0)] = sqrt(0.36) = 0.6.
5. So r13.2 = -0.4/0.6 = -2/3.
6. Note that the partial correlation is larger in magnitude than r13 itself, because removing X2 sharpens the relationship. Option (d) is correct.

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27. 2025 · Q15 Correlation Ratio · Theoretical

QUESTION

If η_{YX} is the correlation ratio of Y on X, which statement(s) is/are correct? I. η_{YX}^2 is independent of change of origin and scale. II. $\eta_{YX} = \eta_{XY}$ III. $\eta_{YX}^2 \geq r^2$, where r is the correlation coefficient between X and Y.

OPTIONS

(a) I and II only

(b) III only

(c) I and III only

(d) I, II and III

ANSWER (AI-derived — no official key exists for this paper)

(c) I and III only

EXAM SHORTCUT (AI-derived explanation)

The correlation ratio is invariant under change of origin and scale (I true) and always dominates r^2 (III true), but it is NOT symmetric: eta_YX and eta_XY generally differ (II false).

TIPS & TRICKS (AI-derived explanation)

- eta_YX^2 measures the proportion of Y's variance explained by ANY function of X, whereas r^2 measures only the LINEAR part - hence eta^2 >= r^2 always.
- Equality eta^2 = r^2 holds exactly when the regression of Y on X is linear - a useful test for linearity.
- Asymmetry: eta_YX is built from the conditional means of Y given X, and eta_XY from those of X given Y. There is no reason for them to agree, unlike r which is symmetric.
- Invariance under linear transformation follows because eta^2 is a ratio of variances, and both numerator and denominator scale by the same factor.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The correlation ratio is defined by eta_YX^2 = Var[E(Y|X)]/Var(Y), the proportion of the variance of Y explained by the conditional means.
2. Statement I: a change of origin or scale in X does not alter the conditioning information, and a change in Y scales both the numerator and the denominator by the same factor. So eta^2 is invariant. TRUE.
3. Statement II: eta_YX is built from E(Y|X) while eta_XY is built from E(X|Y); these are different quantities and there is no general reason for them to be equal, unlike the symmetric correlation coefficient r. FALSE.
4. Statement III: r^2 measures only the linear part of the relationship, whereas eta_YX^2 captures the whole regression function, so eta_YX^2 >= r^2 always, with equality exactly when the regression of Y on X is linear. TRUE.
5. Only statements I and III are correct.
6. Hence option (c) is correct.

---

28. 2025 · Q19 t-Distribution · Numerical

QUESTION

What would be the lowest value of r^2 in a sample of 27 pairs of observations that would be significant at the 5% level? [$t_{25}(0.05)=2.06$]

OPTIONS

(a) 0.1451

(b) 0.2451

(c) 0.3551

(d) 0.4451

ANSWER (AI-derived — no official key exists for this paper)

(a) 0.1451

EXAM SHORTCUT (AI-derived explanation)

t^2 = r^2(n-2)/(1-r^2) with t = 2.06 and n - 2 = 25 gives 25r^2 = 4.2436(1 - r^2), so r^2 = 4.2436/29.2436 = 0.1451.

TIPS & TRICKS (AI-derived explanation)

- Start from the test statistic t = r sqrt(n-2)/sqrt(1-r^2) and SQUARE it to work directly in r^2, which is what the question asks for.
- Rearranged formula: r^2 = t^2/(t^2 + n - 2) = 4.2436/(4.2436 + 25). Memorising this compact form makes the item a one-liner.
- n = 27 pairs gives n - 2 = 25 degrees of freedom, which matches the supplied t_25 value - a confirmation that the set-up is right.
- t^2 = 2.06^2 = 4.2436; keep four decimals, since the options differ in the second decimal of r^2.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. To test H0: rho = 0 the statistic is t = r sqrt(n - 2)/sqrt(1 - r^2) with n - 2 degrees of freedom.
2. With n = 27, the degrees of freedom are 25 and the 5% critical value is t = 2.06.
3. Squaring: t^2 = r^2 (n-2)/(1 - r^2), so 4.2436 = 25 r^2/(1 - r^2).

4. Rearranging: $4.2436(1 - r^2) = 25 r^2$, i.e. $4.2436 = r^2(25 + 4.2436) = 29.2436 r^2$.
5. Hence $r^2 = 4.2436/29.2436 = 0.14512$.
6. The lowest significant value of r^2 is therefore 0.1451, option (a).

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29. 2025 · Q42 Distribution of the Correlation Coefficient · Numerical

COMMON DATA FOR THIS GROUP

Shared data for Q41–Q44. Let (X_1,X_2,X_3) be a random vector with joint pmf:

$$P(X_1=x_1,X_2=x_2,X_3=x_3) = \begin{cases} \frac{50!}{x_1!\ x_2!\ x_3!\ (50-x_1-x_2-x_3)!}\left(\frac{1}{4}\right)^{x_1}\left(\frac{1}{4}\right)^{x_2}\left(\frac{1}{4}\right)^{x_3}\left(\frac{1}{4}\right)^{50-x_1-x_2-x_3} & \text{if } x_1+x_2+x_3\le 50 \\ 0 & \text{otherwise} \end{cases}$$

QUESTION

The correlation between X_1 and X_2 is:

OPTIONS

(a) -1/4

(b) -1/3

(c) 1/4

(d) 1/3

ANSWER (AI-derived — no official key exists for this paper)

(b) -1/3

EXAM SHORTCUT (AI-derived explanation)

For a multinomial, Corr(X_i, X_j) = -sqrt[p_i p_j/((1-p_i)(1-p_j))]. With p_i = p_j = 1/4 this is -(1/4)/(3/4) = -1/3.

TIPS & TRICKS (AI-derived explanation)

- Multinomial moments: Var(X_i) = n p_i(1-p_i) and Cov(X_i, X_j) = -n p_i p_j. The correlation then has n cancelling out.
- The correlation is always NEGATIVE, since the counts must sum to n - options (c) and (d) are eliminated on sign alone.
- With equal probabilities the formula simplifies to -p/(1-p) = -(1/4)/(3/4) = -1/3.
- Substituting numbers: Cov = -50(1/16) = -25/8 and each Var = 50(3/16) = 75/8, so the ratio is -25/75 = -1/3.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For the multinomial distribution with n trials, Var(X_i) = n p_i (1 - p_i) and Cov(X_i, X_j) = -n p_i p_j for i not equal to j.
2. Here n = 50 and p_1 = p_2 = 1/4, so Cov(X1, X2) = -50(1/4)(1/4) = -50/16 = -25/8.
3. Var(X1) = Var(X2) = 50(1/4)(3/4) = 150/16 = 75/8.
4. Corr(X1, X2) = Cov/sqrt(Var(X1)Var(X2)) = (-25/8)/(75/8) = -1/3.
5. Equivalently, the general formula gives -sqrt[(1/4)(1/4)/((3/4)(3/4))] = -(1/4)/(3/4) = -1/3.
6. Option (b) is correct.

30. 2026 · Q22 Distribution of the Correlation Coefficient · Numerical

QUESTION

How many pairs of observations (n) must be in a sample so that an observed correlation coefficient of 0.6 gives a calculated t greater than 2.72?

OPTIONS

(a) $n<10$

(b) $10<n<13$

(c) $13<n<15$

(d) $n>15$

ANSWER (AI-derived — no official key exists for this paper)

(d) $n>15$

EXAM SHORTCUT (AI-derived explanation)

With r = 0.6, sqrt(1 - r^2) = 0.8, so t = 0.6 sqrt(n-2)/0.8 = 0.75 sqrt(n-2). Setting 0.75 sqrt(n-2) > 2.72 gives sqrt(n-2) > 3.627, n - 2 > 13.15, so n > 15.15, i.e. n > 15.

TIPS & TRICKS (AI-derived explanation)

- Test statistic for H0: rho = 0 is t = r sqrt(n-2)/sqrt(1 - r^2) with n - 2 degrees of freedom - memorise it exactly.
- r = 0.6 gives the Pythagorean pair 0.6/0.8 = 0.75, so the algebra stays in clean decimals.
- Solve the inequality for n rather than testing each option: n > 2 + (2.72/0.75)^2 = 2 + 13.15 = 15.15.
- Since n must be a whole number, n >= 16, which is exactly the statement n > 15; do not round 15.15 down to 15.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. To test the significance of an observed correlation coefficient, t = r sqrt(n - 2)/sqrt(1 - r^2) with (n - 2) degrees of freedom.
2. With r = 0.6: r^2 = 0.36, so 1 - r^2 = 0.64 and sqrt(1 - r^2) = 0.8.
3. Therefore t = 0.6 sqrt(n - 2)/0.8 = 0.75 sqrt(n - 2).
4. Require t > 2.72: 0.75 sqrt(n - 2) > 2.72, so sqrt(n - 2) > 2.72/0.75 = 3.62667.
5. Squaring: n - 2 > 13.1527, hence n > 15.1527.
6. Since n is an integer, n must be at least 16, i.e. n > 15.
7. Hence option (d) is correct.

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31. 2026 · Q23 Partial Correlation · Theoretical

QUESTION

$b_{13.2}$ is the regression coefficient of X_1 on X_3 (holding X_2), and $b_{31.2}$ is that of X_3 on X_1 (holding X_2). What is $b_{13.2} \times b_{31.2}$?

OPTIONS

(a) $r_{12.3}^{\ 2}$

(b) $r_{13.2}^{\ 2}$

(c) $r_{23.1}^{2}$

(d) r^{2}

ANSWER **(AI-derived — no official key exists for this paper)**

(b) $r_{13.2}^{2}$

EXAM SHORTCUT **(AI-derived explanation)**

Exactly as in the two-variable case, the product of the two reciprocal regression coefficients is the square of the corresponding correlation - here the partial correlation r_{13.2}. Answer (b).

TIPS & TRICKS **(AI-derived explanation)**

- Read the subscripts: both coefficients involve variables 1 and 3 with 2 held constant, so the answer must carry the subscript 13.2.
- Simple-case analogue: b_yx * b_xy = r^{2}. The partial version simply carries the conditioning subscript through.
- Explicit forms: b_{13.2} = r_{13.2} *sigma_{1.2}/sigma_{3.2} and b_{31.2} = r_{13.2}* sigma_{3.2}/sigma_{1.2}; the partial standard deviations cancel on multiplication.
- Note r_{13.2} = r_{31.2} (partial correlation is symmetric in the two primary subscripts), which is why the single symbol suffices.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. The partial regression coefficient of X_1 on X_3 holding X_2 fixed is b_{13.2} = r_{13.2} * (sigma_{1.2}/sigma_{3.2}), where sigma_{1.2} and sigma_{3.2} are the partial (residual) standard deviations.
2. Similarly b_{31.2} = r_{13.2} * (sigma_{3.2}/sigma_{1.2}), using the symmetry r_{31.2} = r_{13.2}.
3. Multiplying: b_{13.2} *b_{31.2} = r_{13.2}^2* (sigma_{1.2}/sigma_{3.2})(sigma_{3.2}/sigma_{1.2}) = r_{13.2}^2.
4. This is the exact analogue of b_yx * b_xy = r^{2} in simple linear regression.
5. Options (a) and (c) carry the wrong variable pair, and option (d) refers to an unspecified simple correlation.
6. Hence option (b) is correct.

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32. 2026 · Q25 Intraclass Correlation · Conceptual

QUESTION

If $r(2x,y)=0.25$, what is $r(4x\text{-}3,\ \frac{7\text{-}4y}{8})$?

OPTIONS

(a) 1/4

(b) 1/8

(c) -1/4

(d) -1/8

ANSWER **(AI-derived — no official key exists for this paper)**

(c) -1/4

EXAM SHORTCUT (AI-derived explanation)

Correlation is unchanged by any linear change of scale and origin except that a negative multiplier flips the sign. 4x - 3 keeps the sign; (7 - 4y)/8 = 7/8 - y/2 reverses it. So the answer is -0.25.

TIPS & TRICKS (AI-derived explanation)

- Rule: r(a + bX, c + dY) = sign(bd) * r(X, Y). Only the signs of the multipliers matter; their magnitudes and the additive constants do not.
- First simplify the given information: r(2x, y) = 0.25 means r(x, y) = 0.25, since 2 > 0.
- Rewrite (7 - 4y)/8 as 7/8 - (1/2)y to expose the multiplier -1/2 clearly; leaving it as a compound fraction is what makes candidates miss the sign.
- The magnitude never changes, so options (b) and (d) with 1/8 are wrong on principle - a scale change can never shrink a correlation.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The correlation coefficient is invariant under a change of origin and positive change of scale, and reverses sign under a negative change of scale: r(a + bX, c + dY) = sign(b)*sign(d)*r(X, Y).
2. Given r(2x, y) = 0.25. Since the multiplier 2 is positive, r(x, y) = 0.25.
3. First variable: 4x - 3 has multiplier b = 4 > 0, so the sign is preserved.
4. Second variable: (7 - 4y)/8 = 7/8 - (1/2) y has multiplier d = -1/2 < 0, so the sign is reversed.
5. Therefore r(4x - 3, (7 - 4y)/8) = sign(4) *sign(-1/2)* 0.25 = -0.25 = -1/4.
6. Hence option (c) is correct.

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33. 2026 · Q27 Partial Correlation · Theoretical

QUESTION

$R_{i.jk}$ is the multiple correlation coefficient, r_{ij} the simple correlation coefficient, and σ_i the standard deviation of X_i. Consider: I. $R_{1.23}^2 = \sigma_1^2 \sigma_2^2 (1-r_{12}^2)(1-r_{13})^2$ II. $R_{1.23}^2 \geq r_{12}^2$ III. $R_{1.23}^2 \geq r_{13.2}^2$ Which are correct?

OPTIONS

(a) I and II only

(b) II and III only

(c) I and III only

(d) I, II and III

ANSWER (AI-derived — no official key exists for this paper)

(b) II and III only

EXAM SHORTCUT (AI-derived explanation)

Statement I is dimensionally impossible - a squared correlation cannot carry sigma^2 factors - so drop it. II and III are both standard consequences of 1 - R^2_{1.23} = (1 - r_12^2)(1 - r^2_{13.2}). Answer (b).

TIPS & TRICKS (AI-derived explanation)

- Key identity: 1 - R^2_{1.23} = (1 - r_12^2)(1 - r^2_{13.2}). Almost every multiple-correlation inequality follows from it in one step.

- Correlations are unit-free, so any formula for R^2 containing sigma_1^2 sigma_2^2 must be wrong - reject statement I on sight.
- The correct formula is R^2_{1.23} = (r_12^2 + r_13^2 - 2 r_12 r_13 r_23)/(1 - r_23^2).
- Remember the ordering 0 <= R^2_{1.23} <= 1 and R_{1.23} >= |r_1j| for every j: adding a regressor can never reduce the multiple correlation.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. Statement I: R^2_{1.23} = sigma_1^2 sigma_2^2 (1 - r_12^2)(1 - r_13)^2. This is false. R^2 is a unit-free quantity lying in [0,1], so it cannot involve the variances; the correct expression is R^2_{1.23} = (r_12^2 + r_13^2 - 2 r_12 r_13 r_23)/(1 - r_23^2).
2. Use the standard identity 1 - R^2_{1.23} = (1 - r_12^2)(1 - r^2_{13.2}). Writing a = r_12^2 and b = r^2_{13.2}, both in [0,1], this gives R^2_{1.23} = 1 - (1 - a)(1 - b) = a + b - ab.
3. Statement II: R^2_{1.23} - r_12^2 = (a + b - ab) - a = b(1 - a) >= 0, since 0 <= a <= 1 and b >= 0. So R^2_{1.23} >= r_12^2 is correct.
4. Statement III: R^2_{1.23} - r^2_{13.2} = (a + b - ab) - b = a(1 - b) >= 0, since 0 <= b <= 1 and a >= 0. So R^2_{1.23} >= r^2_{13.2} is correct.
5. Only statements II and III hold.
6. Hence option (b) is correct.

S9 · STATISTICAL METHODS

Standard Errors & Large Sample Tests

6 questions · 2018: 0 2019: 1 2020: 0 2021: 0 2022: 1 2023: 3 2024: 0 2025: 0 2026: 1

Weight. 6 questions (3.7% of Statistical Methods) · appeared in 4 of 9 papers · peak year 2023 (3)

Examiner's pattern. Standard-error arithmetic: SE of a proportion, SE of a mean, SE of r, plus confidence intervals and sample-size determination. Usually one or two easy marks — the formulae are short and the numbers are clean.

Must-know. SE(phat) = sqrt(pq/n); SE(xbar) = sigma/sqrt(n); SE(r) = (1 - r^2)/sqrt(n); 95% CI = xbar +/- 1.96sigma/sqrt(n).

1. 2019 · Q23 Large Sample Tests & Standard Errors · Theoretical

QUESTION

Let W_1 and W_2 denote the standard errors of the sample means $\overline{X}_1$ and $\overline{X}_2$ of two independent random samples. The standard error of $T=(\overline{X}_1-\overline{X}_2)$ is

OPTIONS

(a) 0

(b) $|W_1-W_2|$

(c) $\sqrt{|W_1^2-W_2^2|}$

(d) $\sqrt{W_1^2+W_2^2}$

ANSWER (AI-derived — no official key exists for this paper)

(d) $\sqrt{W_1^2+W_2^2}$

EXAM SHORTCUT (AI-derived explanation)

For independent samples, variances ADD even for a difference: Var(Xbar1 - Xbar2) = W1^2 + W2^2, so the standard error is sqrt(W1^2 + W2^2).

TIPS & TRICKS (AI-derived explanation)

- The minus sign is squared away: Var(A - B) = Var(A) + Var(B) when A and B are independent. Standard errors never subtract.
- Standard errors combine in quadrature, like the sides of a right triangle - never linearly. That kills option (b) at once.
- Option (a) 0 would require the two sample means to be identical every time, and option (c) can even become imaginary in spirit (it needs an absolute value to stay real) - both are structurally impossible.
- Same formula in raw terms: SE = sqrt(s1^2/n1 + s2^2/n2), which is exactly sqrt(W1^2 + W2^2).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. W1 and W2 are the standard errors of Xbar1 and Xbar2, so Var(Xbar1) = W1^2 and Var(Xbar2) = W2^2.
2. The two samples are independent, so Cov(Xbar1, Xbar2) = 0.
3. Hence Var(T) = Var(Xbar1 - Xbar2) = Var(Xbar1) + Var(Xbar2) - 2Cov = W1^2 + W2^2.
4. The standard error is the square root of this variance: SE(T) = sqrt(W1^2 + W2^2).

5. Hence option (d) is correct.

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2. 2022 · Q59 Sampling Distribution of Sample Mean · Numerical

QUESTION

The 95% CI for population mean, sample size 100 from $N(\mu, 25)$, sample mean 5.96 (Z=1.96):

OPTIONS

(a) $[5.47, 6.45]$

(b) $[4.98, 6.94]$

(c) $[5, 6]$

(d) $[4.1, 6.99]$

ANSWER (AI-derived — no official key exists for this paper)

(b) $[4.98, 6.94]$

EXAM SHORTCUT (AI-derived explanation)

SE = sqrt(25)/sqrt(100) = 5/10 = 0.5, so the margin is 1.96(0.5) = 0.98 and the interval is 5.96 +- 0.98 = [4.98, 6.94].

TIPS & TRICKS (AI-derived explanation)

- Convert the variance to a standard deviation first: 25 gives sigma = 5, not 25. Using the variance directly inflates the interval enormously.
- SE = sigma/sqrt(n) = 5/10 = 0.5. The clean numbers are deliberate - the margin 1.96 x 0.5 = 0.98 is exact.
- The interval must be SYMMETRIC about the sample mean 5.96; check that 5.96 is the midpoint of your chosen option. For (b), (4.98 + 6.94)/2 = 5.96 exactly.
- Option (a) [5.47, 6.45] has the right midpoint but a margin of 0.49 - the result of forgetting a factor of 2 or halving the SE.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The population is N(mu, 25), so sigma = sqrt(25) = 5, and the sample size is n = 100.
2. The standard error of the sample mean is sigma/sqrt(n) = 5/10 = 0.5.
3. For a 95% confidence interval the multiplier is Z = 1.96, so the margin of error is 1.96 x 0.5 = 0.98.
4. The interval is xbar +- 0.98 = 5.96 +- 0.98.
5. Lower limit 5.96 - 0.98 = 4.98; upper limit 5.96 + 0.98 = 6.94.
6. Hence the interval is [4.98, 6.94], option (b).

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3. 2023 · Q33 Distribution of the Correlation Coefficient · Numerical

QUESTION

If the correlation coefficient between heights of fathers and sons from a random sample of 900 is calculated as 0.67, what is the standard error?

OPTIONS

(a) 0.018

(b) 0.04

(c) 0.18

(d) 1.8

ANSWER (AI-derived — no official key exists for this paper)

(a) 0.018

EXAM SHORTCUT (AI-derived explanation)

SE(r) = (1 - r^2)/sqrt(n) = (1 - 0.4489)/30 = 0.5511/30 = 0.0184, i.e. about 0.018.

TIPS & TRICKS (AI-derived explanation)

- Standard error of a sample correlation: SE(r) = (1 - r^2)/sqrt(n). The numerator is 1 - r SQUARED, not 1 - r.
- sqrt(900) = 30 makes the arithmetic clean: 0.5511/30 = 0.01837.
- The SE shrinks like 1/sqrt(n), so with n = 900 it must be small - roughly 0.02 - which identifies option (a) at a glance.
- For inference on a non-zero rho, prefer Fisher's z transformation with variance 1/(n-3); this simple SE formula is used chiefly for a quick significance check.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The standard error of the sample correlation coefficient is SE(r) = (1 - r^2)/sqrt(n).
2. Here r = 0.67, so r^2 = 0.4489 and 1 - r^2 = 0.5511.
3. The sample size is n = 900, so sqrt(n) = 30.
4. SE(r) = 0.5511/30 = 0.01837.
5. Rounded, this is 0.018.
6. Option (a) is correct.

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4. 2023 · Q34 t-Distribution · Numerical

QUESTION

The mean height obtained from a random sample of size 100 is 64 inches. The standard deviation of the height distribution is known to be 3 inches. What is the 95% confidence interval for mean height of the population? (Assume $Z_{0.025}$=2)

OPTIONS

(a) (63.4, 64.6)

(b) (63.1, 64.1)

(c) (62.8, 65.8)

(d) (62, 66)

ANSWER (AI-derived — no official key exists for this paper)

(a) (63.4, 64.6)

EXAM SHORTCUT (AI-derived explanation)

SE = 3/sqrt(100) = 0.3, so the margin is 2(0.3) = 0.6 and the interval is 64 +- 0.6 = (63.4, 64.6).

TIPS & TRICKS (AI-derived explanation)

- SE = sigma/sqrt(n) = 3/10 = 0.3. The population SD is given, so the normal multiplier applies rather than t.
- The question supplies Z = 2 rather than 1.96 to keep the arithmetic exact - use the value given, not the one you remember.
- The interval must be SYMMETRIC about the sample mean 64: check that (63.4 + 64.6)/2 = 64. Option (b) fails this test at once.
- Interval width 2 x 2 x 0.3 = 1.2 inches; options (c) and (d) are three and five times too wide, corresponding to forgetting the sqrt(n).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The sample mean is xbar = 64 inches, the population standard deviation is sigma = 3 inches and the sample size is n = 100.
2. The standard error of the mean is sigma/sqrt(n) = 3/10 = 0.3.
3. For a 95% confidence interval with sigma known, the multiplier supplied is Z(0.025) = 2.
4. Margin of error = 2 x 0.3 = 0.6.
5. The interval is 64 +- 0.6, i.e. from 63.4 to 64.6.
6. Option (a) is correct.

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5. **2023 · Q35** Large Sample Tests & Standard Errors · Numerical

QUESTION

In an art school, 380 children out of 800 were found to be singers. If E is the standard error of proportion of singers, then what is $1280000E^2$ equal to?

OPTIONS

(a) 39.9

(b) 399

(c) 3990

(d) 4399

ANSWER (AI-derived — no official key exists for this paper)

(b) 399

EXAM SHORTCUT (AI-derived explanation)

p = 380/800 = 0.475, so E^2 = pq/n = 0.475(0.525)/800 = 0.00031172. Multiplying by 1,280,000 gives 399.

TIPS & TRICKS (AI-derived explanation)

- SE of a proportion: E = sqrt(pq/n), so E^2 = pq/n - the question asks for E^2, which spares you the square root entirely.
- pq = 0.475 x 0.525 = 0.249375. Note that pq is always close to 0.25 when p is near 1/2, a useful check.
- 1,280,000/800 = 1600, so the answer is 1600 x 0.249375 = 399 - factor the constant before multiplying.
- The tidy result 399 confirms the arithmetic; a value near 400 is exactly what 1600 x 0.25 would give.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The sample proportion of singers is p = 380/800 = 0.475, so q = 1 - p = 0.525, with n = 800.
2. The standard error of a proportion is E = sqrt(pq/n), hence E^2 = pq/n.

3. pq = 0.475 x 0.525 = 0.249375.
4. E^2 = 0.249375/800 = 0.000311719.
5. 1,280,000 E^2 = 1,280,000 x 0.000311719 = 1600 x 0.249375 = 399.0.
6. Option (b) is correct.

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6. 2026 · Q34 Sampling Distribution of Sample Mean · Conceptual

QUESTION

Sampling without replacement: $V(\bar{x})=\frac{\sigma^2}{n}\left(\frac{N\text{-}n}{N\text{-}1}\right)$. Under which condition(s) is the factor $\frac{N\text{-}n}{N\text{-}1}$ ignored? I. n is small compared to N II. $n=N$ III. Population is infinite

OPTIONS

(a) I only

(b) III only

(c) I and III only

(d) I, II and III

ANSWER **(AI-derived — no official key exists for this paper)**

(c) I and III only

EXAM SHORTCUT **(AI-derived explanation)**

The finite population correction tends to 1 when n is negligible relative to N, and when N is infinite. When n = N it equals 0 - the opposite of ignorable. So I and III only, option (c).

TIPS & TRICKS **(AI-derived explanation)**

- Evaluate the factor at the limits: n/N -> 0 gives (N-n)/(N-1) -> 1 (ignorable); n = N gives (N-N)/(N-1) = 0 (a complete census, zero sampling variance).
- Rule of thumb used in practice: drop the fpc when the sampling fraction n/N is below about 5%.
- n = N means the whole population is surveyed, so V(xbar) = 0 exactly - the factor is doing its most important work there, not being ignored.
- For an infinite population, sampling without replacement is equivalent to sampling with replacement, so V(xbar) = sigma^2/n with no correction.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. The finite population correction (fpc) is (N - n)/(N - 1); it may be dropped only when it is (approximately) equal to 1.
2. Statement I: if n is small compared with N, then (N - n)/(N - 1) is approximately 1 - n/N, which is close to 1. The factor may be ignored. Correct.
3. Statement II: if n = N, then (N - n)/(N - 1) = 0, so V(xbar) = 0. The whole population has been enumerated, so there is no sampling error at all; the factor is emphatically not ignorable. Incorrect.
4. Statement III: for an infinite population, letting N tend to infinity gives (N - n)/(N - 1) -> 1, so V(xbar) = sigma^2/n with no correction. Correct.
5. Only statements I and III are right.
6. Hence option (c) is correct.

S10 · STATISTICAL METHODS

Sampling Distributions: Sample Mean, Sample Variance, t, Chi-square, F

23 questions · 2018: 4 2019: 3 2020: 1 2021: 1 2022: 2 2023: 2 2024: 4 2025: 3 2026: 3

Weight. 23 questions (14.0% of Statistical Methods) · appeared in 9 of 9 papers · peak year 2018 (4)

Examiner's pattern. Constructing t and F from independent normals and chi-squares is the signature item — the C(X1+X2)/sqrt(X3^2+X4^2+X5^2) shell appeared in 2018, 2019 AND 2025 with only the indices changed. Also recurring: ns^2/sigma^2 ~ chi-square(n-1), E(X)E(1/X) for an F variate (asked verbatim in 2018 and 2019), and the MGF of s^2.

Must-know. t_k = Z/sqrt(V/k) with Z and V independent; F(m,n) = (U/m)/(V/n); (n-1)s^2/sigma^2 ~ chi2(n-1); for F(m,n): E(X) = n/(n-2), E(1/X) = m/(m-2); X/Y of two chi-squares gives Beta.

1. **2018 · Q34** Measures of Location · Numerical

QUESTION

A normal population has mean 0.1, SD 2.1. What is the probability that the mean of a sample of size 900 is negative? [Given $P(0<Z<1.43)=0.4236$]

OPTIONS

(a) 0.0764

(b) 0.6408

(c) 0.6074

(d) 0.7046

ANSWER **(AI-derived — no official key exists for this paper)**

(a) 0.0764

EXAM SHORTCUT **(AI-derived explanation)**

SE = 2.1/sqrt(900) = 2.1/30 = 0.07. z = (0 - 0.1)/0.07 = -1.43. P(Z < -1.43) = 0.5 - 0.4236 = 0.0764.

TIPS & TRICKS **(AI-derived explanation)**

- sqrt(900) = 30 is the only arithmetic that matters; 2.1/30 = 0.07 makes the z-score a clean -1.43, which is exactly the value the examiner supplies.
- The supplied figure P(0 < Z < 1.43) = 0.4236 is a CENTRAL area, not a tail. The left tail is 0.5 - 0.4236 = 0.0764 - subtracting from 0.5, never from 1.
- Option (b) 0.6408 = 0.5 + 0.1408 and option (d) 0.7046 are built from misreading the table; option (c) 0.6074 is 0.5 + 0.1074. All assume the wrong tail.
- Sanity check: the population mean 0.1 is positive and the sample is large, so a negative sample mean should be UNLIKELY - the answer must be well below 0.5, which only (a) is.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. The population has mean mu = 0.1 and standard deviation sigma = 2.1; the sample size is n = 900.
2. The standard error of the sample mean is sigma/sqrt(n) = 2.1/30 = 0.07.
3. We need P(Xbar < 0). Standardising: z = (0 - 0.1)/0.07 = -1.4286, which rounds to -1.43.
4. P(Xbar < 0) = P(Z < -1.43) = P(Z > 1.43) by symmetry.
5. Using the supplied value P(0 < Z < 1.43) = 0.4236, we get P(Z > 1.43) = 0.5 - 0.4236 = 0.0764.

6. Hence option (a) is correct.

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2. **2018 · Q35** Chi-square Distribution · Numerical

QUESTION

$(X_1, \ldots, X_5)$ are independent standard normal. The variable $\frac{C(X_1+X_2)}{(X_3^2+X_4^2+X_5^2)^{1/2}}$ has a t-distribution if C equals

OPTIONS

(a) $\frac{\sqrt{3}}{2}$

(b) $\sqrt{\frac{3}{2}}$

(c) $\frac{3}{2}$

(d) $\sqrt{\frac{2}{3}}$

ANSWER **(AI-derived — no official key exists for this paper)**

(b) $\sqrt{\frac{3}{2}}$

EXAM SHORTCUT **(AI-derived explanation)**

Build t = Z/sqrt(chi^2_3/3). Here Z = (X1+X2)/sqrt(2) and chi^2_3 = X3^2+X4^2+X5^2, so t = sqrt(3)(X1+X2)/(sqrt(2) sqrt(sum)). Hence C = sqrt(3)/sqrt(2) = sqrt(3/2).

TIPS & TRICKS **(AI-derived explanation)**

- The t-statistic template is ALWAYS standard normal divided by sqrt(chi-square/its own d.f.). Identify the two pieces and their d.f. before touching the algebra.
- X1 + X2 has variance 2, so it must be divided by sqrt(2) to become standard normal. Forgetting this gives sqrt(3) and no matching option.
- The d.f. is 3 (three squared normals), so the divisor inside the square root is 3 - that is where the sqrt(3) in the numerator comes from.
- sqrt(3/2) = sqrt(6)/2 = 1.2247 while sqrt(3)/2 = 0.866; option (a) is the trap for candidates who take the square root of 3 but not of 2.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. X1, ..., X5 are independent N(0,1).
2. X1 + X2 ~ N(0, 2), so Z = (X1 + X2)/sqrt(2) ~ N(0, 1).
3. X3^2 + X4^2 + X5^2 ~ chi-square with 3 degrees of freedom, and it is independent of X1 + X2.
4. By definition, t = Z / sqrt(chi-square_3 / 3) follows Student's t with 3 d.f.
5. Substituting: t = [(X1+X2)/sqrt(2)] / sqrt((X3^2+X4^2+X5^2)/3) = sqrt(3)(X1+X2) / [sqrt(2) sqrt(X3^2+X4^2+X5^2)].
6. Comparing with C(X1+X2)/(X3^2+X4^2+X5^2)^(1/2) gives C = sqrt(3)/sqrt(2) = sqrt(3/2), option (b).

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3. **2018 · Q36** Chi-square Distribution · Numerical

QUESTION

If $X \sim F(m,n)$ with $m>2, n>2$, then $E(X)\ E(1/X)$ equals

OPTIONS

(a) $\frac{n(n\text{-}2)}{m(m\text{-}2)}$

(b) $\frac{m(m\text{-}2)}{n(n\text{-}2)}$

(c) $\frac{mn}{(m\text{-}2)(n\text{-}2)}$

(d) $\frac{m(n\text{-}2)}{n(m\text{-}2)}$

ANSWER (AI-derived — no official key exists for this paper)

(c) $\frac{mn}{(m\text{-}2)(n\text{-}2)}$

EXAM SHORTCUT (AI-derived explanation)

E(X) = n/(n-2) for X ~ F(m,n). Since 1/X ~ F(n,m), swapping the roles gives E(1/X) = m/(m-2). The product is mn/((m-2)(n-2)).

TIPS & TRICKS (AI-derived explanation)

- The reciprocal property 1/F(m,n) ~ F(n,m) is the key. Once you have it, E(1/X) needs no integration - just swap m and n in the mean formula.
- Remember the mean of F(m,n) involves only the DENOMINATOR d.f.: E = n/(n-2), n > 2. Candidates who use m/(m-2) for E(X) land on option (b), the mirror-image trap.
- The conditions m > 2 and n > 2 in the stem exist precisely so that both expectations are finite - they are a hint about which formula to use.
- By Jensen's inequality E(X)E(1/X) >= 1 always. Option (c) gives mn/((m-2)(n-2)) > 1 for all admissible m, n, which is the only option guaranteed to satisfy this.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For X ~ F(m, n) with n > 2, the mean is E(X) = n/(n - 2).
2. A standard property of the F distribution is that 1/X ~ F(n, m), i.e. the degrees of freedom interchange.
3. Applying the mean formula to 1/X ~ F(n, m), whose denominator d.f. is now m, gives E(1/X) = m/(m - 2), valid since m > 2.
4. Multiplying: E(X) E(1/X) = [n/(n-2)] [m/(m-2)] = mn/((m-2)(n-2)).
5. Hence option (c) is correct.

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4. **2018 · Q38** Chi-square Distribution · Theoretical

QUESTION

Consider the following statements: 1. The normal distribution is a special case of the chi-square distribution with 1 degree of freedom. 2. All moments of order less than n of the t-distribution with n df exist. 3. If t follows Student's t with n df, then t^2 follows $F(1,n)$.

OPTIONS

(a) 1 only

(b) 1 and 2 only

(c) 2 and 3 only

(d) 1, 2 and 3

ANSWER (AI-derived — no official key exists for this paper)

(c) 2 and 3 only

EXAM SHORTCUT (AI-derived explanation)

Statement 1 reverses the relationship - the SQUARE of a standard normal is chi-square(1), not the other way round - so it is false. Statements 2 and 3 are standard results, giving 2 and 3 only.

TIPS & TRICKS (AI-derived explanation)

- Correct direction: Z ~ N(0,1) implies Z^2 ~ chi-square(1). The normal is not a special case of the chi-square; the chi-square family is built FROM normals.
- Moment rule for t with n d.f.: $E(t^k)$ exists only for k < n. The t with 1 d.f. is the Cauchy, which has no mean at all - a good anchor for remembering the rule.
- t_n^2 ~ F(1, n) is worth memorising alongside F(m,n) = 1/F(n,m) and chi-square(n)/n as n grows tending to 1. These three identities cover most sampling-distribution items.
- Where one statement asserts a containment between distribution families, check the DIRECTION of the containment - reversal is the single most common falsification device.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement 1: the chi-square with 1 degree of freedom is the distribution of Z^2 where Z ~ N(0,1). The normal itself is not a chi-square; the chi-square is non-negative and skewed while the normal is symmetric on the whole line. FALSE.
2. Statement 2: for Student's t with n degrees of freedom, the tail behaves like |t|^(-(n+1)), so E(|t|^k) is finite exactly when k < n. Hence all moments of order less than n exist. TRUE.
3. Statement 3: write t = Z/sqrt(V/n) with Z ~ N(0,1) and V ~ chi-square(n) independent. Then t^2 = Z^2/(V/n) = (Z^2/1)/(V/n), a ratio of independent chi-squares divided by their degrees of freedom, which is F(1, n). TRUE.
4. Only statements 2 and 3 are correct, so option (c) is right.

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5. 2019 · Q25 Sampling Distribution of Sample Mean · Numerical

QUESTION

Let $X_1,\dots,X_5$ be a random sample of size 5 from a standard normal population. If $\overline{X}=\frac{1}{5}\sum_{i=1}^{5} X_i$ and $T=\sum_{i=1}^{5}(X_i-\overline{X})^2$, then $E(T^2\overline{X}^2)$ is equal to

OPTIONS

(a) 3

(b) 3.6

(c) 4.8

(d) 5.2

ANSWER (AI-derived — no official key exists for this paper)

(c) 4.8

EXAM SHORTCUT (AI-derived explanation)

Xbar and T are independent, T ~ chi-square(4) and Xbar ~ N(0, 1/5). So E(T^2 $Xbar^2$) = E(T^2) E($Xbar^2$) = (2.4 + 4^2)(1/5) = 24/5 = 4.8.

TIPS & TRICKS (AI-derived explanation)

- Independence of Xbar and the corrected sum of squares in normal sampling is the key that turns E(T^2 $Xbar^2$) into a product of two easy expectations.
- For chi-square with v d.f.: mean v, variance 2v, so E(T^2) = 2v + v^2. With v = 4 that is 8 + 16 = 24.
- Xbar has variance $sigma^2$/n = 1/5, and since its mean is 0, E($Xbar^2$) = 1/5 directly.
- Note T uses n - 1 = 4 degrees of freedom, not 5 - subtracting one for the estimated mean is where most marks are lost here.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For a random sample of size n = 5 from N(0,1), the sample mean Xbar and the corrected sum of squares T = sum $(X_i - Xbar)^2$ are independent.
2. T follows a chi-square distribution with n - 1 = 4 degrees of freedom, and Xbar ~ N(0, 1/5).
3. By independence, E(T^2 $Xbar^2$) = E(T^2) . E($Xbar^2$).
4. For chi-square with 4 d.f.: E(T) = 4 and Var(T) = 8, so E(T^2) = Var(T) + $[E(T)]^2$ = 8 + 16 = 24.
5. Since E(Xbar) = 0, E($Xbar^2$) = Var(Xbar) = 1/5.
6. Therefore E(T^2 $Xbar^2$) = 24 x (1/5) = 4.8, option (c).

---

6. 2019 · Q28 Chi-square Distribution · Numerical

QUESTION

Suppose $X_1, \ldots, X_5$ are independent, each standard normal. A constant C such that $\frac{C(X_1+X_2)}{(X_3^2+X_4^2+X_5^2)^{1/2}}$ has a t-distribution, has value

OPTIONS

(a) $\frac{\sqrt{3}}{2}$

(b) $\sqrt{\frac{3}{2}}$

(c) $\frac{3}{2}$

(d) $\sqrt{\frac{2}{3}}$

ANSWER (AI-derived — no official key exists for this paper)

(b) $\sqrt{\frac{3}{2}}$

EXAM SHORTCUT (AI-derived explanation)

Standardise the numerator: (X1+X2)/sqrt2 ~ N(0,1). The denominator squared is chi-square(3), so divide by sqrt3. Hence C = sqrt3/sqrt2 = sqrt(3/2).

TIPS & TRICKS (AI-derived explanation)

- t = N(0,1) / sqrt(chi-square_v / v). Write that template first and then match the pieces of the given expression to it.

- Var(X1 + X2) = 2, so the numerator needs a sqrt2 divisor; three squared normals give v = 3, so the denominator needs a sqrt3 divisor. Combining gives sqrt(3/2).
- This item is an exact repeat of 2018 Q35 - a clear signal that the sampling-distribution constructions are recycled across papers.
- Numerically sqrt(3/2) = 1.2247 whereas sqrt3/2 = 0.866; if you find a value below 1 you have divided instead of multiplied.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. X1, ..., X5 are independent N(0,1) variables.
2. X1 + X2 ~ N(0, 2), so Z = (X1 + X2)/sqrt(2) is standard normal.
3. $X3^2 + X4^2 + X5^2$ ~ chi-square with 3 degrees of freedom, independent of X1 + X2.
4. By definition t = Z/sqrt(chi-square_3/3) = [(X1+X2)/sqrt2] . sqrt(3)/sqrt($X3^2+X4^2+X5^2$).
5. Rewriting: t = sqrt(3/2) (X1+X2)/($X3^2+X4^2+X5^2$)^(1/2).
6. Hence C = sqrt(3/2), option (b).

---

7. 2019 · Q29 Chi-square Distribution · Numerical

QUESTION

If X is an $F(m,n)$ random variable, $m>2$, $n>2$, then $E(X)\cdot\ E(\frac{1}{X})$ equals

OPTIONS

(a) $\frac{n(n\text{-}2)}{m(m\text{-}2)}$

(b) $\frac{m(m\text{-}2)}{n(n\text{-}2)}$

(c) $\frac{mn}{(m\text{-}2)(n\text{-}2)}$

(d) $\frac{m(n\text{-}2)}{n(m\text{-}2)}$

ANSWER (AI-derived — no official key exists for this paper)

(c) $\frac{mn}{(m\text{-}2)(n\text{-}2)}$

EXAM SHORTCUT (AI-derived explanation)

E(X) = n/(n-2) and, since 1/X ~ F(n,m), E(1/X) = m/(m-2). Their product is mn/((m-2)(n-2)).

TIPS & TRICKS (AI-derived explanation)

- The reciprocal of an F(m,n) variable is F(n,m) - this single property converts E(1/X) into a mean formula with the degrees of freedom swapped.
- The mean of F(m,n) uses only the DENOMINATOR degrees of freedom: n/(n-2). Using m/(m-2) for E(X) produces option (b), the mirror trap.
- Jensen's inequality guarantees E(X)E(1/X) >= 1; only option (c) exceeds 1 for all admissible m, n greater than 2.
- This item repeats 2018 Q36 verbatim - F-distribution moment identities are among the most reused facts in this paper.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For X ~ F(m, n) with n > 2, the mean is E(X) = n/(n - 2).

2. Since $1/X \sim F(n, m)$, the mean formula applied with the roles of m and n exchanged gives $E(1/X) = m/(m - 2)$, valid because $m > 2$.
3. Multiplying the two expectations: $E(X)\, E(1/X) = [n/(n-2)][m/(m-2)] = mn/((m-2)(n-2))$.
4. Hence option (c) is correct.

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8. **2020 · Q4** Measures of Location · Numerical

QUESTION

If $X \sim F(5,10)$: 1. $E(X)>1$. 2. Mode of $X>1$.

OPTIONS

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

ANSWER **(AI-derived — no official key exists for this paper)**

(a) 1 only

EXAM SHORTCUT **(AI-derived explanation)**

$E(X) = n/(n-2) = 10/8 = 1.25 > 1$, so statement 1 holds. Mode $= [(m-2)/m][n/(n+2)] = (3/5)(10/12) = 0.5 < 1$, so statement 2 fails.

TIPS & TRICKS **(AI-derived explanation)**

- Two F formulas to memorise: mean = $n/(n-2)$ using only the DENOMINATOR d.f., and mode = $[(m-2)/m][n/(n+2)]$ using both.
- The F distribution is positively skewed, so mode < median < mean. Since the mean is only 1.25, a mode below 1 is entirely expected - that intuition answers statement 2 before any arithmetic.
- The mean of $F(m,n)$ always exceeds 1 whenever $n > 2$, because $n/(n-2) > 1$. Statement 1 is therefore true for every admissible F distribution.
- The mode is less than 1 for all m, n, since $[(m-2)/m] < 1$ and $[n/(n+2)] < 1$ - a general fact worth carrying.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. For $X \sim F(m, n)$ with $m = 5$ and $n = 10$, the mean is $E(X) = n/(n - 2)$, valid for $n > 2$.
2. $E(X) = 10/(10 - 2) = 10/8 = 1.25$, which is greater than 1. Statement 1 is TRUE.
3. The mode of $F(m, n)$ for $m > 2$ is Mode $= [(m - 2)/m] . [n/(n + 2)]$.
4. Substituting: $[(5-2)/5][10/12] = (3/5)(5/6) = 1/2 = 0.5$.
5. Since $0.5 < 1$, statement 2 is FALSE.
6. Only statement 1 is correct, option (a).

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9. **2021 · Q33** Chi-square Distribution · Theoretical

QUESTION

Consider the following statements: 1. Mean and variance of the chi-square distribution are not the same. 2. The sum of two chi-square variates is also a chi-square variate.

OPTIONS

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

ANSWER (AI-derived — no official key exists for this paper)

(a) 1 only

Source note. AI-derived key with a caveat. Statement 1 is unambiguously true (mean n, variance 2n). Statement 2 states the additive property WITHOUT the independence requirement; as printed it is false, since the sum of two dependent chi-square variables need not be chi-square (e.g. Z^2 and (-Z)^2 sum to 2Z^2, which is not chi-square). Some textbooks state the reproductive property loosely and would treat statement 2 as true, which would make (c) the key; (a) is selected here as the mathematically correct reading.

EXAM SHORTCUT (AI-derived explanation)

Chi-square(n) has mean n and variance 2n, so statement 1 is true. The additive property needs INDEPENDENCE, which statement 2 omits, so as printed it is false.

TIPS & TRICKS (AI-derived explanation)

- Chi-square with v d.f.: mean v, variance 2v, so the two coincide only at v = 0. Statement 1 is safe.
- Reproductive property: if U ~ chi-square(m) and V ~ chi-square(n) are INDEPENDENT then U + V ~ chi-square(m+n). The independence clause is not decorative.
- Counterexample for the dependent case: with Z ~ N(0,1), both Z^2 and $(-Z)^2$ are chi-square(1), but their sum $2Z^2$ has mean 2 and variance 8, whereas chi-square(2) has variance 4.
- Whenever a reproductive or additive property is quoted, check whether independence is stated - its omission is a favourite falsification device.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement 1: for the chi-square distribution with n degrees of freedom, the mean is n and the variance is 2n. These differ for every n >= 1, so the mean and variance are not the same. TRUE.
2. Statement 2: the reproductive property of the chi-square distribution states that the sum of two INDEPENDENT chi-square variates with m and n degrees of freedom is chi-square with m + n degrees of freedom.
3. Without independence the conclusion fails. Take Z ~ N(0,1) and set $U = Z^2$ and $V = (-Z)^2$; each is chi-square with 1 d.f., but $U + V = 2Z^2$ has mean 2 and variance 8.
4. A chi-square variable with mean 2 would have 2 degrees of freedom and variance 4, not 8, so $2Z^2$ is not chi-square.
5. Hence statement 2, as printed without the independence condition, is FALSE.
6. Only statement 1 is correct, option (a).

10. 2022 · Q12 Sampling Distribution of Sample Mean · Numerical

QUESTION

Let $X_1,\ldots,X_5$ be a random sample of size 5 from a standard normal population. Let $\overline{X}=\frac{\sum_{i=1}^{5} X_i}{5}$ and $T=\sum_{i=1}^{5}(X_i\text{-}\overline{X})^2$. What is $E(T^2\overline{X}^2)$ equal to?

OPTIONS

(a) 4.0

(b) 4.2

(c) 24.2

(d) 25.2

ANSWER **(AI-derived — no official key exists for this paper)**

(c) 24.2

Source note. SOURCE NOTE: as printed the expression E(T^2 Xbar^2) evaluates, using the independence of Xbar and T with T ~ chi-square(4), to E(T^2) E(Xbar^2) = 24 x 0.2 = 4.8, which is NOT among the four printed options (the identical item in the 2019 paper, Q25, did list 4.8 and that is its answer). The only option consistent with these same quantities is 24.2 = E(T^2) + E(Xbar^2), i.e. the reading E(T^2 + Xbar^2); the option set (4.0 = E(T), 4.2 = E(T) + E(Xbar^2), 24.2, 25.2) is clearly built around sums. The stem is preserved verbatim and (c) is selected as the only internally consistent key.

EXAM SHORTCUT **(AI-derived explanation)**

Xbar and T are independent with T ~ chi-square(4) and Xbar ~ N(0, 1/5). So E(T^2) = 2(4) + 4^2 = 24 and E(Xbar^2) = 1/5 = 0.2, giving 24.2 for the sum.

TIPS & TRICKS **(AI-derived explanation)**

- In normal sampling Xbar and the corrected sum of squares are INDEPENDENT, and T = sum (X_i - Xbar)^2 ~ chi-square(n-1) - here 4 degrees of freedom, not 5.
- For chi-square with v d.f.: E = v, Var = 2v, so E(T^2) = 2v + v^2 = 8 + 16 = 24.
- E(Xbar^2) = Var(Xbar) = sigma^2/n = 1/5 = 0.2, since E(Xbar) = 0.
- The four options are 4.0, 4.2, 24.2 and 25.2 - all built from the two quantities 4, 24, 0.2 and 1, which is a strong hint that the intended expression is a sum rather than a product.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. For a sample of size 5 from N(0,1), Xbar and T = sum (X_i - Xbar)^2 are independent, with T ~ chi-square with 4 degrees of freedom and Xbar ~ N(0, 1/5).
2. For chi-square with 4 d.f.: E(T) = 4 and Var(T) = 8, so E(T^2) = Var(T) + [E(T)]^2 = 8 + 16 = 24.
3. Since E(Xbar) = 0, E(Xbar^2) = Var(Xbar) = 1/5 = 0.2.
4. The product reading gives E(T^2 Xbar^2) = E(T^2) E(Xbar^2) = 24 x 0.2 = 4.8, which does not appear among the options.
5. The sum reading gives E(T^2) + E(Xbar^2) = 24 + 0.2 = 24.2, which is option (c) and is the only choice these quantities can produce.
6. Hence (c) is selected, with the discrepancy flagged rather than the stem altered.

11. 2022 · Q17 Chi-square Distribution · Numerical

QUESTION

If X is normally distributed with mean μ, variance 1, and Y^2 is independently distributed as central χ^2 with f degrees of freedom, what is the value of

$$\int_{-\infty}^{\infty} \int_{0}^{\infty} y^{f-2} \; e^{\frac{-(x^2+y^2)}{2}+\mu x} \; dy^2 \; dx \; ?$$

OPTIONS

(a) $\Gamma(\frac{f}{2})2^{\frac{\mu+1}{2}}$

(b) $\Gamma(\frac{f}{2})2^{\frac{f+1}{2}}\sqrt{\pi} \; e^{\frac{\mu^2}{2}}$

(c) $\sqrt{\pi} \; e^{\frac{\mu^2}{2}}$

(d) $\Gamma(\frac{f}{2})2^{\frac{f+1}{2}}\sqrt{\pi} \; e^{-\frac{\mu^2}{2}}$

ANSWER (AI-derived — no official key exists for this paper)

(b) $\Gamma(\frac{f}{2})2^{\frac{f+1}{2}}\sqrt{\pi} \; e^{\frac{\mu^2}{2}}$

EXAM SHORTCUT (AI-derived explanation)

Substituting u = y^2 turns the y-integral into a Gamma integral equal to 2^(f/2) Gamma(f/2); completing the square in x gives sqrt(2 pi) e^(mu^2/2). Multiplying: Gamma(f/2) 2^((f+1)/2) sqrt(pi) e^(mu^2/2).

TIPS & TRICKS (AI-derived explanation)

- Read the differential carefully: it is d(y^2), not dy. Setting u = y^2 makes the y-part a standard Gamma integral with no Jacobian fuss.
- The y-integral becomes the integral of u^(f/2 - 1) e^(-u/2) du = 2^(f/2) Gamma(f/2) - the Gamma(shape f/2, scale 2) normalising constant, i.e. exactly the chi-square with f d.f.
- Complete the square in the exponent: -x^2/2 + mu x = -(x - mu)^2/2 + mu^2/2, so the x-integral is sqrt(2 pi) e^(+mu^2/2). The sign of mu^2/2 is POSITIVE, which separates options (b) and (d).
- Combining 2^(f/2) with the sqrt(2) from sqrt(2 pi) gives 2^((f+1)/2) sqrt(pi) - track the powers of 2 to the end.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Handle the y-integral first, noting that the differential is d(y^2). Put u = y^2, so the integrand y^(f-2) e^(-y^2/2) becomes u^((f-2)/2) e^(-u/2).
2. The integral from 0 to infinity of u^(f/2 - 1) e^(-u/2) du is the Gamma(shape f/2, scale 2) normalising constant, namely 2^(f/2) Gamma(f/2).
3. For the x-integral, complete the square: -x^2/2 + mu x = -(x - mu)^2/2 + mu^2/2.
4. So the integral over x is e^(mu^2/2) times the integral of e^(-(x-mu)^2/2) dx = e^(mu^2/2) sqrt(2 pi).
5. Multiplying the two parts: 2^(f/2) Gamma(f/2) . sqrt(2 pi) e^(mu^2/2) = Gamma(f/2) 2^((f+1)/2) sqrt(pi) e^(mu^2/2).
6. Hence option (b) is correct.

12. 2023 · Q36 Chi-square Distribution · Factual

QUESTION

If a sample of size n is taken from $N(\mu,\sigma^2)$ and let $\bar{x}$ and s^2 be sample mean and sample variance respectively, then distribution of $\frac{(n-1)s^2}{\sigma^2}$ is:

OPTIONS

(a) Chi-square with $(n-1)$ degrees of freedom

(b) Chi-square with n degrees of freedom

(c) Student's-t with $(n-1)$ degrees of freedom

(d) Student's-t with n degrees of freedom

ANSWER (AI-derived — no official key exists for this paper)

(a) Chi-square with $(n-1)$ degrees of freedom

EXAM SHORTCUT (AI-derived explanation)

In normal sampling, (n-1)s^2/sigma^2 follows a chi-square distribution with n - 1 degrees of freedom.

TIPS & TRICKS (AI-derived explanation)

- One degree of freedom is lost because the sample mean is estimated from the data - hence n - 1, not n.
- The statistic is non-negative and skewed, so it cannot be a t distribution; t variables range over the whole line. That kills options (c) and (d) immediately.
- Companion facts: xbar and s^2 are INDEPENDENT in normal sampling, and their combination gives the t statistic (xbar - mu)/(s/sqrt n) ~ t with n-1 d.f.
- This result underpins both the confidence interval for sigma^2 and the chi-square test of a specified variance.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let x_1, ..., x_n be a random sample from N(mu, sigma^2), with sample mean xbar and sample variance s^2 = sum (x_i - xbar)^2/(n-1).
2. Then sum (x_i - xbar)^2 = (n-1)s^2, and dividing by sigma^2 standardises the sum of squared deviations.
3. Because the deviations are taken about the ESTIMATED mean xbar rather than the true mean mu, one linear constraint is imposed and one degree of freedom is lost.
4. The classical result is that (n-1)s^2/sigma^2 follows a chi-square distribution with n - 1 degrees of freedom, independently of xbar.
5. The statistic is non-negative and skewed, consistent with a chi-square rather than a t distribution.
6. Option (a) is correct.

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13. 2023 · Q38 Chi-square Distribution · Factual

QUESTION

Let $x_1, x_2, x_3, \ldots, x_n$ be a random sample from a Normal population with mean μ_0 and variance σ^2. If $\bar{X}$ is the sample mean, then the distribution of $\frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}}$ is:

OPTIONS

(a) $N(0,1)$

(b) t with n degrees of freedom

(c) t with $(n\text{-}1)$ degrees of freedom

(d) χ^2 with $(n\text{-}1)$ degrees of freedom

ANSWER (AI-derived — no official key exists for this paper)

(a) $N(0,1)$

EXAM SHORTCUT (AI-derived explanation)

The population standard deviation sigma is used, not the sample s, so the standardised sample mean is exactly standard normal.

TIPS & TRICKS (AI-derived explanation)

- sigma in the denominator gives N(0,1); s in the denominator gives t with n - 1 degrees of freedom. Check which symbol appears before choosing.
- For a NORMAL parent the result is exact for every n, not merely asymptotic - no appeal to the CLT is needed.
- xbar ~ N(mu, sigma^2/n), so subtracting mu and dividing by sigma/sqrt(n) standardises it exactly.
- The t distribution arises only because s is itself random; replacing sigma by s adds variability and fattens the tails.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The sample comes from a normal population with mean mu0 and variance sigma^2, so the sample mean is exactly normal: Xbar ~ N(mu0, sigma^2/n).
2. Subtracting the mean gives Xbar - mu0 ~ N(0, sigma^2/n).
3. Dividing by the standard deviation of Xbar, namely sigma/sqrt(n), standardises it.
4. Since sigma is the KNOWN population standard deviation and not a sample estimate, no additional randomness is introduced.
5. Hence (Xbar - mu0)/(sigma/sqrt(n)) ~ N(0,1) exactly, for every sample size n.
6. Option (a) is correct.

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14. 2024 · Q2 Sampling Distribution of Sample Variance · Numerical

COMMON DATA FOR THIS GROUP

Questions 2–3 are based on: let $X_1, X_2, X_3, \ldots, X_n$ be a random sample from $N(\mu, \sigma^2)$ distribution and

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2, \quad \overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

QUESTION

What is the moment-generating function of S^2?

OPTIONS

(a) $\left(1-\frac{2(n-1)t}{\sigma^2}\right)^{-\frac{n-1}{2}}$

(b) $\left(1-\frac{(n-1)t}{\sigma^2}\right)^{-\frac{n-1}{2}}$

(c) $\left(1-\frac{2\sigma^2 t}{n-1}\right)^{-\frac{n-1}{2}}$

(d) $\left(1-\frac{\sigma^2 t}{n-1}\right)^{-\frac{n-1}{2}}$

ANSWER (AI-derived — no official key exists for this paper)

(c) $\left(1-\frac{2\sigma^2 t}{n-1}\right)^{-\frac{n-1}{2}}$

EXAM SHORTCUT (AI-derived explanation)

(n-1)S^2/sigma^2 ~ chi-square(n-1), so S^2 = [sigma^2/(n-1)] times a chi-square. Replacing t by t sigma^2/(n-1) in (1-2t)^(-(n-1)/2) gives (1 - 2 sigma^2 t/(n-1))^(-(n-1)/2).

TIPS & TRICKS (AI-derived explanation)

- Start from the sampling result (n-1)S^2/sigma^2 ~ chi-square with n - 1 d.f., then treat S^2 as a constant multiple of that chi-square.
- Scaling rule for MGFs: if M_V(t) is the MGF of V then the MGF of cV is M_V(ct). Here c = sigma^2/(n-1).
- chi-square MGF is (1 - 2t)^(-k/2) with k = n - 1, so the exponent -(n-1)/2 is common to all four options - the discriminator is the bracket.
- Dimension check: t multiplies S^2, which has the units of sigma^2, so the product sigma^2 t must appear in the NUMERATOR. Options (a) and (b) have sigma^2 in the denominator and are wrong.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For a random sample from N(mu, sigma^2), the standard result is that (n-1)S^2/sigma^2 follows a chi-square distribution with n - 1 degrees of freedom.
2. Write V = (n-1)S^2/sigma^2, so S^2 = [sigma^2/(n-1)] V.
3. The moment generating function of a chi-square variable with k degrees of freedom is M_V(t) = (1 - 2t)^(-k/2), here with k = n - 1.
4. For a constant multiple, M_(cV)(t) = M_V(ct) with c = sigma^2/(n-1).
5. Hence M_(S^2)(t) = [1 - 2t sigma^2/(n-1)]^(-(n-1)/2).
6. Option (c) is correct.

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15. 2024 · Q3 Sampling Distribution of Sample Variance · Numerical

COMMON DATA FOR THIS GROUP

Questions 2–3 are based on: let $X_1, X_2, X_3, \ldots, X_n$ be a random sample from $N(\mu, \sigma^2)$ distribution and

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i-\overline{X})^2, \qquad \overline{X}=\frac{1}{n}\sum_{i=1}^{n}X_i$$

QUESTION

If

$$r = \frac{1}{2}\,\frac{\Gamma\left(\frac{n-3}{2}\right)}{\Gamma\left(\frac{n-1}{2}\right)}, \qquad n>3$$

then what is the value of $E\left(\frac{1}{S^2}\right)$?

OPTIONS

(a) $\frac{(n-1)r}{\sigma^2}$

(b) $\frac{nr}{\sigma^2}$

(c) $\frac{2nr}{\sigma^2}$

(d) $\frac{nr}{2\sigma^2}$

ANSWER (AI-derived — no official key exists for this paper)

(a) $\frac{(n-1)r}{\sigma^2}$

EXAM SHORTCUT (AI-derived explanation)

E(1/chi-square(k)) = 1/(k-2), so with k = n-1, E(1/S^2) = (n-1)/[sigma^2(n-3)]. Since the Gamma ratio gives r = 1/(n-3), this is (n-1)r/sigma^2.

TIPS & TRICKS (AI-derived explanation)

- Key moment: for V ~ chi-square(k) with k > 2, E(1/V) = 1/(k - 2). Here k = n - 1, so the denominator is n - 3, which explains the condition n > 3.
- Simplify the given r first: Gamma((n-1)/2) = ((n-3)/2) Gamma((n-3)/2), so the ratio is 2/(n-3) and r = 1/(n-3).
- Because S^2 = sigma^2 V/(n-1), the reciprocal brings a factor (n-1)/sigma^2 out in front - track that constant carefully.
- The recurrence Gamma(z+1) = z Gamma(z) is the only Gamma identity needed; applying it once collapses the whole expression.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let V = (n-1)S^2/sigma^2 ~ chi-square with n - 1 degrees of freedom, so S^2 = sigma^2 V/(n-1) and 1/S^2 = (n-1)/(sigma^2 V).
2. For a chi-square variable with k > 2 degrees of freedom, E(1/V) = 1/(k - 2). With k = n - 1 this is 1/(n - 3), which requires n > 3.
3. Hence E(1/S^2) = (n-1)/[sigma^2 (n-3)].
4. Now simplify the given constant: using Gamma(z+1) = z Gamma(z) with z = (n-3)/2, we get Gamma((n-1)/2) = ((n-3)/2) Gamma((n-3)/2).
5. So r = (1/2) Gamma((n-3)/2)/Gamma((n-1)/2) = (1/2)(2/(n-3)) = 1/(n-3).
6. Therefore E(1/S^2) = (n-1)r/sigma^2, option (a).

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16. 2024 · Q7 t-Distribution · Numerical

COMMON DATA FOR THIS GROUP

Questions 7–10 are based on: let $X_1, X_2, X_3, \ldots, X_7$ be iid random variables, each having the $N(0,1)$ distribution.

QUESTION

What is the distribution of the following?

$$\frac{\sqrt{2}\,(X_5 - X_6 + X_7)}{\sqrt{(X_1+X_2+X_3)^2 + 3X_4^2}}$$

OPTIONS

(a) $N(0,1)$

(b) $t_{(2)}$

(c) $t_{(3)}$

(d) $t_{(7)}$

ANSWER (AI-derived — no official key exists for this paper)

(b) $t_{(2)}$

EXAM SHORTCUT (AI-derived explanation)

The numerator X5 - X6 + X7 is N(0,3); the denominator squared satisfies [(X1+X2+X3)^2 + 3X4^2]/3 ~ chi-square(2). Assembling the t template gives t with 2 degrees of freedom.

TIPS & TRICKS (AI-derived explanation)

- Divide the denominator's contents by 3: (X1+X2+X3)^2/3 is the square of a standard normal, and 3X4^2/3 = X4^2 is another - two independent chi-square(1) terms, so chi-square(2).
- X5 - X6 + X7 has variance 1 + 1 + 1 = 3 regardless of the signs, so dividing by sqrt(3) standardises it.
- t = Z/sqrt(chi-square_v/v) with v = 2; the constants sqrt(3) cancel and leave exactly the sqrt(2) shown in the stem - a strong confirmation.
- Degrees of freedom = number of independent squared normals in the denominator, here 2, not the sample size 7. Option (d) is the trap.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Numerator: X5 - X6 + X7 is normal with mean 0 and variance 1 + 1 + 1 = 3, so Z = (X5 - X6 + X7)/sqrt(3) ~ N(0,1).
2. Denominator: X1 + X2 + X3 ~ N(0,3), so (X1+X2+X3)/sqrt(3) is standard normal and (X1+X2+X3)^2/3 ~ chi-square(1).
3. Also X4 ~ N(0,1) gives X4^2 ~ chi-square(1), and these two are independent.
4. Hence W = [(X1+X2+X3)^2 + 3X4^2]/3 = (X1+X2+X3)^2/3 + X4^2 ~ chi-square with 2 degrees of freedom, independent of the numerator.
5. The t statistic is Z/sqrt(W/2) = [(X5-X6+X7)/sqrt3] . sqrt(2)/sqrt(W) = sqrt(2)(X5-X6+X7)/sqrt(3W), and 3W is exactly the expression under the printed radical.
6. So the statistic is t with 2 degrees of freedom, option (b).

---

17. 2024 · Q14 F-Distribution · Theoretical

QUESTION

Suppose X and Y are independent random variables having chi-square distributions with m and n degrees of freedom. Which of the following statements is/are correct? I. $\frac{nX}{mY}$ follows $F_{m,n}$. II. $\frac{mY}{nX}$ follows $F_{n,m}$. Select the answer using the code given below:

OPTIONS

(a) I only

(b) II only

(c) Both I and II

(d) Neither I nor II

ANSWER (AI-derived — no official key exists for this paper)

(c) Both I and II

EXAM SHORTCUT (AI-derived explanation)

F(m,n) is defined as (X/m)/(Y/n) = nX/(mY), so statement I holds; its reciprocal mY/(nX) is F(n,m), so statement II holds too.

TIPS & TRICKS (AI-derived explanation)

- Definition: F with (m,n) degrees of freedom is the ratio of two independent chi-squares EACH DIVIDED BY ITS OWN degrees of freedom.
- Write it out: (X/m)/(Y/n) = (X/m)(n/Y) = nX/(mY). Matching this against statement I confirms it immediately.
- Reciprocal property: 1/F(m,n) = F(n,m). Statement II is exactly the reciprocal of statement I, so the two stand or fall together.
- Because the two statements are reciprocals, only options (c) and (d) are logically possible - and since the definition holds, it must be (c).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. X ~ chi-square(m) and Y ~ chi-square(n) are independent.
2. By definition, an F variable with (m, n) degrees of freedom is the ratio of the two chi-squares each divided by its own degrees of freedom: F = (X/m)/(Y/n).
3. Simplifying, (X/m)(n/Y) = nX/(mY), which is exactly the expression in statement I. TRUE.
4. Taking reciprocals, mY/(nX) = 1/F, and the standard reciprocal property gives 1/F(m,n) = F(n,m).
5. Directly: mY/(nX) = (Y/n)/(X/m), which is the ratio of the chi-squares in the opposite order, hence F with (n, m) degrees of freedom. TRUE.
6. Both statements are correct, option (c).

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18. 2025 · Q10 Chi-square Distribution · Theoretical

QUESTION

Let X follow Chi-square distribution with n degrees of freedom. Which statement(s) is/are correct? I. $\frac{X\text{-}n}{\sqrt{2n}}$ is asymptotically $N(0,1)$. II. $\frac{X}{n}$ is asymptotically normal with mean 1 and variance $\frac{2}{n}$.

OPTIONS

(a) I only

(b) II only

(c) Both I and II

(d) Neither I nor II

ANSWER (AI-derived — no official key exists for this paper)

(c) Both I and II

EXAM SHORTCUT (AI-derived explanation)

chi-square(n) is a sum of n i.i.d. chi-square(1) terms with mean 1 and variance 2, so the CLT gives both statements: (X-n)/sqrt(2n) tends to N(0,1) and X/n is approximately N(1, 2/n).

TIPS & TRICKS (AI-derived explanation)

- Decompose: chi-square(n) = sum of n independent squared standard normals, each with mean 1 and variance 2. The CLT then applies directly.
- Statement I is the standardised form and statement II is the same result divided by n - they are algebraically equivalent, so both stand or fall together.
- Check the scaling: dividing X by n divides the mean by n (giving 1) and the variance by n^2 (giving $2n/n^2$ = 2/n). Statement II is consistent.
- Because the two statements are equivalent, only options (c) and (d) are logically possible - and the CLT confirms (c).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A chi-square variable with n degrees of freedom can be written as X = $Z1^2$ + $Z2^2$ + ... + Zn^2 with the Zi independent standard normal.
2. Each Zi^2 has mean 1 and variance 2, and they are i.i.d.
3. By the Central Limit Theorem, (X - n)/sqrt(2n) converges in distribution to N(0,1). Statement I is TRUE.
4. Dividing by n: X/n has mean 1 and variance $2n/n^2$ = 2/n.
5. Since X/n = 1 + sqrt(2/n) [(X-n)/sqrt(2n)], it is asymptotically normal with mean 1 and variance 2/n. Statement II is TRUE and is simply a restatement of I.
6. Both statements are correct, option (c).

---

19. 2025 · Q56 Chi-square Distribution · Conceptual

QUESTION

Let $(X_1, \ldots, X_{53})$ be a random sample from $N(0,1)$. If for some $\alpha_i, \beta_i, \gamma_i$: $\sum_{i=1}^{53} \alpha_i X_i = 0$, $\sum_{i=1}^{53} \beta_i X_i = 0$, $\sum_{i=1}^{53} \gamma_i X_i = 0$, the distribution of $\sum_{i=1}^{53} X_i^2$ is:

OPTIONS

(a) χ^2_{52}

(b) χ^2_{50}

(c) χ^2_{49}

(d) χ^2_{53}

ANSWER (AI-derived — no official key exists for this paper)

(b) χ^2_{50}

EXAM SHORTCUT (AI-derived explanation)

Each independent linear constraint removes one degree of freedom, so 53 squared standard normals subject to 3 constraints give chi-square with 53 - 3 = 50 degrees of freedom.

TIPS & TRICKS (AI-derived explanation)

- Rule: degrees of freedom = number of variables MINUS number of independent linear restrictions. Count the restrictions carefully - here three.
- This is the same principle that makes sum $(X\_i - Xbar)^2$ a chi-square with n - 1 d.f.: one constraint, one degree lost.
- 53 - 3 = 50; option (a) subtracts only one and option (c) subtracts four - both mis-count the restrictions.

- Option (d) with 53 d.f. would be the unconstrained case, which the stem explicitly rules out.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For a random sample of 53 independent standard normal variables, the unconstrained sum of squares would follow a chi-square distribution with 53 degrees of freedom.
2. The three stated conditions are independent linear restrictions on the X_i.
3. Each independent linear constraint confines the vector to a subspace of one lower dimension, and hence removes one degree of freedom from the sum of squares.
4. Three such constraints therefore reduce the degrees of freedom by 3.
5. Hence the sum of squares follows chi-square with 53 - 3 = 50 degrees of freedom.
6. This is the same principle by which sum (X_i - Xbar)^2 has n - 1 degrees of freedom under the single constraint sum (X_i - Xbar) = 0. Option (b) is correct.

---

20. 2025 · Q59 t-Distribution · Numerical

QUESTION

$X_1, \ldots, X_5$ are independent standard normal. A constant C such that $\frac{C(X_4+X_5)}{(X_1^2+X_2^2+X_3^2)^{1/2}}$ has a t-distribution has value:

OPTIONS

(a) $\sqrt{3/2}$

(b) $\sqrt{3}/2$

(c) $3/2$

(d) $\sqrt{2/3}$

ANSWER (AI-derived — no official key exists for this paper)

(a) $\sqrt{3/2}$

EXAM SHORTCUT (AI-derived explanation)

(X4 + X5)/sqrt2 is standard normal and X1^2 + X2^2 + X3^2 is chi-square(3), so t = sqrt(3)(X4+X5)/(sqrt2 sqrt(sum)), giving C = sqrt(3/2).

TIPS & TRICKS (AI-derived explanation)

- t = N(0,1)/sqrt(chi-square_v/v). Identify the two pieces and their degrees of freedom before doing any algebra.
- X4 + X5 has variance 2, so it needs a sqrt(2) divisor; the three squared normals give v = 3, so a sqrt(3) multiplier appears.
- Combining: C = sqrt(3)/sqrt(2) = sqrt(3/2) = 1.2247. Option (b) sqrt(3)/2 = 0.866 takes the square root of 3 but not of 2.
- This is a repeat of 2018 Q35 and 2019 Q28 with the variables relabelled - the construction is clearly a favourite.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. X1, ..., X5 are independent N(0,1).
2. X4 + X5 ~ N(0, 2), so Z = (X4 + X5)/sqrt(2) is standard normal.
3. X1^2 + X2^2 + X3^2 ~ chi-square with 3 degrees of freedom, independent of X4 and X5.

4. By definition t = Z/sqrt(chi-square_3/3) = [(X4+X5)/sqrt2] sqrt(3)/sqrt(X1^2+X2^2+X3^2).
5. Rewriting, t = sqrt(3/2) (X4+X5)/(X1^2+X2^2+X3^2)^(1/2).
6. Hence C = sqrt(3/2), option (a).

---

21. 2026 · Q24 Chi-square Distribution · Theoretical

QUESTION

$(X_1, \ldots, X_n)$ is a random sample from $N(\mu, \sigma^2)$. What is the distribution of $\frac{ns^2}{\sigma^2}$, where $s^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2$?

OPTIONS

(a) $\chi^2_{(n)}$

(b) $\chi^2_{(2n)}$

(c) $\chi^2_{(n-1)}$

(d) $\chi^2_{(2n-2)}$

ANSWER (AI-derived — no official key exists for this paper)

(c) $\chi^2_{(n-1)}$

EXAM SHORTCUT (AI-derived explanation)

Since s^2 is defined with divisor n, n s^2 = sum of (X_i - Xbar)^2, and that sum divided by sigma^2 is chi-square with n - 1 df (one df lost to estimating the mean by Xbar). Answer (c).

TIPS & TRICKS (AI-derived explanation)

- Track the divisor: with s^2 = (1/n) sum, the chi-square variable is n s^2/sigma^2; with S^2 = (1/(n-1)) sum, it is (n-1)S^2/sigma^2. Both equal the same raw sum over sigma^2 and both have n - 1 df.
- One degree of freedom is always lost because Xbar is estimated from the same data; if the true mean mu were used instead, sum (X_i - mu)^2/sigma^2 would have n df - that is the trap in option (a).
- Fisher's lemma also gives independence of Xbar and s^2 in the normal case, which is what makes the t-statistic work.
- Consequence to remember: E(s^2) = (n-1)sigma^2/n, so s^2 with divisor n is biased downward, while S^2 with divisor n - 1 is unbiased.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Here s^2 = (1/n) sum from i = 1 to n of (X_i - Xbar)^2, so n s^2 = sum of (X_i - Xbar)^2.
2. Decompose sum (X_i - mu)^2 = sum (X_i - Xbar)^2 + n(Xbar - mu)^2.
3. Dividing by sigma^2: sum (X_i - mu)^2/sigma^2 is chi-square with n df, and n(Xbar - mu)^2/sigma^2 is chi-square with 1 df, the two terms on the right being independent (Fisher's lemma).
4. By the additivity of independent chi-square variables, sum (X_i - Xbar)^2/sigma^2 = n s^2/sigma^2 must be chi-square with n - 1 degrees of freedom.
5. Hence option (c) is correct.

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22. 2026 · Q33 Sampling Distribution of Sample Mean · Numerical

QUESTION

A normal population has mean 0.1, sd 2.1. For a random sample of size 900, what is $P(\bar{x}<0)$? [Given $P(0<Z<1.43)=0.4236$]

OPTIONS

(a) 0.0764

(b) 0.4236

(c) 0.5764

(d) 0.9236

ANSWER (AI-derived — no official key exists for this paper)

(a) 0.0764

EXAM SHORTCUT (AI-derived explanation)

SE = 2.1/sqrt(900) = 2.1/30 = 0.07, so z = (0 - 0.1)/0.07 = -1.43. P(Z < -1.43) = 0.5 - 0.4236 = 0.0764.

TIPS & TRICKS (AI-derived explanation)

- sqrt(900) = 30 exactly, so the standard error is a clean 0.07 - the numbers are chosen so that z lands on the tabulated 1.43.
- The supplied value 0.4236 is P(0 < Z < 1.43), a CENTRAL area, so subtract it from 0.5 to get the tail; adding it to 0.5 gives the distractor 0.9236.
- Draw the picture: 0 lies below the mean 0.1, so the required probability must be less than 0.5 - that alone eliminates options (c) and (d).
- Standard error uses sigma/sqrt(n), not sigma; using 2.1 directly would give z = -0.048 and a probability near 0.48.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The population is normal with mu = 0.1 and sigma = 2.1, and the sample size is n = 900.
2. The sampling distribution of xbar is normal with mean 0.1 and standard error sigma/sqrt(n) = 2.1/30 = 0.07.
3. Standardise: z = (0 - 0.1)/0.07 = -0.1/0.07 = -1.4286, approximately -1.43.
4. P(xbar < 0) = P(Z < -1.43) = 0.5 - P(0 < Z < 1.43) = 0.5 - 0.4236 = 0.0764.
5. Hence option (a) is correct.

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23. 2026 · Q35 Sampling Distribution of Sample Variance · Numerical

QUESTION

A random sample of size 5 is from a normal population with mean 2.5, variance 36. What is the probability that $S^2=\frac{1}{n-1}\sum_{i=1}^{n}(X_i-\bar{X})^2$ lies between 27 and 45? [Given $P(\chi_4^{\ 2}\geq 3)=0.55783$, $P(\chi_4^{\ 2}\leq 5)=0.71270$]

OPTIONS

(a) 0.15487

(b) 0.84513

(c) 0.27053

(d) 0.34513

ANSWER **(AI-derived — no official key exists for this paper)**

(c) 0.27053

EXAM SHORTCUT **(AI-derived explanation)**

(n-1)S^2/$sigma^2$ = 4S^2/36 = S^2/9 is chi-square with 4 df. S^2 in (27, 45) becomes chi-square in (3, 5). P = P(chi <= 5) - P(chi <= 3) = 0.71270 - (1 - 0.55783) = 0.27053.

TIPS & TRICKS **(AI-derived explanation)**

- Transform the limits, not the statistic: divide each bound by 9, since S^2/9 is the chi-square variable. 27/9 = 3 and 45/9 = 5.
- Degrees of freedom are n - 1 = 4, not n = 5; the mean 2.5 is irrelevant because S^2 uses deviations from Xbar.
- Convert the given tail probability first: P(chi^2_4 <= 3) = 1 - 0.55783 = 0.44217. Mixing an upper-tail value with a lower-tail value is the main trap here.
- Final subtraction: 0.71270 - 0.44217 = 0.27053. The distractor 0.15487 is 0.71270 - 0.55783, i.e. what you get by forgetting to complement.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. For a random sample of size n = 5 from a normal population with variance $sigma^2$ = 36, the quantity (n - 1)S^2/$sigma^2$ = 4S^2/36 = S^2/9 follows a chi-square distribution with n - 1 = 4 degrees of freedom.
2. Convert the bounds: S^2 = 27 gives 27/9 = 3, and S^2 = 45 gives 45/9 = 5.
3. So P(27 < S^2 < 45) = P(3 < chi^2_4 < 5) = P(chi^2_4 <= 5) - P(chi^2_4 <= 3).
4. From the data, P(chi^2_4 <= 5) = 0.71270, and P(chi^2_4 <= 3) = 1 - P(chi^2_4 >= 3) = 1 - 0.55783 = 0.44217.
5. Therefore the required probability = 0.71270 - 0.44217 = 0.27053.
6. Hence option (c) is correct.

S11 · STATISTICAL METHODS

Tests of Significance & Small Sample Tests

8 questions · 2018: 0 2019: 3 2020: 1 2021: 0 2022: 1 2023: 0 2024: 0 2025: 0 2026: 3

Weight. 8 questions (4.9% of Statistical Methods) · appeared in 4 of 9 papers · peak year 2019 (3)

Examiner's pattern. Conceptual rather than computational: interpreting Type I and Type II errors, the validity conditions for the chi-square test, choosing the right test for a variance, and getting the degrees of freedom right for an F-test of equal variances.

Must-know. Type I = reject a true H0 (= alpha); chi-square validity needs independent observations and expected frequencies >= 5; F-test for two variances uses df (n1-1, n2-1); test for sigma^2 uses chi-square.

1. 2019 · Q22 Tests of Significance · Conceptual

QUESTION

Consider the following statements: 1. A type I error of 0.05 means that 5 times out of 100, one can reject a true null hypothesis. 2. The smaller the type I error rate, the more correct is the inference. 3. Type I error is sensitive to the number of subjects in a sample. 4. Type I error is completely under the control of the experimenter. Which of the above statements is/are correct?

OPTIONS

(a) 1 and 3 only

(b) 2 and 4 only

(c) 4 only

(d) 1, 2, 3 and 4

ANSWER (AI-derived — no official key exists for this paper)

(c) 4 only

Source note. AI-derived key with a caveat. Statement 4 is unambiguously true and statements 2 and 3 are unambiguously false, which forces the key to (c) 'only 4' among the printed options. Statement 1, however, is also a standard textbook gloss on alpha = 0.05 and is defensible as true; the option set contains no '1 and 4' choice, so the printed options appear incomplete. (c) is selected as the only internally consistent option, and the reasoning is set out below.

EXAM SHORTCUT (AI-derived explanation)

alpha is CHOSEN by the experimenter, so statement 4 is certainly true. Statement 2 ignores the alpha-beta trade-off and statement 3 confuses type I with type II error, both false. That leaves (c).

TIPS & TRICKS (AI-derived explanation)

- Fix the trade-off in your mind: lowering alpha raises beta, so a smaller type I error rate does NOT make the inference 'more correct' - it merely shifts which mistake you are more likely to make.
- Sample size drives POWER and the type II error rate; the nominal type I error rate is fixed at whatever alpha the experimenter declares. That single sentence disposes of statement 3.
- Type I error = rejecting a true H0 (a 'false positive'); type II error = failing to reject a false H0 (a 'false negative'). Label them before reading any statement.
- Statement 1 is the standard verbal reading of alpha = 0.05 and can reasonably be argued to be true - note the ambiguity, choose the internally consistent option, and move on rather than losing time.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement 1: alpha = 0.05 means P(reject H0 | H0 true) = 0.05, i.e. in 5 of every 100 repetitions of the experiment a true null hypothesis would be rejected. This is the standard interpretation and is defensible as TRUE.
2. Statement 2: reducing alpha increases the probability beta of a type II error, so a smaller type I error rate does not make the inference more correct - it only trades one kind of error for the other. FALSE.
3. Statement 3: the nominal type I error rate is fixed by the chosen significance level and does not change with n; it is the type II error rate (and hence the power) that is sensitive to sample size. FALSE.
4. Statement 4: the experimenter selects the significance level before the test, so the type I error rate is entirely under their control. TRUE.
5. Since statements 2 and 3 are false, options (b) and (d) are eliminated, as is (a) which contains statement 3.
6. The only internally consistent choice is option (c), 4 only.

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2. 2019 · Q24 Tests of Significance · Numerical

QUESTION

Two independent groups A and B consist of 100 people each with a disease. A serum is given to group A but not B; otherwise treated identically. In groups A and B, 75 and 65 people respectively recover. Given $Z_{0.05}$=1.645, $Z_{0.10}$=1.28. Consider the following statements: 1. The results are due to chance only at 5% level of significance. 2. The serum is effective at 10% level of significance. 3. Assuming the serum has no effect, 70 people are expected to recover in group A. 4. The standard deviation of the difference in proportions is given by 0.06. Which of the above statements are correct?

OPTIONS

(a) 1, 3 and 4

(b) 1, 2 and 3

(c) 1, 2 and 4

(d) 2, 3 and 4

ANSWER (AI-derived — no official key exists for this paper)

(b) 1, 2 and 3

Source note. AI-derived key with a caveat. Statements 1, 2 and 3 are exactly true: the pooled test statistic is Z = 1.543, which is below Z(0.05) = 1.645 but above Z(0.10) = 1.28, and the expected recoveries under H0 are 100 x 0.70 = 70. Statement 4 asserts the standard error is 0.06, whereas the exact pooled value is 0.0648 (unpooled 0.0644); since no option contains all four statements, statement 4 is treated as the intended false one, giving (b). If instead 0.06 is accepted as a rounding, the printed option set has no fully correct choice.

EXAM SHORTCUT (AI-derived explanation)

Pooled p = 140/200 = 0.70, SE = sqrt(0.7 x 0.3 x 0.02) = 0.0648, Z = 0.10/0.0648 = 1.54. Below 1.645 but above 1.28: not significant at 5%, significant at 10%. Expected recoveries in A under H0 = 70.

TIPS & TRICKS (AI-derived explanation)

- For a difference of proportions under H0, pool first: p_hat = (x1 + x2)/(n1 + n2) = 140/200 = 0.70, then SE = sqrt(p_hat q_hat (1/n1 + 1/n2)).

- One Z value answers several statements at once - compute Z = 1.54 and then simply compare it with each supplied critical value.
- 'Results are due to chance' is the language of FAILING to reject H0; 'the serum is effective' is the language of rejecting it. The same Z can do both at different alpha levels, which is exactly what this item exploits.
- Expected frequency under H0 is n x pooled p = 100 x 0.70 = 70 - a one-line calculation that confirms statement 3 without any testing.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Sample proportions: p1 = 75/100 = 0.75 for group A and p2 = 65/100 = 0.65 for group B, with n1 = n2 = 100.
2. Under H0 (the serum has no effect) the pooled proportion is p_hat = (75 + 65)/200 = 0.70, so statement 3's expected recoveries in group A are 100 x 0.70 = 70. TRUE.
3. Standard error of the difference: SE = sqrt(p_hat(1 - p_hat)(1/n1 + 1/n2)) = sqrt(0.70 x 0.30 x 0.02) = sqrt(0.0042) = 0.0648.
4. Test statistic: Z = (0.75 - 0.65)/0.0648 = 0.10/0.0648 = 1.543.
5. Since 1.543 < Z(0.05) = 1.645, H0 is not rejected at the 5% level, so the difference can be attributed to chance - statement 1 is TRUE. Since 1.543 > Z(0.10) = 1.28, H0 is rejected at the 10% level, so the serum is effective at that level - statement 2 is TRUE.
6. Statement 4 claims the standard error is 0.06, but the exact value is 0.0648 (0.0644 unpooled). Taking statement 4 as the inexact one, the correct set is 1, 2 and 3, option (b).

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3. 2019 · Q27 Chi-square Distribution · Conceptual

QUESTION

Which of the following is/are **not** needed for the validity of the χ^2 test? 1. The sample observations should be independent. 2. Constraints on the cell frequencies should not be linear. 3. N, the total frequency, should be reasonably large. 4. None of the theoretical frequencies should be less than 5.

OPTIONS

(a) 1 and 2

(b) 2 only

(c) 2 and 3

(d) 4

ANSWER (AI-derived — no official key exists for this paper)

(b) 2 only

EXAM SHORTCUT (AI-derived explanation)

The genuine conditions are independence, a large total N, and expected frequencies of at least 5. Item 2 states the OPPOSITE of the real requirement - constraints on the cell frequencies must be linear - so only item 2 is not needed.

TIPS & TRICKS (AI-derived explanation)

- The four standard validity conditions for chi-square: (i) observations independent, (ii) N reasonably large, (iii) every expected frequency at least 5, (iv) the constraints on the cell frequencies are LINEAR.

- Read the question direction carefully: it asks which are NOT needed. Items that are genuine requirements must be excluded from the answer.
- Item 2 inverts requirement (iv). Inversion of a real condition is the examiner's favourite way of manufacturing a false statement.
- When an expected frequency falls below 5, pool adjacent classes (and reduce the degrees of freedom accordingly) rather than abandoning the test.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The conditions for the valid application of the chi-square test are: the sample observations must be independent; the total frequency N must be reasonably large; each theoretical (expected) frequency should be at least 5; and the constraints imposed on the cell frequencies must be LINEAR.
2. Item 1 (independence) is a genuine requirement, so it IS needed.
3. Item 2 asserts that the constraints should NOT be linear. This is the negation of the actual condition, so it is not a requirement at all.
4. Item 3 (N reasonably large) is a genuine requirement, so it IS needed.
5. Item 4 (no theoretical frequency below 5) is a genuine requirement, so it IS needed.
6. Only item 2 is not needed, option (b).

---

4. 2020 · Q18 Measures of Dispersion · Conceptual

QUESTION

The df of the F-statistic for testing equality of variances of two normal populations (unknown means), from independent samples of sizes 20 and 37, are

OPTIONS

(a) 18 and 35

(b) 19 and 36

(c) 20 and 37

(d) 21 and 38

ANSWER (AI-derived — no official key exists for this paper)

(b) 19 and 36

EXAM SHORTCUT (AI-derived explanation)

Each sample variance loses one degree of freedom for its estimated mean, so the d.f. are 20 - 1 = 19 and 37 - 1 = 36.

TIPS & TRICKS (AI-derived explanation)

- Rule: the variance-ratio F test uses (n1 - 1, n2 - 1) degrees of freedom when both population means are UNKNOWN and estimated from the samples.
- If the means were KNOWN, no degree of freedom would be lost and the d.f. would be n1 and n2 - the reason the question specifies 'unknown means'.
- Order matters only for reading the table: place the larger variance in the numerator so that F >= 1, then use the corresponding d.f. first.
- Option (a) subtracts 2 from each sample size, confusing this with the pooled t-test's n1 + n2 - 2 - keep the two adjustments apart.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. To test the equality of two normal variances we use the ratio of the two sample variances, F = $s1^2/s2^2$.
2. Each sample variance is computed about its own SAMPLE mean, because the population means are unknown, so one degree of freedom is lost per sample.
3. For the first sample of size 20, the numerator d.f. is 20 - 1 = 19.
4. For the second sample of size 37, the denominator d.f. is 37 - 1 = 36.
5. Hence the statistic follows F(19, 36) under the null hypothesis.
6. Option (b) is correct.

---

5. 2022 · Q54 Chi-square Distribution · Conceptual

QUESTION

To test $H_0 : \sigma^2 = \sigma_0^{\ 2}$ in $N(0, \sigma^2)$, the appropriate test would be:

OPTIONS

(a) t-test

(b) Chi-square test

(c) F-test

(d) Normal test

ANSWER (AI-derived — no official key exists for this paper)

(b) Chi-square test

EXAM SHORTCUT (AI-derived explanation)

A hypothesis about a SINGLE normal variance is tested with the chi-square statistic sum(X_i^2)/$sigma\_0^2$ (here the mean is known to be 0).

TIPS & TRICKS (AI-derived explanation)

- Match test to parameter: one mean with unknown sigma = t-test; one variance = chi-square; ratio of TWO variances = F-test; one mean with known sigma or large n = normal test.
- Because the mean is known to be 0, the statistic is sum(X_i^2)/$sigma\_0^2$ with n degrees of freedom - no degree of freedom is lost for estimating the mean.
- If the mean had to be estimated, the statistic would be (n-1)s^2/$sigma\_0^2$ with n - 1 degrees of freedom. Knowing which applies is the finer point of this item.
- Option (c) F-test is the trap: F compares TWO variances, whereas here a single variance is compared with a fixed constant.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The hypothesis concerns the variance of a single normal population, H0: $sigma^2$ = $sigma\_0^2$.
2. For a sample from N(0, $sigma^2$) with the mean known to be 0, the statistic sum(X_i^2)/$sigma\_0^2$ follows a chi-square distribution with n degrees of freedom under H0.
3. (If the mean had to be estimated, the statistic would be (n-1)s^2/$sigma\_0^2$ with n - 1 degrees of freedom - still a chi-square test.)
4. A t-test concerns a mean when the variance is unknown, and a normal test concerns a mean when the variance is known.

5. An F-test compares the variances of TWO populations, not a single variance with a specified value.
6. Hence the appropriate procedure is the chi-square test, option (b).

---

6. 2026 · Q31 Sampling Distribution of Sample Mean · Numerical

QUESTION

Normal populations π_1, π_2 have equal means, but the sd of π_2 is twice that of π_1 (sd σ). In samples of size 2000 from each, the difference of sample means is, in all probability:

OPTIONS

(a) less than 0.05σ

(b) less than 0.15σ

(c) greater than 0.05σ

(d) greater than 0.15σ

ANSWER (AI-derived — no official key exists for this paper)

(b) less than 0.15σ

EXAM SHORTCUT (AI-derived explanation)

SE of the difference = sqrt(sigma^2/2000 + 4 sigma^2/2000) = sigma sqrt(5/2000) = sigma/20 = 0.05 sigma. 'In all probability' means within 3 SE, i.e. less than 0.15 sigma.

TIPS & TRICKS (AI-derived explanation)

- Variances add for a difference of independent means: sigma_1^2/n_1 + sigma_2^2/n_2, even though the means are subtracted.
- The second sd is 2 sigma, so its variance is 4 sigma^2 - squaring the doubled sd is the step most often skipped.
- 5/2000 = 1/400, so the square root is exactly 1/20 - the numbers are chosen to make the SE come out as 0.05 sigma.
- Exam convention: 'in all probability' (or 'practically certain') means within 3 standard errors, covering about 99.7% of the sampling distribution; 'probable error' would instead be 0.6745 SE.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let sigma_1 = sigma and sigma_2 = 2 sigma, with n_1 = n_2 = 2000, and equal population means so E(xbar_1 - xbar_2) = 0.
2. V(xbar_1 - xbar_2) = sigma_1^2/n_1 + sigma_2^2/n_2 = sigma^2/2000 + 4 sigma^2/2000 = 5 sigma^2/2000 = sigma^2/400.
3. Standard error = sigma/20 = 0.05 sigma.
4. By the central limit theorem the difference of sample means is approximately normal with mean 0 and SE 0.05 sigma.
5. 'In all probability' means within three standard errors: |xbar_1 - xbar_2| < 3(0.05 sigma) = 0.15 sigma, an event of probability about 0.997.
6. Hence the difference is in all probability less than 0.15 sigma, and option (b) is correct.

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7. 2026 · Q32 t-Distribution · Numerical

QUESTION

Mean height of 50 male students with above-average athletic participation: 68.2 in, sd 2.5 in. 50 students with no interest: mean 67.5 in, sd 2.8 in. For the observed difference to be significant at 5% (critical Z = 1.645), the sample size of each group must be increased by:

OPTIONS

(a) at least 78

(b) at most 78

(c) at least 28

(d) at most 28

ANSWER (AI-derived — no official key exists for this paper)

(c) at least 28

EXAM SHORTCUT (AI-derived explanation)

Difference 0.7, and SE = sqrt(14.09/n). Need 0.7 sqrt(n) / 3.7537 >= 1.645, so sqrt(n) >= 8.821 and n >= 77.8, i.e. n = 78. The increase over 50 is at least 28.

TIPS & TRICKS (AI-derived explanation)

- Keep the sample means and sds fixed and treat only n as unknown - that is the whole idea of a 'how much must n increase' question.
- sigma_1^2 + sigma_2^2 = 6.25 + 7.84 = 14.09 with equal n, so SE = sqrt(14.09/n) and the test statistic is 0.7 sqrt(n)/sqrt(14.09).
- Solve for n directly: n >= 14.09 (1.645/0.7)^2 = 14.09 * 5.5225 = 77.81, so n = 78 (round UP, since a larger n only helps).
- Read the question's final step: it asks for the INCREASE, 78 - 50 = 28, not the new size 78. That is why both 78 and 28 appear as options, and 'at least' is right because more data only strengthens significance.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Observed difference of means: d = 68.2 - 67.5 = 0.7 inches, held fixed as n changes.
2. With equal sample sizes n in each group, SE(xbar_1 - xbar_2) = sqrt(s_1^2/n + s_2^2/n) = sqrt((6.25 + 7.84)/n) = sqrt(14.09/n).
3. Test statistic: Z = 0.7 / sqrt(14.09/n) = 0.7 sqrt(n)/sqrt(14.09) = 0.7 sqrt(n)/3.7537.
4. Significance at the 5% level (one-tailed, critical value 1.645) requires 0.7 sqrt(n)/3.7537 >= 1.645.
5. So sqrt(n) >= 1.645 * 3.7537/0.7 = 6.1748/0.7 = 8.8211, giving n >= 77.81.
6. Since n must be an integer, n = 78 at minimum.
7. Required increase in each group = 78 - 50 = 28, and any larger increase also gives significance, so the increase must be at least 28.
8. Hence option (c) is correct.

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8. 2026 · Q36 t-Distribution · Conceptual

QUESTION

The t-statistic for a sample is 3. If the sample variance is doubled, the new t-statistic becomes:

OPTIONS

(a) 6

(b) 3/2

(c) $3/\sqrt{2}$

(d) $1/\sqrt{2}$

ANSWER (AI-derived — no official key exists for this paper)

(c) $3/\sqrt{2}$

EXAM SHORTCUT (AI-derived explanation)

t is inversely proportional to the sample standard deviation s. Doubling the variance multiplies s by sqrt 2, so t becomes 3/sqrt 2.

TIPS & TRICKS (AI-derived explanation)

- Write t = (xbar - mu) sqrt(n)/s and note that only s changes - the numerator, n and mu are untouched.
- Doubling the VARIANCE multiplies the SD by sqrt 2, not by 2; option (b) 3/2 is the trap for those who halve instead.
- Option (a) 6 comes from multiplying rather than dividing - always check the direction: more variability means a smaller (less significant) t.
- General scaling rule: if s^2 is multiplied by k, then t is divided by sqrt(k).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The one-sample t-statistic is t = (xbar - mu)/(s/sqrt(n)) = (xbar - mu) sqrt(n)/s, where s^2 is the sample variance.
2. Only the sample variance is altered; the sample mean, hypothesised mean and sample size are unchanged.
3. If the sample variance is doubled, the new sample standard deviation is sqrt(2 s^2) = s sqrt(2).
4. Hence the new statistic is t_new = (xbar - mu) sqrt(n)/(s sqrt 2) = t/sqrt(2).
5. With t = 3, the new value is 3/sqrt(2), approximately 2.121.
6. Hence option (c) is correct.

S12 · STATISTICAL METHODS

Non-parametric Tests

13 questions · 2018: 1 2019: 0 2020: 5 2021: 3 2022: 3 2023: 0 2024: 0 2025: 0 2026: 1

Weight. 13 questions (7.9% of Statistical Methods) · appeared in 5 of 9 papers · peak year 2020 (5)

Examiner's pattern. Named-test recognition plus two computations. 'Which non-parametric test is analogous to the chi-square goodness-of-fit test?' was asked verbatim in 2018 and 2022 — free marks. Computation appears for the KS statistic, Mann-Whitney U and its variance, and run-test moments.

Must-know. E(U) = mn/2, Var(U) = mn(m+n+1)/12; runs: E(R) = 2 n1 n2/n + 1; KS = sup|Fn(x) - F(x)|; the KS test is the analogue of chi-square goodness of fit.

1. 2018 · Q37 Wilcoxon Test · Factual

QUESTION

Which non-parametric test is analogous to the chi-square goodness-of-fit test?

OPTIONS

(a) Mann–Whitney test

(b) Kolmogorov–Smirnov test

(c) Wilcoxon test

(d) Median test

ANSWER (AI-derived — no official key exists for this paper)

(b) Kolmogorov–Smirnov test

EXAM SHORTCUT (AI-derived explanation)

Both the chi-square goodness-of-fit test and the Kolmogorov-Smirnov test compare an observed distribution with a completely specified theoretical one. K-S is the distribution-free analogue.

TIPS & TRICKS (AI-derived explanation)

- Sort the non-parametric tests by what they compare: goodness of fit to a specified distribution = chi-square and K-S (one sample); two independent samples = Mann-Whitney and median test; paired/matched data = Wilcoxon signed-rank and sign test; randomness = run test.
- Chi-square needs GROUPED data with adequate expected frequencies; K-S works directly on the ungrouped empirical CDF and is therefore preferred for small samples and continuous data.
- K-S uses the maximum absolute deviation D = sup |F_n(x) - F_0(x)| - remember it as a 'largest vertical gap between two CDFs' test.
- Mann-Whitney (option a) and Wilcoxon (option c) are location-comparison tests, and the median test (option d) compares medians - none of them tests fit to a specified distribution.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The chi-square goodness-of-fit test compares observed frequencies with the frequencies expected under a fully specified theoretical distribution.
2. Among the listed non-parametric procedures, the Kolmogorov-Smirnov one-sample test performs the same task: it compares the empirical distribution function F_n(x) with a hypothesised F_0(x).
3. The K-S statistic is D = sup over x of |F_n(x) - F_0(x)|, the largest vertical distance between the two curves, and a large D leads to rejection of the hypothesised distribution.

4. The Mann-Whitney and Wilcoxon tests compare locations of samples, and the median test compares medians; none of them tests agreement with a specified distribution.
5. Hence the Kolmogorov-Smirnov test is the analogue of the chi-square goodness-of-fit test, option (b).

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2. 2020 · Q1 Non-parametric Tests · Numerical

QUESTION

The value of the Kolmogorov–Smirnov statistic to test whether $0.32, 0.16, 0.98, 0.16, 0.08$ arise from $U(0,2)$ is

OPTIONS

(a) 0.16

(b) 0.52

(c) 0.64

(d) 0.76

ANSWER **(AI-derived — no official key exists for this paper)**

(c) 0.64

EXAM SHORTCUT **(AI-derived explanation)**

Under U(0,2), F0(x) = x/2. Sort the data (0.08, 0.16, 0.16, 0.32, 0.98) and compare the step function with F0; the largest gap occurs just after x = 0.32, where Fn = 0.8 and F0 = 0.16, giving D = 0.64.

TIPS & TRICKS **(AI-derived explanation)**

- Always evaluate the gap on BOTH sides of each jump: |i/n - F0(x_i)| and |(i-1)/n - F0(x_i)|. The maximum of all these is D.
- Handle the tie at 0.16 by jumping the empirical CDF straight from 0.2 to 0.6 - two observations share the point, so the step has height 2/5.
- The uniform CDF on (0,2) is x/2, so the theoretical values 0.04, 0.08, 0.08, 0.16, 0.49 are trivial to write down; most of the work is bookkeeping, not arithmetic.
- The data are heavily concentrated near 0 while U(0,2) spreads to 2, so a LARGE D is expected - options (a) 0.16 and (b) 0.52 understate the mismatch.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. Under H0 the distribution is U(0,2), so F0(x) = x/2 for 0 <= x <= 2.
2. Sort the sample: 0.08, 0.16, 0.16, 0.32, 0.98, with theoretical values F0 = 0.04, 0.08, 0.08, 0.16, 0.49.
3. The empirical CDF takes the values 0.2 after 0.08, 0.6 after the tied pair at 0.16, 0.8 after 0.32 and 1.0 after 0.98.
4. Compute the deviations at each point (both before and after the jump): at 0.08, |0.2 - 0.04| = 0.16; at 0.16, |0.6 - 0.08| = 0.52; at 0.32, |0.8 - 0.16| = 0.64; at 0.98, |1.0 - 0.49| = 0.51.
5. The pre-jump deviations (0.04, 0.12, 0.44, 0.31) are all smaller.
6. The Kolmogorov-Smirnov statistic is the largest of these, D = 0.64, option (c).

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3. 2020 · Q3 Small Sample Tests · Conceptual

QUESTION

In a clinical trial, n persons had cream A on one randomly chosen arm and cream B on the other. Which test examines the difference (response is continuous)?

OPTIONS

(a) Two-sample Kolmogorov–Smirnov test

(b) Two-sample t-test if normality can be assumed

(c) Paired t-test if normality can be assumed

(d) Test for randomness

ANSWER (AI-derived — no official key exists for this paper)

(c) Paired t-test if normality can be assumed

EXAM SHORTCUT (AI-derived explanation)

Each person receives BOTH creams, one per arm, so the observations are naturally paired within a person. With a continuous response and normality, the paired t-test is the right procedure.

TIPS & TRICKS (AI-derived explanation)

- Look for what shares a subject: two measurements on the SAME person are paired, and pairing must be preserved by analysing the within-person differences.
- Pairing removes between-person variation, so the paired t-test is far more powerful here than the two-sample t-test, which wrongly treats the 2n measurements as independent.
- The randomisation of which arm gets which cream protects against a left/right arm effect; it does not turn the design into an independent-samples one.
- If normality could not be assumed, the correct non-parametric analogues would be the Wilcoxon signed-rank test or the sign test - both paired procedures, not the Mann-Whitney.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Each of the n persons receives cream A on one arm and cream B on the other, so both treatments are observed on the same individual.
2. The two responses from one person are therefore dependent - a matched pair - and the natural analysis works with the n within-person differences d_i = A_i - B_i.
3. The two-sample Kolmogorov-Smirnov test and the two-sample t-test both assume two INDEPENDENT samples, which is violated here.
4. A test for randomness addresses the order of a sequence and is irrelevant to comparing two treatments.
5. Given a continuous response and an assumption of normality for the differences, the appropriate procedure is the paired t-test on the d_i.
6. Hence option (c) is correct.

---

4. 2020 · Q5 Run Test · Numerical

QUESTION

In 30 tosses of a coin, the sequence obtained is:

$$\mathrm{H\ T\ T\ H\ T\ H\ H\ H\ T\ H\ H\ T\ T\ H\ T\ H\ T\ H\ H\ T\ H\ T\ T\ H\ T\ H\ H\ T\ H\ T}$$

The number of runs is

OPTIONS

(a) 30

(b) 22

(c) 20

(d) 28

ANSWER **(AI-derived — no official key exists for this paper)**

(b) 22

EXAM SHORTCUT **(AI-derived explanation)**

A run is a maximal block of identical symbols. Count the changeovers: the sequence switches 21 times in 30 tosses, so there are 22 runs.

TIPS & TRICKS **(AI-derived explanation)**

- Runs = number of changes of symbol + 1. Counting CHANGES is faster and far less error-prone than bracketing blocks.
- Mark the sequence in groups as you read: H | TT | H | T | HHH | T | HH | TT | H | T | H | T | HH | T | H | TT | H | T | HH | T | H | T.
- Sanity range: with 30 tosses the number of runs must lie between 2 and 30; a well-mixed fair sequence gives roughly n/2 + 1 = 16 or more, so 22 is plausible while 28 and 30 would be near-perfect alternation.
- For the run test, the expected number of runs is 2 n1 n2/n + 1; here n1 = 16 heads and n2 = 14 tails give about 15.9, so 22 is on the high side, indicating more alternation than chance.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. A run is a maximal unbroken sequence of identical outcomes.
2. Reading the 30 tosses in order and grouping: H | TT | H | T | HHH | T | HH | TT | H | T | H | T | HH | T | H | TT | H | T | HH | T | H | T.
3. Counting the groups: 1 H, 2 TT, 3 H, 4 T, 5 HHH, 6 T, 7 HH, 8 TT, 9 H, 10 T, 11 H, 12 T, 13 HH, 14 T, 15 H, 16 TT, 17 H, 18 T, 19 HH, 20 T, 21 H, 22 T.
4. The group lengths sum to 1+2+1+1+3+1+2+2+1+1+1+1+2+1+1+2+1+1+2+1+1+1 = 30, confirming that every toss has been counted.
5. There are therefore 22 runs.
6. Option (b) is correct.

---

5. **2020 · Q6** Sign Test · Conceptual

QUESTION

A manufacturer claims a product's lifetime exceeds 250 hours, to be verified from 24 observed lifetimes. Which test can be used?

OPTIONS

(a) Sign test

(b) Mann–Whitney U test

(c) Median test

(d) Kolmogorov–Smirnov test

ANSWER **(AI-derived — no official key exists for this paper)**

(a) Sign test

EXAM SHORTCUT **(AI-derived explanation)**

One sample, a hypothesised value for the location (250 hours), and no distributional assumption - that is exactly the one-sample SIGN test.

TIPS & TRICKS **(AI-derived explanation)**

- Match the test to the design: ONE sample against a stated value = sign test (or Wilcoxon signed-rank); TWO independent samples = Mann-Whitney U or median test; fit to a distribution = K-S or chi-square.
- The sign test simply counts how many of the 24 lifetimes exceed 250 and refers that count to a Binomial(24, 0.5) under H0 - no distributional assumption at all.
- Mann-Whitney and the median test both need two samples, so options (b) and (c) are ruled out by the design alone.
- The Kolmogorov-Smirnov test would require a fully specified distribution, not just a claim about one quantile, so option (d) is also inapplicable.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. The manufacturer's claim concerns the location of a single population - that the lifetime typically exceeds 250 hours - and there is only one sample, of 24 observations.
2. The sign test is the standard one-sample non-parametric procedure for a hypothesised median or location value: count how many observations exceed 250 and compare that count with Binomial(n, 0.5) under H0.
3. The Mann-Whitney U test and the median test both compare TWO independent samples, so they do not apply.
4. The Kolmogorov-Smirnov test compares the empirical CDF with a fully specified theoretical distribution, which is not what is claimed here.
5. Hence the sign test is appropriate, option (a).

---

6. **2020 · Q19** Sign Test · Numerical

QUESTION

To test whether the upper 60th percentile of a continuous distribution equals 25 or less, the sign test (sample size 10) rejects H_0 if the number of observations larger than 25 is ≥ 7. Given $(0.6)^7 \approx 0.028$, the size of the test is approximately

OPTIONS

(a) 0.04

(b) 0.05

(c) 0.10

(d) 0.38

ANSWER **(AI-derived — no official key exists for this paper)**

(d) 0.38

EXAM SHORTCUT (AI-derived explanation)

Under H0 the count of observations above 25 is Binomial(10, 0.6). The size is P(X >= 7) = 0.2150 + 0.1209 + 0.0403 + 0.0060 = 0.382.

TIPS & TRICKS (AI-derived explanation)

- The size of a test is the rejection probability computed UNDER H0, so use p = 0.6 (the null quantile), not 0.5.
- Sum all four upper terms: P(7), P(8), P(9) and P(10). Stopping at P(X = 7) = 0.215 gives a badly understated size.
- The supplied hint $(0.6)^7$ = 0.028 feeds every term: P(7) = 120(0.028)(0.064), P(8) = 45(0.0168)(0.16), and so on.
- Since the mean of Binomial(10, 0.6) is 6, a cut-off of 7 is barely above the mean, so a LARGE size is expected - options (a), (b) and (c) are implausibly small.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Under H0 the upper 60th percentile equals 25, so P(X > 25) = 0.6 and the number of observations exceeding 25 in a sample of 10 is Binomial(10, 0.6).
2. The test rejects H0 when this count is at least 7, so the size is P(B >= 7) with B ~ Binomial(10, 0.6).
3. P(B = 7) = C(10,7)$(0.6)^7(0.4)^3$ = 120(0.02799)(0.064) = 0.2150.
4. P(B = 8) = C(10,8)$(0.6)^8(0.4)^2$ = 45(0.016796)(0.16) = 0.1209; P(B = 9) = 10$(0.6)^9$(0.4) = 0.0403; P(B = 10) = $(0.6)^{10}$ = 0.0060.
5. Adding: 0.2150 + 0.1209 + 0.0403 + 0.0060 = 0.3822.
6. The size of the test is approximately 0.38, option (d).

---

7. 2021 · Q35 Order Statistics - Minimum, Maximum & Median · Numerical

QUESTION

Samples of sizes 4 and 3 from X and Y combine as $x_3<x_2<y_1<y_2<x_4<y_3<x_1$. The Mann–Whitney statistic for identicality of distributions is

OPTIONS

(a) 5

(b) 7

(c) 10

(d) 11

ANSWER (AI-derived — no official key exists for this paper)

(a) 5

EXAM SHORTCUT (AI-derived explanation)

Rank the combined sample: X occupies ranks 1, 2, 5, 7 so R_X = 15. Then U = R_X - n1(n1+1)/2 = 15 - 10 = 5.

TIPS & TRICKS (AI-derived explanation)

- Two equivalent counts: U = R1 - n1(n1+1)/2 = 5, and its complement n1 n2 - U = 12 - 5 = 7. Both appear among the options, so state which convention you are using.

- Direct counting confirms it: for each X, count the Y observations that precede it - 0, 0, 2, 3, totalling 5.
- Check U1 + U2 = n1 n2 = 4 x 3 = 12. Here 5 + 7 = 12, so the two values are consistent and one of them is the answer.
- For the test itself the SMALLER of the two U values is compared with the tabulated critical value, which is why 5 is the statistic normally reported.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The combined ordering is x3 < x2 < y1 < y2 < x4 < y3 < x1, so the ranks are x3 = 1, x2 = 2, y1 = 3, y2 = 4, x4 = 5, y3 = 6, x1 = 7.
2. The X sample (n1 = 4) has ranks 1, 2, 5, 7, so R_X = 15; the Y sample (n2 = 3) has ranks 3, 4, 6, so R_Y = 13. (Total 28 = 7 x 8/2, as required.)
3. The Mann-Whitney statistic for the X sample is U = R_X - n1(n1 + 1)/2 = 15 - (4)(5)/2 = 15 - 10 = 5.
4. Equivalently, count for each X observation the number of Y observations preceding it: 0, 0, 2, 3, giving 5.
5. The complementary value is n1 n2 - U = 12 - 5 = 7, and 5 + 7 = 12 = n1 n2 confirms the arithmetic.
6. The statistic used for the test of identical distributions is the smaller value, U = 5, option (a).

---

8. **2021 · Q36** Measures of Dispersion · Numerical

QUESTION

The variance of the Mann–Whitney statistic for independent samples of sizes 15 and 20 is

OPTIONS

(a) 28

(b) 532

(c) 875

(d) 900

ANSWER (AI-derived — no official key exists for this paper)

(d) 900

EXAM SHORTCUT (AI-derived explanation)

Var(U) = n1 n2 (n1 + n2 + 1)/12 = 15 x 20 x 36/12 = 300 x 3 = 900.

TIPS & TRICKS (AI-derived explanation)

- Memorise the pair: E(U) = n1 n2/2 and Var(U) = n1 n2(n1 + n2 + 1)/12. Both are needed for the normal approximation.
- Cancel before multiplying: 36/12 = 3, so the product is just 15 x 20 x 3 = 900. Multiplying first invites arithmetic slips.
- Note the +1 in (n1 + n2 + 1) = 36; using 35 gives 875, which is planted as option (c).
- For the large-sample test, z = (U - n1n2/2)/sqrt(Var(U)) = (U - 150)/30 here - the standard deviation of 30 is a memorable by-product.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For independent samples of sizes n1 and n2, the Mann-Whitney statistic U has, under the null hypothesis of identical distributions, mean n1 n2/2 and variance n1 n2 (n1 + n2 + 1)/12.
2. Here n1 = 15 and n2 = 20, so n1 + n2 + 1 = 36.

3. Var(U) = 15 x 20 x 36/12.
4. Simplify 36/12 = 3, giving Var(U) = 15 x 20 x 3 = 300 x 3 = 900.
5. Hence the variance is 900, option (d).

---

9. 2021 · Q40 Run Test · Numerical

QUESTION

Ordered lifetimes (days) of a component: 198,211,216,219,224,225,230,236,243,252,253,253,262,264,268,271,272,275,282,284,288,291,294,295. 1. The sample is random with median lifetime 257.50. 2. Mean number of runs is 13 with standard deviation 1.396.

OPTIONS

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

ANSWER (AI-derived — no official key exists for this paper)

(d) Neither 1 nor 2

Source note. AI-derived key with a caveat. The median is indeed 257.50 (mean of the 12th and 13th ordered values 253 and 262) and E(R) = 13 is correct for n1 = n2 = 12. However the printed standard deviation 1.396 does not match the exact value sqrt(2 n1 n2 (2 n1 n2 - N)/(N^2 (N-1))) = sqrt(5.739) = 2.396, so statement 2 as printed is false; and the sequence as presented is sorted, giving only 2 runs against an expected 13 (z = -4.6), so the randomness claim in statement 1 fails. Both statements therefore fail as printed, giving (d). If 1.396 were a misprint for 2.396 the key would be (b).

EXAM SHORTCUT (AI-derived explanation)

n1 = n2 = 12 about the median 257.50, so E(R) = 2n1n2/N + 1 = 13 and SD(R) = sqrt(5.739) = 2.396 - not the printed 1.396. The sorted sequence has only 2 runs, so randomness is rejected.

TIPS & TRICKS (AI-derived explanation)

- Runs test about the median: E(R) = 2 n1 n2/(n1 + n2) + 1 and Var(R) = 2 n1 n2 (2 n1 n2 - n1 - n2)/[$(n1+n2)^2$ (n1+n2-1)]. Compute both before judging either statement.
- Find the median first: with N = 24 it is the average of the 12th and 13th values, (253 + 262)/2 = 257.50.
- A perfectly sorted sequence always yields exactly 2 runs - the theoretical minimum - so any ordered data set fails a runs test for randomness by construction.
- Numerical care: 2 n1 n2 = 288 and 2 n1 n2 - N = 264, so Var(R) = 288 x 264/(576 x 23) = 5.739 and SD = 2.396.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. There are N = 24 observations. The median is the average of the 12th and 13th ordered values: (253 + 262)/2 = 257.50, so that part of statement 1 is correct.
2. Twelve observations fall below the median and twelve above, so n1 = n2 = 12.
3. Expected number of runs: E(R) = 2 n1 n2/(n1 + n2) + 1 = 2(144)/24 + 1 = 12 + 1 = 13, matching statement 2's mean.

4. Variance: Var(R) = 2 n1 n2 (2 n1 n2 - N)/[N^2 (N-1)] = 288(264)/[576(23)] = 76032/13248 = 5.7391, so SD(R) = 2.396, not the printed 1.396. Statement 2 is therefore FALSE as printed.
5. The sequence as given is in ascending order, so all twelve values below the median precede all twelve above, producing exactly R = 2 runs. The standardised value is (2 - 13)/2.396 = -4.59, which decisively rejects randomness, so statement 1 is FALSE.
6. Neither statement holds as printed, option (d).

---

10. 2022 · Q53 Kolmogorov-Smirnov Test · Factual

QUESTION

Which non-parametric test is analogous to a Chi-square test of goodness of fit?

OPTIONS

(a) Mann-Whitney test

(b) Kolmogorov-Smirnov test

(c) Wilcoxon test

(d) Median test

ANSWER (AI-derived — no official key exists for this paper)

(b) Kolmogorov-Smirnov test

EXAM SHORTCUT (AI-derived explanation)

Both the chi-square goodness-of-fit test and the Kolmogorov-Smirnov test compare an observed distribution with a fully specified theoretical one.

TIPS & TRICKS (AI-derived explanation)

- Sort the non-parametric tests by design: one sample against a specified distribution = chi-square and K-S; two independent samples = Mann-Whitney, median test; paired = Wilcoxon signed-rank, sign test.
- K-S works on the ungrouped empirical CDF and needs no minimum expected frequencies, which makes it preferable to chi-square for small continuous samples.
- The K-S statistic D = sup |F_n(x) - F_0(x)| is the largest vertical gap between the empirical and hypothesised CDFs.
- This item repeats 2018 Q37 verbatim - the goodness-of-fit pairing is clearly worth memorising.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The chi-square goodness-of-fit test compares observed class frequencies with those expected under a fully specified theoretical distribution.
2. The Kolmogorov-Smirnov one-sample test performs the same task in a distribution-free way, comparing the empirical distribution function F_n(x) with the hypothesised F_0(x).
3. Its statistic is D = sup over x of |F_n(x) - F_0(x)|, and a large D leads to rejection of the hypothesised distribution.
4. The Mann-Whitney and Wilcoxon tests compare locations of samples, and the median test compares medians; none tests agreement with a specified distribution.
5. Hence the analogue of the chi-square goodness-of-fit test is the Kolmogorov-Smirnov test, option (b).

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11. 2022 · Q57 Mann-Whitney Test · Conceptual

QUESTION

Which one of the following is correct regarding non-parametric methods?

OPTIONS

(a) Sign test uses rank only

(b) Wilcoxon test uses both sign and rank

(c) Mann-Whitney test is a one-sample test

(d) Kruskal-Wallis test is a two-sample test

ANSWER (AI-derived — no official key exists for this paper)

(b) Wilcoxon test uses both sign and rank

EXAM SHORTCUT (AI-derived explanation)

The Wilcoxon signed-rank test uses both the SIGN of each difference and the RANK of its magnitude - the only correct statement here.

TIPS & TRICKS (AI-derived explanation)

- Sign test uses signs ONLY, discarding magnitudes; the Wilcoxon signed-rank test adds the ranks of the magnitudes, which is why it is more powerful.
- Mann-Whitney U is a TWO-sample test for independent samples; Kruskal-Wallis is its k-sample generalisation (the non-parametric one-way ANOVA).
- Sample-count map: sign and Wilcoxon signed-rank = one sample or paired; Mann-Whitney and median test = two independent samples; Kruskal-Wallis = k samples.
- Options (a), (c) and (d) each misstate one of these design facts, so knowing the map answers the item without any computation.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Option (a): the sign test records only whether each observation lies above or below the hypothesised value, using the SIGNS and discarding the magnitudes entirely. It does not use ranks. FALSE.
2. Option (b): the Wilcoxon signed-rank test ranks the absolute differences and then attaches to each rank the sign of the corresponding difference, summing the positive (or negative) signed ranks. It therefore uses BOTH sign and rank. TRUE.
3. Option (c): the Mann-Whitney U test compares two INDEPENDENT samples, so it is a two-sample test, not a one-sample test. FALSE.
4. Option (d): the Kruskal-Wallis test is the non-parametric analogue of one-way ANOVA and compares k independent samples, k >= 2 in general. Calling it a two-sample test is wrong. FALSE.
5. Only the statement about the Wilcoxon test is correct.
6. Hence option (b) is correct.

---

12. 2022 · Q58 Chi-square Distribution · Numerical

QUESTION

A person has 10 black, 8 red, 5 green, 12 orange, 15 blue balls. The χ^2 value testing H_0: colours occur with equal frequency, is:

OPTIONS

(a) 6.9

(b) 5.8

(c) 5.6

(d) 5.4

ANSWER **(AI-derived — no official key exists for this paper)**

(b) 5.8

EXAM SHORTCUT **(AI-derived explanation)**

Total 50 balls over 5 colours gives expected 10 each. Deviations are 0, -2, -5, 2, 5, so chi-square = (0 + 4 + 25 + 4 + 25)/10 = 58/10 = 5.8.

TIPS & TRICKS **(AI-derived explanation)**

- Under 'equal frequency' the expected count is simply N/k = 50/5 = 10, the same for every class - which makes the denominator constant and the sum easy.
- With a common expectation E, chi-square = [sum of $(O - E)^2$]/E. Compute the squared deviations first and divide once at the end.
- Check the total: 10 + 8 + 5 + 12 + 15 = 50, so the expected counts are right; a wrong total invalidates everything downstream.
- Degrees of freedom = k - 1 = 4, whose 5% critical value is 9.49; since 5.8 < 9.49 the hypothesis of equal frequencies is not rejected.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. The observed frequencies are 10, 8, 5, 12 and 15, totalling N = 50 balls across k = 5 colours.
2. Under H0 of equal frequency, each expected frequency is E = 50/5 = 10.
3. Deviations O - E: 0, -2, -5, 2, 5.
4. Squared deviations: 0, 4, 25, 4, 25, summing to 58.
5. Since every expected frequency is 10, chi-square = 58/10 = 5.8.
6. Option (b) is correct. (With 4 degrees of freedom the 5% critical value is 9.49, so H0 would not be rejected.)

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13. 2026 · Q37 Non-parametric Tests · Conceptual

QUESTION

The Kruskal-Wallis test is exactly equivalent to one-way ANOVA when:

OPTIONS

(a) data are normal

(b) population distributions are identical and normal

(c) sample sizes are equal

(d) all observations are replaced by their ranks

ANSWER **(AI-derived — no official key exists for this paper)**

(d) all observations are replaced by their ranks

EXAM SHORTCUT (AI-derived explanation)

Kruskal-Wallis IS the one-way ANOVA computed on ranks - the H statistic is a monotone function of the between-groups sum of squares of the ranks. So the equivalence holds exactly when observations are replaced by their ranks. Answer (d).

TIPS & TRICKS (AI-derived explanation)

- Most classical non-parametric tests are the corresponding parametric test applied to ranks: Kruskal-Wallis to one-way ANOVA, Wilcoxon rank-sum to the two-sample t, Spearman's rho to Pearson's r, Friedman to two-way ANOVA.
- Kruskal-Wallis is distribution-free, so normality (options a, b) is exactly what it does NOT require - those options invert the point of the test.
- Equal sample sizes (option c) are convenient but neither necessary nor sufficient for the equivalence.
- H = (12/[N(N+1)]) * sum (R_i^2/n_i) - 3(N+1), which is the between-groups sum of squares of the ranks scaled by the known variance of the ranks; H is compared with chi-square on k - 1 df.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The Kruskal-Wallis H test is the rank analogue of the one-way analysis of variance for comparing k independent samples.
2. Its statistic is H = 12/[N(N+1)] * sum over i of (R_i^2/n_i) - 3(N + 1), where R_i is the rank sum of the i-th group and N is the total sample size.
3. Because the ranks 1, 2, ..., N have a known total sum and known total sum of squares, the total sum of squares of the ranks is a constant. Consequently H is a strictly increasing function of the between-groups sum of squares of the ranks, which is exactly what one-way ANOVA's F is based on.
4. Therefore performing a one-way ANOVA on the ranks gives a test equivalent to Kruskal-Wallis: the two procedures coincide precisely when all observations are replaced by their ranks.
5. Options (a) and (b) are wrong because Kruskal-Wallis is a distribution-free test that does not assume normality; option (c) is wrong because equal sample sizes are not what creates the equivalence.
6. Hence option (d) is correct.

S13 · STATISTICAL METHODS

Order Statistics - Minimum, Maximum, Range, Median

16 questions · 2018: 2 2019: 2 2020: 3 2021: 1 2022: 1 2023: 0 2024: 3 2025: 1 2026: 3

Weight. 16 questions (9.8% of Statistical Methods) · appeared in 8 of 9 papers · peak year 2020 (3)

Examiner's pattern. Steady 1-3 marks every year. The pdf of the minimum, maximum, r-th order statistic and the sample median, plus expectations for uniform and exponential parents. The minimum of iid exponentials being exponential with rate nlambda is the single most reused fact.

Must-know. g_r(x) = n!/((r-1)!(n-r)!) F^(r-1) (1-F)^(n-r) f; min of n iid Exp(lambda) ~ Exp(nlambda); for U(0,1): E[X_(r)] = r/(n+1); P(X_(1) > x) = (1 - F(x))^n.

1. 2018 · Q39

Order Statistics - Minimum, Maximum & Median · Numerical

QUESTION

For a random sample of size 2 from $N(0,\sigma^2)$, what is $E[X_{(1)}]$ (first order statistic)?

OPTIONS

(a) $\frac{2\sigma}{\sqrt{\pi}}$

(b) $\frac{\sigma}{\pi}$

(c) $-\frac{\sigma}{\sqrt{\pi}}$

(d) $-\frac{\sigma}{\sqrt{2\pi}}$

ANSWER **(AI-derived — no official key exists for this paper)**

(c) $-\frac{\sigma}{\sqrt{\pi}}$

EXAM SHORTCUT **(AI-derived explanation)**

min(X1,X2) = [(X1+X2) - |X1-X2|]/2. The first term has mean 0 and X1 - X2 ~ N(0, 2 sigma^2), so E|X1-X2| = sigma sqrt(2) . sqrt(2/pi) = 2 sigma/sqrt(pi). Hence E[X_(1)] = -sigma/sqrt(pi).

TIPS & TRICKS **(AI-derived explanation)**

- Identity worth memorising: min(a,b) = (a + b - |a - b|)/2 and max(a,b) = (a + b + |a - b|)/2. It converts order-statistic expectations into a mean-absolute-deviation computation.
- For W ~ N(0, tau^2), E|W| = tau sqrt(2/pi). Here tau = sigma sqrt(2), so E|X1 - X2| = 2 sigma/sqrt(pi).
- Sign check first: the minimum of two symmetric mean-zero variables must have a NEGATIVE expectation, which immediately eliminates options (a) and (b).
- For a sample of size 2 from any symmetric distribution, E[X_(1)] = -E[X_(2)]; the two order statistics are mirror images about the mean.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. Let X1, X2 be independent N(0, sigma^2) and let X_(1) = min(X1, X2).
2. Use the identity min(X1, X2) = [(X1 + X2) - |X1 - X2|]/2.
3. E(X1 + X2) = 0 since both means are 0.
4. X1 - X2 ~ N(0, 2 sigma^2), so with tau = sigma sqrt(2) and the half-normal mean formula E|W| = tau sqrt(2/pi), we get E|X1 - X2| = sigma sqrt(2) . sqrt(2/pi) = 2 sigma/sqrt(pi).
5. Therefore E[X_(1)] = [0 - 2 sigma/sqrt(pi)]/2 = -sigma/sqrt(pi).

6. Hence option (c) is correct.

---

2. **2018 · Q40** Order Statistics - Minimum, Maximum & Median · Theoretical

QUESTION

If $X_i,\ i=1,\dots,n$ are iid with c.d.f. $F(x)=x,\ 0\le\ x\le 1$, the p.d.f. $g(r)$ of the range $R=X_{(n)}-X_{(1)}$ is

OPTIONS

(a) $n(n-1)r^{n-2}(1-r)$

(b) $n(n-1)r(1-r)^{n-2}$

(c) nr^{n-1}

(d) $(n-1)r^{n-2}$

ANSWER (AI-derived — no official key exists for this paper)

(a) $n(n-1)r^{n-2}(1-r)$

EXAM SHORTCUT (AI-derived explanation)

For a U(0,1) sample the range R ~ Beta(n-1, 2), whose density is n(n-1) r^(n-2)(1-r) on (0,1). Recognise the Beta and read the answer off.

TIPS & TRICKS (AI-derived explanation)

- General range density: g(r) = n(n-1) times the integral over x of f(x) f(x+r) [F(x+r) - F(x)]^(n-2) dx. For U(0,1) the bracket is just r and the integral contributes the factor (1 - r).
- Shape check: R ~ Beta(n-1, 2) means the exponents must be (n-1)-1 = n-2 on r and 2-1 = 1 on (1-r). Only option (a) has that pairing; option (b) has them swapped.
- Normalisation test: the integral of n(n-1) r^(n-2)(1-r) over (0,1) is n(n-1) [1/(n-1) - 1/n] = 1. Option (b) also integrates to 1, so use the shape argument, not normalisation, to separate them.
- Options (c) and (d) are densities of the MAXIMUM and of a related order statistic, not of the range - a classic substitution trap.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The X_i are i.i.d. Uniform(0,1), so f(x) = 1 on (0,1) and F(x) = x on (0,1).
2. The joint density of the smallest and largest order statistics is h(u, v) = n(n-1) f(u) f(v) [F(v) - F(u)]^(n-2) = n(n-1)(v - u)^(n-2) for 0 < u < v < 1.
3. Put R = V - U and integrate out U. For a fixed r in (0,1), U can range over (0, 1 - r).
4. g(r) = integral from 0 to 1-r of n(n-1) r^(n-2) du = n(n-1) r^(n-2) (1 - r), for 0 < r < 1.
5. This is the Beta(n-1, 2) density, and it integrates to n(n-1)[1/(n-1) - 1/n] = 1, confirming correctness.
6. Hence option (a) is correct.

---

3. **2019 · Q30** Order Statistics - Minimum, Maximum & Median · Numerical

QUESTION

What is the probability that the sample median, based on a random sample of size 3 from a distribution with p.d.f. $f(x)=2x,\ 0<x<1$, exceeds $\frac{1}{2}$?

OPTIONS

(a) $\frac{3}{4}$

(b) $\frac{13}{16}$

(c) $\frac{27}{32}$

(d) $\frac{29}{32}$

ANSWER (AI-derived — no official key exists for this paper)

(c) $\frac{27}{32}$

EXAM SHORTCUT (AI-derived explanation)

$F(x) = x^2$, so p = P(X > 1/2) = 1 - 1/4 = 3/4. The median of 3 exceeds 1/2 iff at least 2 observations do: $3(3/4)^2(1/4) + (3/4)^3$ = 27/64 + 27/64 = 27/32.

TIPS & TRICKS (AI-derived explanation)

- For an odd sample size n = 2m+1, the median exceeds c exactly when at least m+1 of the observations exceed c. Convert the order-statistic question into a binomial one.
- Get the CDF first: for f(x) = 2x on (0,1), $F(x) = x^2$, so F(1/2) = 1/4 and p = 3/4 - the single number that drives everything.
- Both binomial terms equal 27/64 here (since 3 x (9/16)(1/4) = 27/64 and $(3/4)^3$ = 27/64), so the total 54/64 = 27/32 falls out neatly - a good sign the set-up is right.
- Option (a) 3/4 is simply p itself, planted for candidates who forget the 'at least two out of three' step.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The density is f(x) = 2x on (0,1), so the CDF is $F(x) = x^2$ for 0 < x < 1.
2. P(X > 1/2) = 1 - F(1/2) = 1 - 1/4 = 3/4; call this p.
3. For a sample of size 3, the sample median is the second order statistic X_(2). It exceeds 1/2 exactly when at least 2 of the 3 observations exceed 1/2.
4. Let B ~ Binomial(3, 3/4) count the observations exceeding 1/2. Then P(median > 1/2) = P(B = 2) + P(B = 3).
5. P(B = 2) = $C(3,2)(3/4)^2(1/4)$ = 3(9/16)(1/4) = 27/64, and P(B = 3) = $(3/4)^3$ = 27/64.
6. Adding: 27/64 + 27/64 = 54/64 = 27/32, option (c).

---

4. **2019 · Q36** Order Statistics - Minimum, Maximum & Median · Numerical

QUESTION

If X and Y are independent $N(0,1)$, what is $E[\text{Max}(X,Y)]$ equal to?

OPTIONS

(a) $\frac{1}{\sqrt{2\pi}}$

(b) $\frac{1}{\sqrt{\pi}}$

(c) $\frac{1}{\sqrt{2}}$

(d) $\frac{2}{\sqrt{\pi}}$

ANSWER (AI-derived — no official key exists for this paper)

(b) $\frac{1}{\sqrt{\pi}}$

EXAM SHORTCUT (AI-derived explanation)

max(X,Y) = [(X+Y) + |X-Y|]/2. The first part has mean 0, and X - Y ~ N(0,2) gives E|X-Y| = sqrt2 . sqrt(2/pi) = 2/sqrt(pi). Half of that is 1/sqrt(pi).

TIPS & TRICKS (AI-derived explanation)

- Identity to memorise: max(a,b) = (a + b + |a - b|)/2. It reduces an order-statistic expectation to a mean absolute deviation.
- For W ~ N(0, tau^2), E|W| = tau sqrt(2/pi). Here tau = sqrt2, so E|X-Y| = 2/sqrt(pi).
- General result: for a sample of 2 from N(mu, sigma^2), E[max] = mu + sigma/sqrt(pi) and E[min] = mu - sigma/sqrt(pi). This is the companion of 2018 Q39.
- Sign and size check: the answer must be positive and modest (about 0.564). Option (d) 2/sqrt(pi) = 1.128 is the value of E|X - Y|, the classic half-way trap.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let X, Y be independent N(0,1) and use max(X, Y) = [(X + Y) + |X - Y|]/2.
2. E(X + Y) = 0 since both means are zero.
3. X - Y ~ N(0, 2), so with tau = sqrt(2) and the half-normal result E|W| = tau sqrt(2/pi), we get E|X - Y| = sqrt(2) sqrt(2/pi) = 2/sqrt(pi).
4. Therefore E[max(X, Y)] = [0 + 2/sqrt(pi)]/2 = 1/sqrt(pi).
5. Numerically this is 0.5642.
6. Hence option (b) is correct.

---

5. **2020 · Q7** Order Statistics - Minimum, Maximum & Median · Theoretical

QUESTION

If $X_{(r)}$ is the rth order statistic from a sample of size n from a continuous c.d.f. $F(x)$, then $F(X_{(r)})$ is the rth order statistic in a sample of size n from the

OPTIONS

(a) Uniform distribution over $(-1,+1)$

(b) Uniform distribution over $(0,1)$

(c) Normal distribution

(d) Exponential distribution

ANSWER (AI-derived — no official key exists for this paper)

(b) Uniform distribution over $(0,1)$

EXAM SHORTCUT (AI-derived explanation)

The probability integral transform makes F(X) ~ U(0,1). Since F is increasing, it preserves order, so F(X_(r)) is the rth order statistic of a U(0,1) sample.

TIPS & TRICKS (AI-derived explanation)

- Probability integral transform: for continuous F, the variable U = F(X) is always Uniform(0,1), whatever the parent distribution. This is the single most useful transformation in order-statistic theory.
- F is non-decreasing, so it maps the rth smallest X to the rth smallest U - order is preserved and the ranks carry over unchanged.
- Consequence worth memorising: F(X_(r)) ~ Beta(r, n - r + 1), which gives E[F(X_(r))] = r/(n+1) - the basis of plotting positions in probability plots.
- The result is distribution-free, which is exactly why non-parametric methods and simulation both rest on it.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let X have a continuous CDF F. The probability integral transform states that U = F(X) follows the Uniform(0,1) distribution.
2. Indeed, for 0 < u < 1, P(F(X) <= u) = P(X <= F^(-1)(u)) = F(F^(-1)(u)) = u, which is the U(0,1) CDF.
3. Applying F to each of X_1, ..., X_n gives a random sample U_1, ..., U_n from U(0,1).
4. Because F is non-decreasing, it preserves the ordering of the sample, so the rth smallest of the U values is F applied to the rth smallest X value: U_(r) = F(X_(r)).
5. Therefore F(X_(r)) is the rth order statistic of a sample of size n from the Uniform(0,1) distribution.
6. Option (b) is correct.

---

6. **2020 · Q10** Order Statistics - Minimum, Maximum & Median · Theoretical

QUESTION

$X_1, \ldots, X_5$ are five independent observations from a continuous distribution with c.d.f. $F(x)$ and p.d.f. $f(x)$. What is the p.d.f. of the sample median?

OPTIONS

(a) $5[F(x)]^4 f(x)$

(b) $5[1-F(x)]^4 f(x)$

(c) $5[1-F(x)]^2[F(x)]^2 f(x)$

(d) $30[1-F(x)]^2[F(x)]^2 f(x)$

ANSWER (AI-derived — no official key exists for this paper)

(d) $30[1-F(x)]^2[F(x)]^2 f(x)$

EXAM SHORTCUT (AI-derived explanation)

For n = 5 the median is the 3rd order statistic, so the constant is 5!/(2! 2!) = 30 and the density is 30 F^2 $(1-F)^2$ f.

TIPS & TRICKS (AI-derived explanation)

- General order-statistic density: g_r(x) = [n!/((r-1)!(n-r)!)] [F(x)]^(r-1) [1 - F(x)]^(n-r) f(x). Substitute r and n and the whole answer follows.
- For the median of 5 observations, r = 3, so the exponents are r - 1 = 2 and n - r = 2 - symmetric, as the median should be.
- The constant 5!/(2!2!) = 120/4 = 30 is what separates (c) from (d); option (c) has the right shape but the wrong normalising constant.

- Options (a) and (b) are the densities of the MAXIMUM and the MINIMUM respectively - useful to recognise as the r = n and r = 1 special cases.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For a sample of size n from a continuous distribution, the density of the rth order statistic is g_r(x) = [n!/((r-1)!(n-r)!)] [F(x)]^(r-1) [1-F(x)]^(n-r) f(x).
2. With n = 5 observations, the sample median is the third order statistic, so r = 3.
3. The exponents are r - 1 = 2 on F(x) and n - r = 2 on 1 - F(x).
4. The constant is 5!/(2! 2!) = 120/(2 x 2) = 30.
5. Hence the density of the sample median is 30 $[F(x)]^2$ $[1 - F(x)]^2$ f(x).
6. Option (d) is correct.

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7. 2020 · Q20 Order Statistics - Minimum, Maximum & Median · Numerical

QUESTION

Ten runners' finishing times (seconds) are i.i.d. $U(10,11)$, independent of each other. The expected time of the winner is

OPTIONS

(a) $10\frac{1}{11}$

(b) $10\frac{1}{10}$

(c) $10\frac{1}{9}$

(d) $10\frac{1}{2}$

ANSWER (AI-derived — no official key exists for this paper)

(a) $10\frac{1}{11}$

EXAM SHORTCUT (AI-derived explanation)

The winner has the MINIMUM time. For U(a,b) with n observations, E[min] = a + (b-a)/(n+1) = 10 + 1/11.

TIPS & TRICKS (AI-derived explanation)

- Uniform order-statistic means: E[X_(r)] = a + (b - a) r/(n+1). For the minimum r = 1 and for the maximum r = n - two formulas in one.
- The winner is the FASTEST, i.e. the smallest time, so r = 1. Reading 'winner' as the maximum gives 10 + 10/11 and no matching option.
- The n + 1 in the denominator is the standard 'plotting position' divisor - the n order statistics divide the interval into n + 1 equal expected gaps.
- With 10 runners on a 1-second interval, the expected winning margin below 10 seconds is 1/11 = 0.0909 s - a small, plausible figure.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The ten finishing times are i.i.d. Uniform(10, 11), and the winner's time is the minimum, X_(1).
2. For a sample of size n from U(a, b), the rth order statistic has mean E[X_(r)] = a + (b - a) r/(n + 1).
3. For the minimum, r = 1, so E[X_(1)] = a + (b - a)/(n + 1).
4. Substituting a = 10, b = 11 and n = 10: E[X_(1)] = 10 + 1/11.

5. Numerically this is 10.0909 seconds.
6. Hence the expected time of the winner is 10 + 1/11, option (a).

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8. **2021 · Q11** Order Statistics - Minimum, Maximum & Median · Numerical

QUESTION

Consider a random sample of size 4 from a population with density $f(x)=2x,\ 0<x<1$ (and 0 otherwise). If the ordered observations are $x_1<x_2<x_3<x_4$, then $P[\frac{1}{2}<x_3]$ is

OPTIONS

(a) $\frac{15}{16}$

(b) $\frac{243}{256}$

(c) $\frac{1}{2}$

(d) $\frac{247}{256}$

ANSWER (AI-derived — no official key exists for this paper)

(b) $\frac{243}{256}$

EXAM SHORTCUT (AI-derived explanation)

F(x) = x^2, so p = P(X > 1/2) = 3/4. The third of four order statistics exceeds 1/2 iff at least 2 observations do: 1 - $(1/4)^4$ - 4(3/4)$(1/4)^3$ = 1 - 13/256 = 243/256.

TIPS & TRICKS (AI-derived explanation)

- Convert the order-statistic event into a binomial count: X_(r) > c iff at most r - 1 observations are <= c, i.e. at least n - r + 1 exceed c. Here n = 4, r = 3, so at least 2 must exceed 1/2.
- Compute the complement: it is far quicker to subtract P(0 exceed) + P(1 exceeds) than to add three terms.
- With p = 3/4, P(0) = $(1/4)^4$ = 1/256 and P(1) = 4(3/4)$(1/4)^3$ = 12/256, totalling 13/256 - clean numbers that confirm the set-up.
- Option (a) 15/16 is the answer to 'at least one exceeds 1/2' and option (c) 1/2 ignores the density's shape - both are planted for partial reasoning.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The density f(x) = 2x on (0,1) has CDF F(x) = x^2, so F(1/2) = 1/4 and p = P(X > 1/2) = 3/4.
2. With n = 4 observations, the third order statistic x_3 exceeds 1/2 exactly when at least 4 - 3 + 1 = 2 of the observations exceed 1/2.
3. Let B ~ Binomial(4, 3/4) count the observations exceeding 1/2. We need P(B >= 2) = 1 - P(B = 0) - P(B = 1).
4. P(B = 0) = $(1/4)^4$ = 1/256.
5. P(B = 1) = C(4,1)(3/4)$(1/4)^3$ = 4(3/4)(1/64) = 12/256.
6. So P(B >= 2) = 1 - 13/256 = 243/256, option (b).

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9. 2022 · Q52 Order Statistics - Minimum, Maximum & Median · Theoretical

QUESTION

Let X_i be i.i.d., $i=1,\ldots,n$, with c.d.f. $F(x)$, p.d.f. $f(x)$. The p.d.f. $g_Z(x)$ where $Z=\min\{X_1,\ldots,X_n\}$ is:

OPTIONS

(a) $n[F(x)]^{n-1}f(x)$

(b) $n[1-F(x)]^{n-1}f(x)$

(c) $[1-F(x)]^{n-1}f(x)$

(d) $[F(x)]^{n-1}f(x)$

ANSWER (AI-derived — no official key exists for this paper)

(b) $n[1-F(x)]^{n-1}f(x)$

EXAM SHORTCUT (AI-derived explanation)

P(Z > x) = $[1 - F(x)]^n$, so differentiating the CDF 1 - $[1-F(x)]^n$ gives g(x) = n[1 - F(x)]^(n-1) f(x).

TIPS & TRICKS (AI-derived explanation)

- Derive from the SURVIVAL function for a minimum: Z > x means every X_i > x, giving $[1 - F(x)]^n$ directly.
- For the MAXIMUM the mirror argument uses the CDF: P(max <= x) = $[F(x)]^n$, so its density is n[F(x)]^(n-1) f(x) - option (a) here.
- The constant n must be present: it is the number of ways the minimum can be attained by one of the n observations. Options (c) and (d) omit it and do not integrate to 1.
- This is the r = 1 case of the general order-statistic density [n!/((r-1)!(n-r)!)][F]^(r-1)[1-F]^(n-r) f.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let Z = min(X_1, ..., X_n) where the X_i are i.i.d. with CDF F and density f.
2. The event Z > x occurs exactly when every X_i > x, so by independence P(Z > x) = $[1 - F(x)]^n$.
3. Hence the CDF of Z is G(x) = 1 - $[1 - F(x)]^n$.
4. Differentiating with respect to x: g(x) = -n[1 - F(x)]^(n-1) . (-f(x)) = n[1 - F(x)]^(n-1) f(x).
5. This is the r = 1 case of the general order-statistic density formula.
6. Hence option (b) is correct.

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10. 2024 · Q1 Order Statistics - Minimum, Maximum & Median · Theoretical

QUESTION

If $X_{(1)}, X_{(2)}, X_{(3)}, X_{(4)}, X_{(5)}$ are order statistics from a population with pdf $f(x)=2e^{-2x}$, $x>0$, then what is the density function of $X_{(1)}$?

OPTIONS

(a) $10e^{-10X_{(1)}}$

(b) $1-e^{-2X_{(1)}}$

(c) $1-e^{-10X_{(1)}}$

(d) $10e^{-2X_{(1)}}$

ANSWER (AI-derived — no official key exists for this paper)

(a) $10e^{-10X_{(1)}}$

EXAM SHORTCUT (AI-derived explanation)

The minimum of n i.i.d. Exponential(lambda) variables is Exponential(n lambda). With n = 5 and lambda = 2 the rate is 10, so the density is 10 e^(-10x).

TIPS & TRICKS (AI-derived explanation)

- Memorise: min of n i.i.d. Exp(lambda) is Exp(n lambda) - the rates ADD, because P(min > x) = [e^(-lambda x)]^n = e^(-n lambda x).
- Options (b) and (c) are cumulative distribution functions, not densities - a density must integrate to 1, and 1 - e^(-2x) does not.
- Option (d) 10 e^(-2x) is not a density either: its integral is 5, not 1. Checking normalisation eliminates three options instantly.
- Companion result: the MAXIMUM of exponentials is NOT exponential; only the minimum inherits the family.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The parent density is f(x) = 2 e^(-2x) for x > 0, i.e. Exponential with rate 2, whose survival function is P(X > x) = e^(-2x).
2. For a sample of size 5, the event X_(1) > x means every one of the five observations exceeds x.
3. By independence, P(X_(1) > x) = [e^(-2x)]^5 = e^(-10x).
4. So the CDF of X_(1) is 1 - e^(-10x), and differentiating gives the density 10 e^(-10x) for x > 0.
5. That is, X_(1) ~ Exponential with rate 5 x 2 = 10.
6. Option (a) is correct.

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11. **2024 · Q17** Order Statistics - Minimum, Maximum & Median · Theoretical

QUESTION

The pdf $g_n(x)$ of $X_{(n)}$ in a random sample of size n from the exponential distribution with pdf $f(x)=e^{-x}$, $x\geq$ 0, is:

OPTIONS

(a) $(1-e^{-x})^{n-1}$

(b) $n(1-e^{-x})^{n-1}e^{-x}$

(c) $(1-e^{-x})^{n}$

(d) $n(1-e^{-x})^{n-1}$

ANSWER (AI-derived — no official key exists for this paper)

(b) $n(1-e^{-x})^{n-1}e^{-x}$

EXAM SHORTCUT (AI-derived explanation)

The density of the maximum is n[F(x)]^(n-1) f(x). With F(x) = 1 - e^(-x) and f(x) = e^(-x) this is n(1-e^(-x))^(n-1) e^(-x).

TIPS & TRICKS (AI-derived explanation)

- Maximum: P(X_(n) <= x) = $[F(x)]^n$, so differentiating gives n[F(x)]^(n-1) f(x). Minimum uses $[1 - F(x)]^n$ instead.
- The answer must contain BOTH the CDF power and the density factor f(x) = e^(-x). Options (a), (c) and (d) are missing one or the other.
- Options (a) and (c) are not densities at all: $(1-e^{(-x)})^n$ is the CDF of the maximum, integrating to infinity as a density.
- The constant n counts the ways the maximum can be attained by one of the n observations - it must be present, which eliminates option (a).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For the exponential density f(x) = e^(-x) on x >= 0, the CDF is F(x) = 1 - e^(-x).
2. The maximum X_(n) satisfies X_(n) <= x exactly when every observation is at most x, so P(X_(n) <= x) = $[F(x)]^n$ = $(1 - e^{(-x)})^n$.
3. Differentiating with respect to x gives the density g_n(x) = n[F(x)]^(n-1) f(x).
4. Substituting: g_n(x) = n(1 - e^(-x))^(n-1) e^(-x) for x >= 0.
5. Check: integrating this over (0, infinity) gives $[(1-e^{(-x)})^n]$ from 0 to infinity = 1, a valid density.
6. Option (b) is correct.

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12. 2024 · Q20

Order Statistics - Minimum, Maximum & Median · Numerical

QUESTION

If $X_{(1)}, X_{(2)}, X_{(3)}, X_{(4)}, X_{(5)}$ are order statistics from $U[0,2]$, then what is the value of the pdf $f_{X_{(1)}}(1)$?

OPTIONS

(a) 1/16

(b) 5/16

(c) 1/32

(d) 5/32

ANSWER (AI-derived — no official key exists for this paper)

(d) 5/32

EXAM SHORTCUT (AI-derived explanation)

For U[0,2], f = 1/2 and F(x) = x/2. The minimum has density $5(1 - x/2)^4$ (1/2), which at x = 1 is 5(1/16)(1/2) = 5/32.

TIPS & TRICKS (AI-derived explanation)

- Density of the minimum: n[1 - F(x)]^(n-1) f(x). Substitute n = 5, F(1) = 1/2 and f = 1/2.
- Careful with the uniform on [0,2]: the density is 1/2, NOT 1, and F(1) = 1/2, not 1. Both appear in the formula.
- Compute in stages: $(1 - 1/2)^4$ = 1/16, then 5 x (1/16) x (1/2) = 5/32.
- Option (b) 5/16 omits the density factor 1/2 and option (c) 1/32 omits the constant 5 - each distractor drops one factor.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For the Uniform[0,2] distribution the density is f(x) = 1/2 and the CDF is F(x) = x/2 for 0 <= x <= 2.

2. The density of the smallest order statistic in a sample of size n is f_(1)(x) = n[1 - F(x)]^(n-1) f(x).
3. With n = 5: f_(1)(x) = 5(1 - x/2)^4 (1/2).
4. At x = 1: 1 - 1/2 = 1/2, so (1/2)^4 = 1/16.
5. f_(1)(1) = 5 x (1/16) x (1/2) = 5/32.
6. Option (d) is correct.

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13. 2025 · Q57 Order Statistics - Minimum, Maximum & Median · Numerical

QUESTION

Let M be the minimum of a random sample of size n from $f(x) = \begin{cases} e^{-(x-\theta)}, & \theta < x < \infty \\ 0, & \text{otherwise} \end{cases}$. What is the distribution of $2n(M-\theta)$?

OPTIONS

(a) $\chi^2_{(2n)}$

(b) Exponential, mean 2

(c) Exponential, mean 0.5

(d) $\chi^2_{(n)}$

ANSWER (AI-derived — no official key exists for this paper)

(b) Exponential, mean 2

EXAM SHORTCUT (AI-derived explanation)

P(M > m) = e^(-n(m - theta)), so M - theta ~ Exp(rate n). Then P(2n(M-theta) > w) = e^(-w/2), an exponential with mean 2.

TIPS & TRICKS (AI-derived explanation)

- Minimum of n shifted exponentials: the rate multiplies by n, so M - theta ~ Exp(n) with mean 1/n.
- Scaling by 2n turns rate n into rate n/(2n) = 1/2, hence mean 2 - the factor 2n is chosen precisely to standardise it.
- Exponential with mean 2 is the same as chi-square with 2 degrees of freedom, which is why such pivots are used to build confidence intervals for theta.
- Options (a) and (d) have degrees of freedom growing with n, but the scaling removes all dependence on n - a structural giveaway.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The density f(x) = e^(-(x - theta)) for x > theta gives the survival function P(X > x) = e^(-(x - theta)).
2. For the minimum of n independent observations, P(M > m) = [e^(-(m - theta))]^n = e^(-n(m - theta)) for m > theta.
3. Hence M - theta follows an exponential distribution with rate n.
4. Let W = 2n(M - theta). Then P(W > w) = P(M - theta > w/(2n)) = e^(-n w/(2n)) = e^(-w/2).
5. That is the survival function of an exponential distribution with rate 1/2, i.e. mean 2 (equivalently chi-square with 2 degrees of freedom).
6. Option (b) is correct.

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14. 2026 · Q26 Order Statistics - Minimum, Maximum & Median · Numerical

QUESTION

Let $Y_1 \le Y_2 \le \cdots \le Y_6$ be order statistics from $F(x)=1-e^{-\lambda x}$, $\lambda=0.5$. What is the cdf of Y_1?

OPTIONS

(a) $1-e^{y/2}$

(b) $1-e^{y/3}$

(c) $1-e^{3y}$

(d) $1-e^{-3y}$

ANSWER (AI-derived — no official key exists for this paper)

(d) $1-e^{-3y}$

EXAM SHORTCUT (AI-derived explanation)

The minimum of n iid exponentials with rate lambda is exponential with rate n*lambda. Here n = 6 and lambda = 0.5, so Y_1 is Exp(3) and its cdf is 1 - e^{-3y}.

TIPS & TRICKS (AI-derived explanation)

- Memorise: min of iid Exp(lambda) is Exp(n lambda) - the hazard rates simply add. This is the exponential's most exam-friendly property.
- Derive it in one line if you forget: P(Y_1 > y) = [P(X > y)]^{n} = (e^{-lambda y})^{n} = e^{-n lambda y}.
- A cdf must increase from 0 to 1, so any option with a positive exponent (e^{y/2}, e^{y/3}, e^{3y}) is impossible - that alone leaves only (d).
- By contrast the maximum has cdf [1 - e^{-lambda y}]^{n}, which is NOT exponential - a common companion question.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The parent distribution is F(x) = 1 - e^{-lambda x} with lambda = 0.5, i.e. exponential with rate 0.5, and the sample size is n = 6.
2. Y_1 is the smallest order statistic, so Y_1 > y if and only if every one of the six observations exceeds y.
3. By independence, P(Y_1 > y) = [1 - F(y)]^{6} = (e^{-0.5 y})^{6} = e^{-3y}, for y > 0.
4. Therefore the cdf is F_{Y_1}(y) = 1 - P(Y_1 > y) = 1 - e^{-3y}, for y > 0.
5. So Y_1 is exponential with rate n*lambda = 6(0.5) = 3.
6. Hence option (d) is correct.

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15. 2026 · Q38 Order Statistics - Minimum, Maximum & Median · Theoretical

QUESTION

If $X \sim U(0,1)$, what is $P(X_{(1)} > x)$?

OPTIONS

(a) $1-x^{n}$

(b) $(1-x)^{n}$

(c) nx^{n-1}

(d) x^{n}

ANSWER (AI-derived — no official key exists for this paper)

(b) $(1\text{-}x)^n$

EXAM SHORTCUT (AI-derived explanation)

The minimum exceeds x precisely when every observation exceeds x. For U(0,1), P(X_i > x) = 1 - x, so by independence P(X_(1) > x) = $(1 - x)^n$.

TIPS & TRICKS (AI-derived explanation)

- Golden rule of order statistics: 'minimum > x' means ALL exceed x, and 'maximum <= x' means ALL are at most x. Convert to these two forms first.
- For the maximum the companion result is P(X_(n) <= x) = x^n, which is option (d) - the standard swap-the-extreme distractor.
- Check the endpoints: at x = 0 the probability must be 1 and at x = 1 it must be 0. $(1 - x)^n$ gives 1 and 0; x^n gives 0 and 1, so it is the wrong extreme.
- Differentiating gives the density of the minimum, n(1-x)^(n-1), i.e. X_(1) is Beta(1, n) - useful if a follow-up asks for its mean 1/(n+1).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let X_1, ..., X_n be a random sample from U(0,1) and let X_(1) = min(X_1, ..., X_n).
2. The event {X_(1) > x} occurs if and only if every observation exceeds x.
3. For a single observation, P(X_i > x) = 1 - x for 0 <= x <= 1.
4. By independence, P(X_(1) > x) = $[P(X\_1 > x)]^n$ = $(1 - x)^n$.
5. (Endpoint check: at x = 0 this is 1 and at x = 1 it is 0, as a survival function must be.)
6. Hence option (b) is correct.

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16. 2026 · Q39 Measures of Dispersion · Theoretical

QUESTION

For a sample of odd size n from $U(0,1)$, the mean and variance of the distribution of the median are respectively:

OPTIONS

(a) 1/2 and 1/4

(b) 1/2 and $1/(4n+8)$

(c) 1/4 and $(n\text{-}3)/(2n+4)$

(d) Both do not exist

ANSWER (AI-derived — no official key exists for this paper)

(b) 1/2 and $1/(4n+8)$

EXAM SHORTCUT (AI-derived explanation)

For odd n the sample median is the ((n+1)/2)-th order statistic from U(0,1), which is Beta(m, m) with m = (n+1)/2. Symmetry gives mean 1/2, and the Beta variance $m^2/[(2m)^2(2m+1)]$ = 1/[4(n+2)] = 1/(4n+8).

TIPS & TRICKS (AI-derived explanation)

- Order statistics from U(0,1): X_(k) ~ Beta(k, n - k + 1), with mean k/(n+1) and variance k(n-k+1)/[$(n+1)^2$ (n+2)].
- For the median of an odd sample, k = (n+1)/2, so both Beta parameters are equal and the distribution is symmetric about 1/2 - the mean follows without computation.
- Plug into the variance formula directly: k(n-k+1) = $[(n+1)/2]^2$, so the variance is $[(n+1)/2]^2$/[$(n+1)^2$(n+2)] = 1/[4(n+2)].
- Sanity check at n = 1: the 'median' is the single observation U(0,1) with variance 1/12, and 1/(4*1+8) = 1/12. Correct.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For a sample of odd size n from U(0,1), the sample median is the k-th order statistic with k = (n + 1)/2.
2. The k-th order statistic of a uniform (0,1) sample follows the Beta distribution with parameters k and n - k + 1.
3. Here n - k + 1 = n - (n+1)/2 + 1 = (n + 1)/2 = k, so the median follows Beta(k, k) with k = (n+1)/2 - a distribution symmetric about 1/2.
4. Mean = k/(k + k) = 1/2.
5. Variance of Beta(a, b) is ab/[$(a+b)^2$ (a + b + 1)]. With a = b = k: variance = k^2/[$(2k)^2$ (2k + 1)] = 1/[4(2k + 1)].
6. Since 2k + 1 = (n + 1) + 1 = n + 2, the variance = 1/[4(n + 2)] = 1/(4n + 8).
7. Check at n = 1: variance = 1/12, matching Var(U(0,1)) = 1/12.
8. Hence option (b) is correct.

S14 · STATISTICAL METHODS

Asymptotic Relative Efficiency

1 question · 2018: 0 2019: 0 2020: 0 2021: 1 2022: 0 2023: 0 2024: 0 2025: 0 2026: 0

Weight. 1 question (0.6% of Statistical Methods) · appeared in 1 of 9 papers · peak year 2021 (1)

Examiner's pattern. Asked exactly once in nine years (2021), comparing the sample mean and sample median as estimators of a normal mean. Learn the one number and move on.

Must-know. ARE(median : mean) at the normal = 2/pi = 0.637. ARE = ratio of asymptotic variances (inverted), so the mean is the more efficient estimator here.

1. 2021 · Q37 Order Statistics - Minimum, Maximum & Median · Theoretical

QUESTION

In estimating μ from $N(\mu, \sigma^2)$, the sample mean is more asymptotically efficient than the sample median for all values of

OPTIONS

(a) μ and σ^2

(b) σ^2, iff $\mu=0$

(c) μ, iff $\sigma^2=1$

(d) σ^2, iff $\mu \neq 0$

ANSWER (AI-derived — no official key exists for this paper)

(a) μ and σ^2

EXAM SHORTCUT (AI-derived explanation)

The asymptotic relative efficiency of the median to the mean under normality is 2/pi = 0.637, a constant free of mu and sigma^2. So the mean is more efficient for all parameter values.

TIPS & TRICKS (AI-derived explanation)

- Asymptotic variances under N(mu, sigma^2): mean has sigma^2/n, median has pi sigma^2/(2n). Their ratio 2/pi is independent of both parameters.
- Because sigma^2 cancels in the ratio, no condition on sigma^2 can possibly matter - options (b), (c) and (d) impose conditions that are structurally irrelevant.
- 0.637 means the median 'wastes' about 36% of the sample: 100 observations analysed by the median carry the information of only 64 analysed by the mean.
- The picture reverses for heavy-tailed parents; for the Laplace distribution the median is the more efficient estimator - useful contrast for a follow-up question.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For a random sample of size n from N(mu, sigma^2), the sample mean has variance sigma^2/n.
2. The sample median is asymptotically normal with variance 1/(4n [f(m)]^2), where f is the density at the median m = mu.
3. For the normal density, f(mu) = 1/(sigma sqrt(2 pi)), so the asymptotic variance of the median is pi sigma^2/(2n).

4. The asymptotic relative efficiency of the median with respect to the mean is (sigma^2/n)/(pi sigma^2/(2n)) = 2/pi = 0.637.
5. This ratio involves neither mu nor sigma^2, so the mean is more efficient whatever their values.
6. Hence the mean is more asymptotically efficient for all mu and sigma^2, option (a).